ASD-STE100 TECHNICAL MATHEMATICAL MANUAL

Linear Algebra: The Math Book

Eigenvalues, SVD, Markov Matrices, Computer Graphics, and Complex FFT

Writing Standard: ASD-STE100 (Simplified Technical English).
Pedagogical Base: Every syllabus equation is derived step-by-step from Grade 12 (+2) mathematics.
Completeness: Contains all 28 Book Examples/Worked Examples and all 255 Problem Set Exercises copied verbatim with complete mathematical solutions.

Syllabus Coverage: Chapters 6, 8, and 10

Sections 6.1, 6.2, 6.4, 6.6, 6.7 · Sections 8.3, 8.7 · Sections 10.1, 10.2, 10.3

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Your +2 Mathematical Toolkit

We derive every equation in this manual from ten Grade 12 (+2) mathematical rules. Refer to labels **T1** through **T10** in the derivations.

T1 2 by 2 Determinant and Singular Matrix Test

For a 2×2 matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $\det(M) = ad - bc$. A homogeneous system $Mx = 0$ has a nonzero solution $x \neq 0$ if and only if $\det(M) = 0$ (M is singular).

T2 Quadratic Formula and Vieta's Root Relations

The roots of $\lambda^2 - b\lambda + c = 0$ are $\lambda = \dfrac{b \pm \sqrt{b^2 - 4c}}{2}$. When factored as $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 = 0$, the sum of the roots is $\lambda_1 + \lambda_2 = b$ and the product of the roots is $\lambda_1\lambda_2 = c$.

T3 2 by 2 Matrix Inverse Formula

When $\det(M) = ad - bc \neq 0$, the inverse of $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is:

$$M^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

T4 Matrix Multiplication by Columns and Transpose Rule

If $S = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$ has columns x_i, then $AS = \begin{bmatrix} Ax_1 & \cdots & Ax_n \end{bmatrix}$. For any matrices A and B, the transpose reverses the product order: $(AB)^T = B^T A^T$ and $(AB)^{-1} = B^{-1}A^{-1}$.

T5 Dot Product, Length, and Perpendicular Vectors

For real vectors $x, y \in \mathbb{R}^n$, the dot product is $x^T y = x_1 y_1 + \cdots + x_n y_n$. The squared length is $\|x\|^2 = x^T x$. Vectors x and y are perpendicular (orthogonal) when $x^T y = 0$.

T6 Infinite Geometric Series

For any number x with $|x| < 1$, the infinite geometric series converges:

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1 - x}.$$

T7 Complex Numbers and Complex Conjugate

The imaginary unit satisfies $i^2 = -1$. A complex number is $z = a + ib$ with real part a and imaginary part b. Its complex conjugate is $\bar{z} = a - ib$. The product is real and nonnegative: $z\bar{z} = (a + ib)(a - ib) = a^2 + b^2 = |z|^2$.

T8 Euler's Formula and Polar Form

From the Taylor series of e^x, $\cos\theta$, and $\sin\theta$, Euler's formula is $e^{i\theta} = \cos\theta + i\sin\theta$. Any complex number is $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$, where $r = \sqrt{a^2+b^2}$ and $\bar{z} = re^{-i\theta}$.

T9 Double-Angle Trigonometric Identities

For any angle θ: $\cos^2\theta + \sin^2\theta = 1$, $\cos(2\theta) = \cos^2\theta - \sin^2\theta$, and $\sin(2\theta) = 2\sin\theta\cos\theta$.

T10 Determinant of a Product and Transpose

For square matrices A and B: $\det(AB) = (\det A)(\det B)$, $\det(A^T) = \det(A)$, and $\det(M^{-1}) = 1/\det(M)$.

MODULE 1

Eigenvalues, Eigenvectors, and Matrix Decompositions

CHAPTER 1

Introduction to Eigenvalues (Section 6.1)

Syllabus Equations: $Ax = \lambda x$, $(A - \lambda I)x = 0$, $\det(A - \lambda I) = 0$, $\sum \lambda_i = \text{trace}(A)$, $\prod \lambda_i = \det(A)$, complex eigenvalues $\lambda = \pm i$.

1.1 Derivation 1: The Characteristic Equation $\det(A - \lambda I) = 0$

GOAL

Find the special numbers λ (eigenvalues) and nonzero vectors x (eigenvectors) that keep the same direction when multiplied by a square matrix A.

FROM YOUR +2 BASE

Use **T1**: A linear system $Mx = 0$ has a nonzero solution $x \neq 0$ if and only if $\det(M) = 0$. Also use the identity matrix property $Ix = x$.

Step 1 Write the condition that Ax is a scalar multiple of x:

$$Ax = \lambda x, \qquad (x \neq 0). \tag{1.1}$$

Step 2 Replace λx with $\lambda I x$ and move the term to the left side:

$$\begin{aligned} Ax - \lambda Ix &= 0 && \text{use } Ix = x \\ (A - \lambda I)x &= 0. && \text{factor out } x \text{ on the right} \end{aligned} \tag{1.2}$$

Step 3 Require a nonzero eigenvector $x \neq 0$. By **T1**, the matrix $M = A - \lambda I$ must be singular:

$$\boxed{\det(A - \lambda I) = 0.} \tag{1.3}$$

RESULT (Eigenvalue and Eigenvector Equations)

1. Solve the characteristic polynomial $\det(A - \lambda I) = 0$ to find the n eigenvalues $\lambda_1, \ldots, \lambda_n$.
2. For each λ, solve $(A - \lambda I)x = 0$ to find the eigenvectors x in the nullspace of $A - \lambda I$.

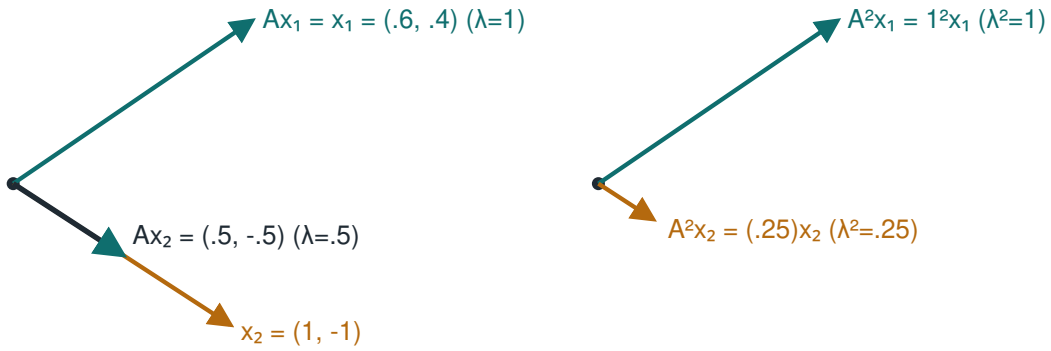


Fig. 1.1 Eigenvectors keep their directions when multiplied by A and A^2. The eigenvalues are squared.

1.2 Derivation 2: Trace and Determinant Checks

GOAL

Prove that the sum of the eigenvalues equals the trace of A and the product of the eigenvalues equals $\det(A)$.

FROM YOUR +2 BASE

Use **T1** and **T2** for a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and polynomial factoring for $n \times n$.

Step 1 Subtract λ from the diagonal of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and compute the determinant:

$$\begin{aligned} \det(A - \lambda I) = \det \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} &= (a-\lambda)(d-\lambda) - bc && \text{by } \textbf{T1} \\ &= \lambda^2 - (a+d)\lambda + (ad - bc) = 0. && \text{expand in powers of } \lambda \end{aligned} \tag{1.4}$$

Step 2 Factor the polynomial using its roots λ_1 and λ_2:

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 = 0. \tag{1.5}$$

Step 3 Match the coefficients of λ^1 and λ^0 by **T2**:

$$\begin{aligned} \lambda_1 + \lambda_2 &= a + d = \operatorname{trace}(A), && \text{sum of diagonal entries} \\ \lambda_1\lambda_2 &= ad - bc = \det(A). && \text{constant term} \end{aligned} \tag{1.6}$$

Step 4 For an $n \times n$ matrix, factor $\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda)$. Set $\lambda = 0$:

$$\boxed{\det(A) = \lambda_1\lambda_2\cdots\lambda_n, \qquad \operatorname{trace}(A) = a_{11} + a_{22} + \cdots + a_{nn} = \lambda_1 + \lambda_2 + \cdots + \lambda_n.} \tag{1.7}$$

WATCH OUT

Row operations (elimination) change the eigenvalues. The pivots of U multiply to give $\det(A)$, but the sum of the pivots is **not** the sum of the eigenvalues.

1.3 Derivation 3: Powers A^k and Shift $A + cI$

GOAL

Find the eigenvalues and eigenvectors of A^2, A^k, and $A + cI$ from $Ax = \lambda x$.

Step 1 Start with $Ax = \lambda x$. Multiply both sides on the left by A:

$$A(Ax) = A(\lambda x) \implies A^2x = \lambda(Ax) = \lambda(\lambda x) = \lambda^2 x. \tag{1.8}$$

Repeat k times to get $\boxed{A^k x = \lambda^k x}$. The eigenvector x stays the same; the eigenvalue is λ^k.

Step 2 Add $cIx = cx$ to $Ax = \lambda x$:

$$Ax + cIx = \lambda x + cx \implies \boxed{(A + cI)x = (\lambda + c)x.} \tag{1.9}$$

Shifting a matrix by cI shifts every eigenvalue by c and does not change the eigenvectors.

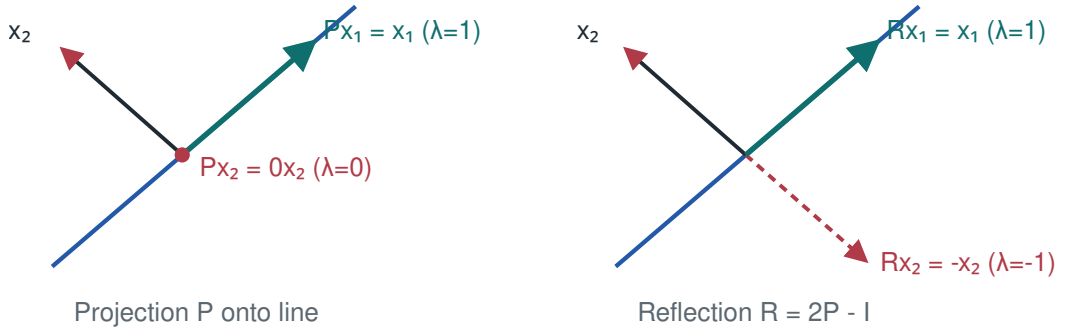


Fig. 1.2 Projection P has eigenvalues $1, 0$. Reflection $R = 2P - I$ has eigenvalues $2(1) - 1 = 1$ and $2(0) - 1 = -1$.

1.4 Section 6.1 In-Text Examples and Worked Examples

BOOK EXAMPLE 6.1.1 Markov Matrix Eigenvalues and Powers

The matrix $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ has two eigenvalues $\lambda = 1$ and $\lambda = 1/2$. Look at $\det(A - \lambda I)$, find the eigenvectors x_1 and x_2, and compute $A^{99} \begin{bmatrix} .8 \\ .2 \end{bmatrix}$.

BOOK EXAMPLE 6.1.2 Projection Matrix

The projection matrix $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ has eigenvalues $\lambda = 1$ and $\lambda = 0$. Find its eigenvectors x_1 and x_2, and project $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ onto Pv.

BOOK EXAMPLE 6.1.3 Reflection Matrix

The reflection matrix $R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues 1 and -1. Relate R to $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ by $R = 2P - I$ and find its eigenvalues and eigenvectors.

BOOK EXAMPLE 6.1.4 Singular Matrix

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is already singular (zero determinant). Find its λ's and x's.

BOOK EXAMPLE 6.1.5 90 Degree Rotation Matrix

The 90° rotation $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has no real eigenvectors. Find its eigenvalues and complex eigenvectors. Check trace and determinant.

BOOK EXAMPLE 6.1 A Worked Example 6.1 A

Find the eigenvalues and eigenvectors of A and A^2 and A^{-1} and $A + 4I$:

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}.$$

Check the trace $\lambda_1 + \lambda_2$ and the determinant $\lambda_1 \lambda_2$ for A and also A^2.

BOOK EXAMPLE 6.1 B Worked Example 6.1 B

Find the eigenvalues and eigenvectors of this 3 by 3 matrix A:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

END-OF-PAGE SOLUTION: Section 6.1 Examples (1–5, 6.1 A, 6.1 B)

Ex. 6.1.1: $\det\begin{bmatrix} .8-\lambda & .3 \\ .2 & .7-\lambda \end{bmatrix} = \lambda^2 - \frac{3}{2}\lambda + \frac{1}{2} = (\lambda-1)(\lambda-\frac{1}{2}) = 0 \implies \lambda_1 = 1, \lambda_2 = \frac{1}{2}$. Nullspace of $A - I$: $x_1 = (.6, .4)$. Nullspace of $A - \frac{1}{2}I$: $x_2 = (1, -1)$. Decompose $\begin{bmatrix} .8 \\ .2 \end{bmatrix} = x_1 + (.2)x_2$. Then $A^{99}\begin{bmatrix} .8 \\ .2 \end{bmatrix} = 1^{99}x_1 + (.2)\left(\frac{1}{2}\right)^{99}x_2 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} + (.2)\left(\frac{1}{2}\right)^{99}\begin{bmatrix} 1 \\ -1 \end{bmatrix} \approx \begin{bmatrix} .6000 \\ .4000 \end{bmatrix}$.

Ex. 6.1.2: Columns add to 1 $\implies \lambda_1 = 1$ with $x_1 = (1, 1)$. Singular $\implies \lambda_2 = 0$ with $x_2 = (1, -1)$. Since $Px_2 = 0$ and $Px_1 = x_1$, $Pv = P\begin{bmatrix} 1 \\ -1 \end{bmatrix} + P\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Ex. 6.1.3: $R = 2P - I$. Eigenvalues of R are $2\lambda(P) - 1$, giving $2(1) - 1 = 1$ and $2(0) - 1 = -1$. Eigenvectors are unchanged: $x_1 = (1, 1)$ and $x_2 = (1, -1)$.

Ex. 6.1.4: $\det(A - \lambda I) = (1-\lambda)(4-\lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0 \implies \lambda_1 = 0, \lambda_2 = 5$. For $\lambda_1 = 0$: $(A - 0I)x = 0 \implies x_1 = (2, -1)$. For $\lambda_2 = 5$: $(A - 5I)x = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} x = 0 \implies x_2 = (1, 2)$.

Ex. 6.1.5: $\det(Q - \lambda I) = \lambda^2 + 1 = 0 \implies \lambda_1 = -i, \lambda_2 = i$. Sum $= 0 = \text{trace}(Q)$; Product $= (-i)(i) = 1 = \det(Q)$. Eigenvectors: $x_1 = (1, i)$ for $\lambda = -i$ and $x_2 = (i, 1)$ for $\lambda = i$.

Ex. 6.1 A: $\det(A - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0 \implies \lambda_1 = 1, \lambda_2 = 3$. Trace $= 2 + 2 = 1 + 3 = 4$; det $= 4 - 1 = 1 \cdot 3 = 3$. Eigenvectors: $x_1 = (1, 1)$ for $\lambda_1 = 1$; $x_2 = (1, -1)$ for $\lambda_2 = 3$. A^2, A^{-1}, and $A + 4I$ keep the same eigenvectors x_1, x_2. A^2 has $\lambda^2 = 1, 9$ (trace $5 + 5 = 10 = 1 + 9$, det $= 25 - 16 = 9$). A^{-1} has $\lambda^{-1} = 1, \frac{1}{3}$. $A + 4I$ has $\lambda + 4 = 5, 7$.

Ex. 6.1 B: $\det(A - \lambda I) = (1-\lambda)[(2-\lambda)(1-\lambda) - 2] = (1-\lambda)(-\lambda)(3-\lambda) = 0 \implies \lambda = 0, 1, 3$. Eigenvectors: $x_1 = (1, 1, 1)$ for $\lambda_1 = 0$; $x_2 = (1, 0, -1)$ for $\lambda_2 = 1$; $x_3 = (1, -2, 1)$ for $\lambda_3 = 3$.

1.5 Problem Set 6.1 (Verbatim Questions and Solutions)

1. The example at the start of the chapter has powers of this matrix A:

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} .70 & .45 \\ .30 & .55 \end{bmatrix} \quad \text{and} \quad A^\infty = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$$

Find the eigenvalues of these matrices. All powers have the same eigenvectors.
(a) Show from A how a row exchange can produce different eigenvalues.
(b) Why is a zero eigenvalue *not* changed by the steps of elimination?

2. Find the eigenvalues and the eigenvectors of these two matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

$A + I$ has the ____ eigenvectors as A. Its eigenvalues are ____ by 1.

3. Compute the eigenvalues and eigenvectors of A and A^{-1}. Check the trace!

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} -1/2 & 1 \\ 1/2 & 0 \end{bmatrix}.$$

A^{-1} has the ____ eigenvectors as A. When A has eigenvalues λ_1 and λ_2, its inverse has eigenvalues ____.

4. Compute the eigenvalues and eigenvectors of A and A^2:

$$A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} 7 & -3 \\ -2 & 6 \end{bmatrix}.$$

A^2 has the same ____ as A. When A has eigenvalues λ_1 and λ_2, A^2 has eigenvalues ____. In this example, why is $\lambda_1^2 + \lambda_2^2 = 13$?

5. Find the eigenvalues of A and B (easy for triangular matrices) and $A + B$:

$$A = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A + B = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}.$$

Eigenvalues of $A + B$ (*are equal to*)(*are not equal to*) eigenvalues of A plus eigenvalues of B.

6. Find the eigenvalues of A and B and AB and BA:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad AB = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}.$$

(a) Are the eigenvalues of AB equal to eigenvalues of A times eigenvalues of B?
(b) Are the eigenvalues of AB equal to the eigenvalues of BA?

7. Elimination produces $A = LU$. The eigenvalues of U are on its diagonal; they are the ____. The eigenvalues of L are on its diagonal; they are all ____. The eigenvalues of A are not the same as ____.

8. (a) If you know that x is an eigenvector, the way to find λ is to ____.
(b) If you know that λ is an eigenvalue, the way to find x is to ____.

9. What do you do to the equation $Ax = \lambda x$, in order to prove (a), (b), and (c)?
(a) λ^2 is an eigenvalue of A^2, as in Problem 4.
(b) λ^{-1} is an eigenvalue of A^{-1}, as in Problem 3.
(c) $\lambda + 1$ is an eigenvalue of $A + I$, as in Problem 2.

10. Find the eigenvalues and eigenvectors for both of these Markov matrices A and A^∞. Explain from those answers why A^{100} is close to A^∞:

$$A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix} \quad \text{and} \quad A^\infty = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

END-OF-PAGE SOLUTION: Problem Set 6.1 (Problems 1–10)

1. Eigenvalues of A: 1 and 0.5. Eigenvalues of A^2: $1^2 = 1$ and $0.5^2 = 0.25$. Eigenvalues of A^∞: 1 and 0. (a) Exchanging rows of A gives $B = \begin{bmatrix} .2 & .7 \\ .8 & .3 \end{bmatrix}$. Then $\det(B - \lambda I) = \lambda^2 - 0.5\lambda - 0.5 = (\lambda - 1)(\lambda + 0.5) = 0$, so $\lambda = 1, -0.5$. (b) $\lambda = 0$ means A is singular ($\det A = 0$). Elimination preserves singularity (U also has a zero pivot and $\det U = 0$), so $\lambda = 0$ remains an eigenvalue.
2. For A: $\det(A - \lambda I) = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \implies \lambda_1 = 5, \lambda_2 = -1$ with $x_1 = (1,1), x_2 = (2,-1)$. For $A + I$: $\lambda_1 = 6, \lambda_2 = 0$ with the same x_1, x_2. Blanks: **same**; **increased**.
3. For A: $\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0 \implies \lambda_1 = 2, \lambda_2 = -1$; $\text{trace}(A) = 0 + 1 = 2 + (-1) = 1$. Eigenvectors: $x_1 = (1,1), x_2 = (2,-1)$. For A^{-1}: $\lambda = 1/2, -1$; $\text{trace}(A^{-1}) = -1/2 + 0 = 1/2 - 1 = -1/2$. Blanks: **same**; $1/\lambda_1$ **and** $1/\lambda_2$.
4. For A: $\lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2) = 0 \implies \lambda_1 = -3, \lambda_2 = 2$ with $x_1 = (3,-2), x_2 = (1,1)$. For A^2: $\lambda = 9, 4$ with the same x_1, x_2. Blanks: **eigenvectors**; λ_1^2 **and** λ_2^2. Reason: $\lambda_1^2 + \lambda_2^2 = 9 + 4 = 13 = 7 + 6 = \text{trace}(A^2)$.
5. $\lambda(A) = 3, 1$; $\lambda(B) = 1, 3$. For $A + B$: $\det(A + B - \lambda I) = (4 - \lambda)^2 - 1 = 0 \implies \lambda(A + B) = 5, 3$. Choice: **are not equal to** (since $3 + 1 = 4$ and $1 + 3 = 4$, or $3 + 3 = 6$ and $1 + 1 = 2$, neither is $5, 3$).
6. $\lambda(A) = 1, 1$; $\lambda(B) = 1, 1$. For AB and BA: $\lambda^2 - 4\lambda + 1 = 0 \implies \lambda = 2 \pm \sqrt{3}$. (a) **No**. (b) **Yes**.
7. Blanks: **pivots**; **1**; **the pivots (or eigenvalues of U or L)**.
8. (a) **multiply Ax and compare with x ($\lambda = (Ax)_i / x_i$)**. (b) **solve** $(A - \lambda I)x = 0$.
9. (a) Multiply $Ax = \lambda x$ by A to get $A^2 x = \lambda Ax = \lambda^2 x$. (b) Multiply by A^{-1} and λ^{-1} to get $A^{-1}x = \lambda^{-1}x$. (c) Add $Ix = x$ to get $(A + I)x = (\lambda + 1)x$.
10. For A: $\lambda_1 = 1$ with $x_1 = (1,2)$ (or $(1/3, 2/3)$), and $\lambda_2 = 0.4$ with $x_2 = (1,-1)$. For A^∞: $\lambda_1 = 1$ with $x_1 = (1/3, 2/3)$ and $\lambda_2 = 0$ with $x_2 = (1,-1)$. Since $0.4^{100} \approx 0$, A^{100} has eigenvalues 1 and $0.4^{100} \approx 0$ with the exact same eigenvectors as A^∞, so $A^{100} \approx A^\infty$.

11. Here is a strange fact about 2 by 2 matrices with eigenvalues $\lambda_1 \neq \lambda_2$: The columns of $A - \lambda_1 I$ are multiples of the eigenvector x_2. Any idea why this should be?

12. Find three eigenvectors for this matrix P (projection matrices have $\lambda = 1$ and 0):

$$P = \begin{bmatrix} .2 & .4 & 0 \\ .4 & .8 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

If two eigenvectors share the same λ, so do all their linear combinations. Find an eigenvector of P with no zero components.

13. From the unit vector $u = (\frac{1}{6}, \frac{1}{6}, \frac{3}{6}, \frac{5}{6})$ construct the rank one projection matrix $P = uu^T$. This matrix has $P^2 = P$ because $u^T u = 1$.
(a) $Pu = u$ comes from $(uu^T)u = u(____)$. Then u is an eigenvector with $\lambda = 1$.
(b) If v is perpendicular to u show that $Pv = 0$. Then $\lambda = 0$.
(c) Find three independent eigenvectors of P all with eigenvalue $\lambda = 0$.

14. Solve $\det(Q - \lambda I) = 0$ by the quadratic formula to reach $\lambda = \cos\theta \pm i \sin\theta$:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \text{rotates the } xy \text{ plane by the angle } \theta. \text{ No real } \lambda\text{'s.}$$

Find the eigenvectors of Q by solving $(Q - \lambda I)x = 0$. Use $i^2 = -1$.

15. Every permutation matrix leaves $x = (1, 1, \ldots, 1)$ unchanged. Then $\lambda = 1$. Find two more λ's (possibly complex) for these permutations, from $\det(P - \lambda I) = 0$:

$$P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

16. *The determinant of A equals the product* $\lambda_1 \lambda_2 \cdots \lambda_n$. Start with the polynomial $\det(A - \lambda I)$ separated into its n factors (always possible). Then set $\lambda = 0$:

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda) \quad \text{so} \quad \det A = ____.$$

Check this rule in Example 1 where the Markov matrix has $\lambda = 1$ and $\frac{1}{2}$.

17. The sum of the diagonal entries (the *trace*) equals the sum of the eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{has} \quad \det(A - \lambda I) = \lambda^2 - (a + d)\lambda + ad - bc = 0.$$

The quadratic formula gives the eigenvalues $\lambda = (a + d + \sqrt{\quad})/2$ and $\lambda = ____$. Their sum is $____$. If A has $\lambda_1 = 3$ and $\lambda_2 = 4$ then $\det(A - \lambda I) = ____$.

18. If A has $\lambda_1 = 4$ and $\lambda_2 = 5$ then $\det(A - \lambda I) = (\lambda - 4)(\lambda - 5) = \lambda^2 - 9\lambda + 20$. Find three matrices that have trace $a + d = 9$ and determinant 20 and $\lambda = 4, 5$.

19. A 3 by 3 matrix B is known to have eigenvalues $0, 1, 2$. This information is enough to find three of these (give the answers where possible):
(a) the rank of B (b) the determinant of $B^T B$ (c) the eigenvalues of $B^T B$ (d) the eigenvalues of $(B^2 + I)^{-1}$.

20. Choose the last rows of A and C to give eigenvalues $4,7$ and $1,2,3$:

$$\textbf{Companion matrices} \qquad A = \begin{bmatrix} 0 & 1 \\ * & * \end{bmatrix} \qquad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ * & * & * \end{bmatrix}.$$

END-OF-PAGE SOLUTION: Problem Set 6.1 (Problems 11–20)

11. By the Cayley-Hamilton theorem (or direct check), $(A-\lambda_2 I)(A-\lambda_1 I)=0$. Therefore every column of $A-\lambda_1 I$ is in the nullspace of $A-\lambda_2 I$, which is the line spanned by x_2.

12. For $\lambda=0$: $x_1=(-2,1,0)$. For $\lambda=1$: $x_2=(1,2,0)$ and $x_3=(0,0,1)$. Combination with no zeros: $x_2+x_3=(1,2,1)$ has $\lambda=1$.

13. (a) $u^Tu=1$. (b) $Pv=(uu^T)v=u(u^Tv)=u(0)=0=0v$. (c) $v_1=(1,-1,0,0)$, $v_2=(3,0,-1,0)$, $v_3=(5,0,0,-1)$.

14. $\det(Q-\lambda I)=(\cos\theta-\lambda)^2+\sin^2\theta=\lambda^2-2\lambda\cos\theta+1=0 \implies \lambda=\cos\theta\pm i\sin\theta$. For $\lambda_1=\cos\theta+i\sin\theta$: $(-i\sin\theta)x_1-(\sin\theta)x_2=0 \implies x=(1,-i)$. For $\lambda_2=\cos\theta-i\sin\theta$: $x=(1,i)$.

15. For P_1: $\det(P_1-\lambda I)=1-\lambda^3=0 \implies \lambda=1,e^{2\pi i/3},e^{4\pi i/3}$ (which are $-\frac{1}{2}\pm i\frac{\sqrt{3}}{2}$). For P_2: $\det(P_2-\lambda I)=(1-\lambda)(\lambda^2-1)=0 \implies \lambda=1,1,-1$.

16. Blank: $\lambda_1\lambda_2\cdots\lambda_n$. In Example 1: $\det A=(.8)(.7)-(.3)(.2)=0.56-0.06=0.5=(1)(1/2)$.

17. Blanks: $(a+d)^2-4(ad-bc)$ (or $(a-d)^2+4bc$); $(a+d-\sqrt{(a+d)^2-4(ad-bc)})/2$; $a+d$; $\lambda^2-7\lambda+12$.

18. $\begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}, \begin{bmatrix} 6 & 2 \\ -1 & 3 \end{bmatrix}$.

19. (a) $\operatorname{rank}(B)=2$ (two nonzero eigenvalues, one zero eigenvalue). (b) $\det(B^TB)=(\det B)^2=0^2=0$. (c) Cannot be determined from $\lambda(B)$ alone unless B is symmetric. (d) Eigenvalues of $(B^2+I)^{-1}$ are $(0^2+1)^{-1}=1$, $(1^2+1)^{-1}=\frac{1}{2}$, $(2^2+1)^{-1}=\frac{1}{5}$.

20. For A: $\lambda^2-11\lambda+28=0 \implies$ last row is $\begin{bmatrix} -28 & 11 \end{bmatrix}$. For C: $(\lambda-1)(\lambda-2)(\lambda-3)=\lambda^3-6\lambda^2+11\lambda-6=0 \implies$ last row is $\begin{bmatrix} 6 & -11 & 6 \end{bmatrix}$.

21. *The eigenvalues of A equal the eigenvalues of A^T.* This is because $\det(A - \lambda I)$ equals $\det(A^T - \lambda I)$. That is true because ____. Show by an example that the eigenvectors of A and A^T are *not* the same.

22. Construct any 3 by 3 Markov matrix M: positive entries down each column add to 1. Show that $M^T(1,1,1) = (1,1,1)$. By Problem 21, $\lambda = 1$ is also an eigenvalue of M. Challenge: A 3 by 3 singular Markov matrix with trace $\frac{1}{2}$ has what λ's?

23. Find three 2 by 2 matrices that have $\lambda_1 = \lambda_2 = 0$. The trace is zero and the determinant is zero. A might not be the zero matrix but check that $A^2 = 0$.

24. This matrix is singular with rank one. Find three λ's and three eigenvectors:

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 2 \end{bmatrix}.$$

25. Suppose A and B have the same eigenvalues $\lambda_1, \ldots, \lambda_n$ with the same independent eigenvectors $x_1, \ldots, x_n$. Then $A = B$. *Reason:* Any vector x is a combination $c_1x_1 + \cdots + c_nx_n$. What is Ax? What is Bx?

26. The block B has eigenvalues $1, 2$ and C has eigenvalues $3, 4$ and D has eigenvalues $5, 7$. Find the eigenvalues of the 4 by 4 matrix A:

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} = \begin{bmatrix} 0 & 1 & 3 & 0 \\ -2 & 3 & 0 & 4 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 1 & 6 \end{bmatrix}.$$

27. Find the rank and the four eigenvalues of A and C:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

28. Subtract I from the previous A. Find the λ's and then the determinants of

$$B = A - I = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad C = I - A = \begin{bmatrix} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}.$$

29. (Review) Find the eigenvalues of A, B, and C:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

30. When $a + b = c + d$ show that $(1,1)$ is an eigenvector and find both eigenvalues: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

END-OF-PAGE SOLUTION: Problem Set 6.1 (Problems 21–30)

21. Blank: $(A-\lambda I)^T = A^T - \lambda I$ **and** $\det M^T = \det M$. Example: $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ has $x = (1,0)$ for $\lambda = 1$, while $A^T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ has $x = (1,1)$ for $\lambda = 1$.

22. Row i of M^T is column i of M, whose entries add to 1. Thus $(M^T(1,1,1))_i = m_{1i} + m_{2i} + m_{3i} = 1$. Challenge: Markov $\Longrightarrow$ $\lambda_1 = 1$; singular $\Longrightarrow$ $\lambda_2 = 0$; trace $= \frac{1}{2}$ $\Longrightarrow$ $\lambda_3 = \frac{1}{2} - 1 - 0 = -\frac{1}{2}$.

23. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$. Each satisfies $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

24. $\lambda_1 = \text{trace}(A) = 2+2+2 = 6$ with $x_1 = (1,2,1)$. Since $\text{rank}(A) = 1$, $\lambda_2 = \lambda_3 = 0$ with $x_2 = (1,-2,0)$ and $x_3 = (1,0,-1)$ in the nullspace $2x + y + 2z = 0$.

25. $Ax = c_1\lambda_1 x_1 + \cdots + c_n\lambda_n x_n$ and $Bx = c_1\lambda_1 x_1 + \cdots + c_n\lambda_n x_n$. Since $Ax = Bx$ for every vector x, $A = B$.

26. $\det(A - \lambda I) = \det(B - \lambda I)\det(D - \lambda I)$. The block C has no effect. Eigenvalues are $1, 2, 5, 7$.

27. For A: $\text{rank}(A) = 1$, $\lambda = 4, 0, 0, 0$. For C: $\text{rank}(C) = 2$, $\lambda = 2, 2, 0, 0$.

28. For $B = A - I$: subtract 1 from $\lambda(A)$ to get $\lambda(B) = 3, -1, -1, -1$ and $\det B = (3)(-1)^3 = -3$. For $C = -B$: $\lambda(C) = -3, 1, 1, 1$ and $\det C = (-3)(1)^3 = -3$.

29. $\lambda(A) = 1, 4, 6$. For B: $\det(B - \lambda I) = (2-\lambda)(\lambda^2 - 3) = 0 \Longrightarrow \lambda(B) = 2, \sqrt{3}, -\sqrt{3}$. For C: $\lambda(C) = 6, 0, 0$.

30. $A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ c+d \end{bmatrix} = (a+b)\begin{bmatrix} 1 \\ 1 \end{bmatrix} \Longrightarrow \lambda_1 = a + b$. From trace $= a + d$, $\lambda_2 = (a+d) - (a+b) = d - b = a - c$.

31. If we exchange rows 1 and 2 *and* columns 1 and 2, the eigenvalues don't change. Find eigenvectors of A and B for $\lambda = 11$. *Rank one gives* $\lambda_2 = \lambda_3 = 0$.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & 4 \end{bmatrix} \quad \text{and} \quad B = PAP^T = \begin{bmatrix} 6 & 3 & 3 \\ 2 & 1 & 1 \\ 8 & 4 & 4 \end{bmatrix}.$$

32. Suppose A has eigenvalues $0, 3, 5$ with independent eigenvectors u, v, w.
(a) Give a basis for the nullspace and a basis for the column space.
(b) Find a particular solution to $Ax = v + w$. Find all solutions.
(c) $Ax = u$ has no solution. If it did then ____ would be in the column space.

33. Suppose u, v are orthonormal vectors in $\mathbb{R}^2$, and $A = uv^T$. Compute $A^2 = uv^Tuv^T$ to discover the eigenvalues of A. Check that the trace of A agrees with $\lambda_1 + \lambda_2$.

34. Find the eigenvalues of this permutation matrix P from $\det(P - \lambda I) = 0$. Which vectors are not changed by the permutation? They are eigenvectors for $\lambda = 1$. Can you find three more eigenvectors?

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

35. *(Challenge)* There are six 3 by 3 permutation matrices P. What numbers can be the *determinants* of P? What numbers can be *pivots*? What numbers can be the *trace* of P? What *four numbers* can be *eigenvalues* of P, as in Problem 15?

36. *(Challenge) Is there a real 2 by 2 matrix (other than I) with* $A^3 = I$? Its eigenvalues must satisfy $\lambda^3 = 1$. They can be $e^{2\pi i/3}$ and $e^{-2\pi i/3}$. What trace and determinant would this give? Construct a rotation matrix as A (which angle of rotation?).

37. *(Challenge)* (a) Find the eigenvalues and eigenvectors of A. They depend on c:

$$A = \begin{bmatrix} .4 & 1-c \\ .6 & c \end{bmatrix}.$$

(b) Show that A has just one line of eigenvectors when $c = 1.6$.
(c) This is a Markov matrix when $c = .8$. Then A^n will approach what matrix A^∞?

END-OF-PAGE SOLUTION: Problem Set 6.1 (Problems 31–37)

31. For A: $x = (1,3,4)$ gives $Ax = 11x$. For B: exchange components 1 and 2 to get $Px = (3,1,4)$, which gives $B(3,1,4) = 11(3,1,4)$.
32. (a) Nullspace basis: $\{u\}$ (since $Au = 0$). Column space basis: $\{v, w\}$ (since $Av = 3v, Aw = 5w$). (b) Particular solution: $x_p = \frac{1}{3}v + \frac{1}{5}w$. Complete solution: $x = \frac{1}{3}v + \frac{1}{5}w + cu$ for any scalar c. (c) Blank: u (which is impossible because u, v, w are independent and $\{v, w\}$ spans $C(A)$).
33. Since u, v are orthonormal, $v^Tu = 0$. Thus $A^2 = u(v^Tu)v^T = 0$, which means $\lambda_1 = 0$ and $\lambda_2 = 0$. Also $\text{trace}(A) = u_1v_1 + u_2v_2 = u^Tv = 0 = \lambda_1 + \lambda_2$.
34. $\det(P - \lambda I) = \lambda^4 - 1 = 0 \implies \lambda = 1, -1, i, -i$. Multiples of $x_1 = (1,1,1,1)$ are unchanged ($\lambda = 1$). Other three eigenvectors: $x_2 = (1,-1,1,-1)$ for $\lambda = -1$; $x_3 = (1,-i,-1,i)$ for $\lambda = i$; $x_4 = (1,i,-1,-i)$ for $\lambda = -i$.
35. Determinants: 1 or -1. Pivots: always 1 (after row exchanges). Trace: $0, 1$, or 3.

Eigenvalues: $1, -1, e^{2\pi i/3}, e^{4\pi i/3}$.
36. $\text{trace}(A) = e^{2\pi i/3} + e^{-2\pi i/3} = 2\cos(2\pi/3) = -1$; $\det(A) = e^{2\pi i/3}e^{-2\pi i/3} = 1$. Rotation by $\theta = 120^\circ$ $(2\pi/3)$: $A = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$.
37. (a) Columns add to $1 \implies \lambda_1 = 1$ with $x_1 = (1-c, 0.6)$. From trace $= 0.4 + c$, $\lambda_2 = c - 0.6$ with $x_2 = (1, -1)$. (b) When $c = 1.6$, $\lambda_1 = \lambda_2 = 1$ and $x_1 = (-1, 0.6) = -0.6(1, -1)$ (Wait: $x_1 = (1 - 1.6, 0.6) = (-0.6, 0.6) = -0.6x_2$), so only one line of eigenvectors $(1, -1)$ exists. (c) When $c = 0.8$, $x_1 = (0.2, 0.6)$ which normalizes to $(1/4, 3/4) = (.25, .75)$. Thus $A^\infty = \begin{bmatrix} .25 & .25 \\ .75 & .75 \end{bmatrix}$.

CHAPTER 2

Diagonalizing a Matrix (Section 6.2)

Syllabus Equations: $AS = S\Lambda$, $A = S\Lambda S^{-1}$, $S^{-1}AS = \Lambda$, $A^k = S\Lambda^k S^{-1}$, $u_k = A^k u_0 = c_1\lambda_1^k x_1 + \cdots + c_n\lambda_n^k x_n$, Fibonacci $F_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2}$, Geometric Multiplicity GM $\leq$ AM.

2.1 Derivation 1: The Diagonalization Formula $A = S\Lambda S^{-1}$

GOAL

Factor an $n \times n$ matrix A with n linearly independent eigenvectors $x_1, \ldots, x_n$ into $A = S\Lambda S^{-1}$, and derive the power formula $A^k = S\Lambda^k S^{-1}$.

FROM YOUR +2 BASE

Use **T4**: Multiplication of a matrix A by a column matrix $S = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$ acts column by column. A matrix S with n linearly independent columns is invertible.

Step 1 Put the n eigenvectors $x_1, \ldots, x_n$ into the columns of the eigenvector matrix S, and multiply A times S:

$$AS = A\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} Ax_1 & \cdots & Ax_n \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \cdots & \lambda_n x_n \end{bmatrix}. \tag{2.1}$$

Step 2 Factor the right side into S times the diagonal eigenvalue matrix Λ:

$$\begin{bmatrix} \lambda_1 x_1 & \cdots & \lambda_n x_n \end{bmatrix} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = S\Lambda. \tag{2.2}$$

Step 3 Because the n eigenvectors are linearly independent, S^{-1} exists. Multiply $AS = S\Lambda$ on the right by S^{-1} or on the left by S^{-1}:

$$\boxed{AS = S\Lambda \iff A = S\Lambda S^{-1} \iff S^{-1}AS = \Lambda.} \tag{2.3}$$

Step 4 Multiply $A = S\Lambda S^{-1}$ by itself k times. Each interior $S^{-1}S = I$ cancels:

$$A^k = (S\Lambda S^{-1})(S\Lambda S^{-1})\cdots(S\Lambda S^{-1}) = \boxed{S\Lambda^k S^{-1}.} \tag{2.4}$$

2.2 Derivation 2: Independence of Eigenvectors for Distinct λ_i

GOAL

Prove that eigenvectors $x_1, \ldots, x_j$ corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_j$ are linearly independent.

Step 1 For $j = 2$, suppose $c_1x_1 + c_2x_2 = 0$.

Step 2 Multiply by A to get $c_1\lambda_1x_1 + c_2\lambda_2x_2 = 0$. Multiply the first equation by λ_2 to get $c_1\lambda_2x_1 + c_2\lambda_2x_2 = 0$.

Step 3 Subtract the two equations to eliminate x_2:

$$c_1(\lambda_1 - \lambda_2)x_1 = 0. \tag{2.5}$$

Since $\lambda_1 \neq \lambda_2$ and $x_1 \neq 0$, we must have $c_1 = 0$, which forces $c_2 = 0$. For j vectors, multiplying by $(A - \lambda_2 I)\cdots(A - \lambda_j I)$ leaves $(\lambda_1 - \lambda_2)\cdots(\lambda_1 - \lambda_j)c_1x_1 = 0 \implies c_1 = 0$.

2.3 Derivation 3: Difference Equations $u_{k+1} = Au_k$ and Binet's Fibonacci Formula

GOAL

Solve $u_{k+1} = Au_k$ in three steps and derive the closed-form formula for the kth Fibonacci number F_k.

Step 1 (General Solution) Expand the starting vector u_0 in the eigenvector basis: $u_0 = c_1x_1 + \cdots + c_nx_n = Sc$, so $c = S^{-1}u_0$. Multiply by A^k using $A^kx_i = \lambda_i^k x_i$:

$$\boxed{u_k = A^ku_0 = S\Lambda^kS^{-1}u_0 = c_1\lambda_1^kx_1 + c_2\lambda_2^kx_2 + \cdots + c_n\lambda_n^kx_n.} \tag{2.6}$$

Step 2 (Fibonacci System) Write $F_{k+2} = F_{k+1} + F_k$ and $F_{k+1} = F_{k+1}$ as a 2×2 system for $u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$ starting at $u_0 = \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$u_{k+1} = \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = Au_k. \tag{2.7}$$

Step 3 (Eigenvalues and Eigenvectors) Solve $\det(A - \lambda I) = \lambda^2 - \lambda - 1 = 0$ by **T2**:

$$\lambda_1 = \frac{1+\sqrt{5}}{2} \approx 1.618, \quad \lambda_2 = \frac{1-\sqrt{5}}{2} \approx -0.618, \quad x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}. \tag{2.8}$$

Step 4 (Match u_0 and take k steps) Write $u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2}(x_1 - x_2)$. Then $u_k = \frac{\lambda_1^kx_1 - \lambda_2^kx_2}{\lambda_1 - \lambda_2}$. Read off the second component F_k:

$$\boxed{F_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right].} \tag{2.9}$$

2.4 Section 6.2 In-Text Examples and Worked Examples

BOOK EXAMPLE 6.2.1 Triangular Matrix Powers

This $A = \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}$ is triangular so the λ's are on the diagonal: $\lambda = 1$ and $\lambda = 6$. Find S, S^{-1}, Λ, and compute $A^k = S\Lambda^k S^{-1}$.

BOOK EXAMPLE 6.2.2 Powers of a Markov Matrix

Diagonalize the Markov matrix $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ as $A = S\Lambda S^{-1}$, compute A^k, and find the limit A^∞ as $k \to \infty$.

BOOK EXAMPLE 6.2.3 Modified Fibonacci Sequence

Start from $u_0 = (1, 0)$. Compute $A^k u_0$ when $A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ has $\lambda_1 = 2$ with $x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\lambda_2 = -1$ with $x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

BOOK EXAMPLE 6.2 A Worked Example 6.2 A: Lucas Numbers

The **Lucas numbers** are like the Fibonacci numbers except they start with $L_1 = 1$ and $L_2 = 3$. Following the rule $L_{k+2} = L_{k+1} + L_k$, the next Lucas numbers are $4, 7, 11, 18$. Show that the Lucas number L_{100} is $\lambda_1^{100} + \lambda_2^{100}$.

BOOK EXAMPLE 6.2 B Worked Example 6.2 B: Rank-One Shift of $5I$

Find the inverse and the eigenvalues and the determinant of A:

$$A = 5 * \text{eye}(4) - \text{ones}(4) = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}.$$

Describe an eigenvector matrix S that gives $S^{-1}AS = \Lambda$.

END-OF-PAGE SOLUTION: Section 6.2 Examples (1–3, 6.2 A, 6.2 B)

Ex. 6.2.1: Eigenvectors are $x_1 = (1, 0)$ for $\lambda_1 = 1$ and $x_2 = (1, 1)$ for $\lambda_2 = 6$. Thus $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Then $A^k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 6^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6^k - 1 \\ 0 & 6^k \end{bmatrix}$.

Ex. 6.2.2: $S = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & .5 \end{bmatrix}$, $S^{-1} = \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}$. $A^k = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & (.5)^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}$.

As $k \to \infty$, $(.5)^k \to 0$, so $A^\infty = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix} = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}$.

Ex. 6.2.3: Step 1: Solve $c_1\begin{bmatrix}2\\1\end{bmatrix}+c_2\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix} \implies c_1=c_2=\frac{1}{3}$. Step 2 & 3: $u_k=\frac{1}{3}2^k\begin{bmatrix}2\\1\end{bmatrix}+\frac{1}{3}(-1)^k\begin{bmatrix}1\\-1\end{bmatrix}$. Second component is $F_k=\frac{2^k-(-1)^k}{3}$.

Ex. 6.2 A: Set $u_k=\begin{bmatrix}L_{k+1}\\L_k\end{bmatrix}$ with $u_1=\begin{bmatrix}L_2\\L_1\end{bmatrix}=\begin{bmatrix}3\\1\end{bmatrix}$. Since $\lambda_1+\lambda_2=1$ and $\lambda_1^2+\lambda_2^2=3$, $u_1=\lambda_1\begin{bmatrix}\lambda_1\\1\end{bmatrix}+\lambda_2\begin{bmatrix}\lambda_2\\1\end{bmatrix}=\lambda_1x_1+\lambda_2x_2$ ($c_1=\lambda_1, c_2=\lambda_2$). Multiply by A^{99}: $u_{100}=A^{99}u_1=\lambda_1^{100}x_1+\lambda_2^{100}x_2$. The second component is $L_{100}=\lambda_1^{100}+\lambda_2^{100}$.

Ex. 6.2 B: ones(4) has rank 1 and trace 4, so its eigenvalues are $4,0,0,0$. Subtracting from $5I$ gives $\lambda(A)=5-4,5-0,5-0,5-0=1,5,5,5$. Thus $\det A=1\cdot5^3=125$. Using the orthogonal Hadamard matrix $S=H=\frac{1}{2}\begin{bmatrix}1&1&1&1\\1&-1&1&-1\\1&1&-1&-1\\1&-1&-1&1\end{bmatrix}=H^{-1}$, we get $A^{-1}=\frac{1}{5}(\text{eye}(4)+\text{ones}(4))$.

2.5 Problem Set 6.2 (Verbatim Questions and Solutions)

1. (a) Factor these two matrices into $A = S\Lambda S^{-1}$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$$

(b) If $A = S\Lambda S^{-1}$ then $A^3 = (\quad)(\quad)(\quad)$ and $A^{-1} = (\quad)(\quad)(\quad)$.

2. If A has $\lambda_1 = 2$ with eigenvector $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\lambda_2 = 5$ with $x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, use $S\Lambda S^{-1}$ to find A. No other matrix has the same λ's and x's.

3. Suppose $A = S\Lambda S^{-1}$. What is the eigenvalue matrix for $A + 2I$? What is the eigenvector matrix? Check that $A + 2I = (\quad)(\quad)(\quad)^{-1}$.

4. True or false: If the columns of S (eigenvectors of A) are linearly independent, then
(a) A is invertible (b) A is diagonalizable (c) S is invertible (d) S is diagonalizable.

5. If the eigenvectors of A are the columns of I, then A is a ____ matrix. If the eigenvector matrix S is triangular, then S^{-1} is triangular. Prove that A is also triangular.

6. Describe all matrices S that diagonalize this matrix A (find all eigenvectors):

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}.$$

Then describe all matrices that diagonalize A^{-1}.

7. Write down the most general matrix that has eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

8. Diagonalize the Fibonacci matrix by completing S^{-1}:

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}.$$

Do the multiplication $S\Lambda^k S^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to find its second component. This is the kth Fibonacci number $F_k = (\lambda_1^k - \lambda_2^k)/(\lambda_1 - \lambda_2)$.

9. Suppose G_{k+2} is the *average* of the two previous numbers G_{k+1} and G_k:

$$\begin{aligned} G_{k+2} &= \tfrac{1}{2}G_{k+1} + \tfrac{1}{2}G_k \\ G_{k+1} &= G_{k+1} \end{aligned} \qquad \text{is} \qquad \begin{bmatrix} G_{k+2} \\ G_{k+1} \end{bmatrix} = \begin{bmatrix} \quad A \quad \end{bmatrix} \begin{bmatrix} G_{k+1} \\ G_k \end{bmatrix}.$$

(a) Find the eigenvalues and eigenvectors of A.
(b) Find the limit as $n \to \infty$ of the matrices $A^n = S\Lambda^n S^{-1}$.
(c) If $G_0 = 0$ and $G_1 = 1$ show that the Gibonacci numbers approach $\frac{2}{3}$.

10. Prove that every third Fibonacci number in $0, 1, 1, 2, 3, \ldots$ is even.

END-OF-PAGE SOLUTION: Problem Set 6.2 (Problems 1–10)

1. (a) First A: $\lambda = 1, 3$, $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, $S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Second A: $\lambda = 0, 4$, $S = \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$, $S^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$. (b) $A^3 = (S)(\Lambda^3)(S^{-1})$ and $A^{-1} = (S)(\Lambda^{-1})(S^{-1})$.

2. $A = S\Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$.

3. Eigenvalue matrix is $\Lambda + 2I$. Eigenvector matrix is S. Check: $S(\Lambda + 2I)S^{-1} = S\Lambda S^{-1} + 2SIS^{-1} = A + 2I$.

4. (a) **False** (λ could be 0). (b) **True**. (c) **True**. (d) **False** (S itself need not be diagonalizable, e.g., $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$).

5. Blank: **diagonal**. Since the product of upper (or lower) triangular matrices S, Λ, and S^{-1} is triangular, $A = S\Lambda S^{-1}$ is triangular.

6. Eigenvectors of A are $(2, 1)$ for $\lambda = 4$ and $(0, 1)$ for $\lambda = 2$. All diagonalizing matrices have the form $S = \begin{bmatrix} 2a & 0 \\ a & b \end{bmatrix}$ or $\begin{bmatrix} 0 & 2a \\ b & a \end{bmatrix}$ with $ab \neq 0$. The exact same matrices S diagonalize A^{-1}.

7. $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \lambda_1 + \lambda_2 & \lambda_1 - \lambda_2 \\ \lambda_1 - \lambda_2 & \lambda_1 + \lambda_2 \end{bmatrix} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$.

8. $S^{-1} = \frac{1}{\lambda_1 - \lambda_2}\begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$. Then $S\Lambda^k S^{-1}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2}\begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \lambda_1^k \\ -\lambda_2^k \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2}\begin{bmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{bmatrix}$.

9. $A = \begin{bmatrix} .5 & .5 \\ 1 & 0 \end{bmatrix}$. (a) $\lambda_1 = 1, \lambda_2 = -0.5$ with $x_1 = (1, 1), x_2 = (1, -2)$. (b) $A^n = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 1^n & 0 \\ 0 & (-.5)^n \end{bmatrix}\frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \to \begin{bmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix}$. (c) $A^\infty \begin{bmatrix} G_1 \\ G_0 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 2/3 \end{bmatrix}$.

10. Modulo 2, the Fibonacci matrix satisfies $A^3 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \pmod 2$. Thus the parity repeats with period 3: even (0), odd (1), odd (1), even (2), etc.

11. True or false: If the eigenvalues of A are $2, 2, 5$ then the matrix is certainly
(a) invertible (b) diagonalizable (c) not diagonalizable.

12. True or false: If the only eigenvectors of A are multiples of $(1, 4)$ then A has
(a) no inverse (b) a repeated eigenvalue (c) no diagonalization $S\Lambda S^{-1}$.

13. Complete these matrices so that $\det A = 25$. Then check that $\lambda = 5$ is repeated—the trace is 10 so the determinant of $A - \lambda I$ is $(\lambda - 5)^2$. Find an eigenvector with $Ax = 5x$. These matrices will not be diagonalizable because there is no second line of eigenvectors.

$$A = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}.$$

14. The matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is not diagonalizable because the rank of $A - 3I$ is ____. Change one entry to make A diagonalizable. Which entries could you change?

15. $A^k = S\Lambda^k S^{-1}$ approaches the zero matrix as $k \to \infty$ if and only if every λ has absolute value less than ____. Which of these matrices has $A^k \to 0$?

$$A_1 = \begin{bmatrix} .6 & .9 \\ .4 & .1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} .6 & .9 \\ .1 & .6 \end{bmatrix}.$$

16. (Recommended) Find Λ and S to diagonalize A_1 in Problem 15. What is the limit of Λ^k as $k \to \infty$? What is the limit of $S\Lambda^k S^{-1}$? In the columns of this limiting matrix you see the ____.

17. Find Λ and S to diagonalize A_2 in Problem 15. What is $(A_2)^{10}u_0$ for these u_0?

$$u_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \text{and} \quad u_0 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{and} \quad u_0 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

18. Diagonalize A and compute $S\Lambda^k S^{-1}$ to prove this formula for A^k:

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \text{has} \quad A^k = \frac{1}{2}\begin{bmatrix} 1+3^k & 1-3^k \\ 1-3^k & 1+3^k \end{bmatrix}.$$

19. Diagonalize B and compute $S\Lambda^k S^{-1}$ to prove this formula for B^k:

$$B = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix} \quad \text{has} \quad B^k = \begin{bmatrix} 5^k & 5^k - 4^k \\ 0 & 4^k \end{bmatrix}.$$

20. Suppose $A = S\Lambda S^{-1}$. Take determinants to prove $\det A = \det \Lambda = \lambda_1\lambda_2 \cdots \lambda_n$. This quick proof only works when A can be ____.

END-OF-PAGE SOLUTION: Problem Set 6.2 (Problems 11–20)

11. (a) **True** ($\det A = 20 \neq 0$). (b) **False** (GM of $\lambda = 2$ could be 1). (c) **False** (GM of $\lambda = 2$ could be 2).
12. (a) **False** (λ could be nonzero). (b) **True** (distinct λ's would give two independent eigenvectors). (c) **True** (2×2 matrix with only one line of eigenvectors cannot be diagonalized).

13. First A: $\begin{bmatrix} 8 & -9 \\ 1 & 2 \end{bmatrix}$ has $x = (3,1)$. Second A: $\begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix}$ has $x = (1,-1)$. Third A: $\begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$ has $x = (1,-1)$.

14. Blank: **1**. Change the $(1,2)$ entry from 1 to 0, or change $(1,1)$ or $(2,2)$ to any number $\neq 3$, or change $(2,1)$ to any number $\neq 0$.

15. Blank: **1**. For A_1: $\lambda = 1, -0.3$ (does not go to 0). For A_2: $\lambda = 0.6 \pm 0.3 = 0.9, 0.3$ (both $|\lambda| < 1$, so $A_2^k \to 0$).

16. $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -.3 \end{bmatrix}$, $S = \begin{bmatrix} 9 & 1 \\ 4 & -1 \end{bmatrix}$, $S^{-1} = \frac{1}{13}\begin{bmatrix} 1 & 1 \\ 4 & -9 \end{bmatrix}$. Limit of Λ^k is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Limit of $S\Lambda^k S^{-1}$ is $\frac{1}{13}\begin{bmatrix} 9 & 9 \\ 4 & 4 \end{bmatrix}$. Blank: **steady state eigenvector (for $\lambda = 1$)**.

17. $\Lambda = \begin{bmatrix} .9 & 0 \\ 0 & .3 \end{bmatrix}$, $S = \begin{bmatrix} 3 & 3 \\ 1 & -1 \end{bmatrix}$. Since $u_0 = (3,1)$ and $(3,-1)$ are the eigenvectors, $(A_2)^{10}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = (.9)^{10}\begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $(A_2)^{10}\begin{bmatrix} 3 \\ -1 \end{bmatrix} = (.3)^{10}\begin{bmatrix} 3 \\ -1 \end{bmatrix}$, and $(A_2)^{10}\begin{bmatrix} 6 \\ 0 \end{bmatrix} = (.9)^{10}\begin{bmatrix} 3 \\ 1 \end{bmatrix} + (.3)^{10}\begin{bmatrix} 3 \\ -1 \end{bmatrix}$.

18. $S = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, $S^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Multiplying $S\Lambda^k S^{-1}$ yields the formula.

19. $S = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = S^{-1}$, $\Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$. Multiplying $S\Lambda^k S^{-1}$ yields $\begin{bmatrix} 5^k & 5^k - 4^k \\ 0 & 4^k \end{bmatrix}$.

20. By **T10**, $\det A = (\det S)(\det \Lambda)(\det S^{-1}) = \det \Lambda = \lambda_1 \cdots \lambda_n$. Blank: **diagonalized**.

21. Show that trace ST = trace TS, by adding the diagonal entries of ST and TS:

$$S = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} q & r \\ s & t \end{bmatrix}.$$

Choose T as ΛS^{-1}. Then $S\Lambda S^{-1}$ has the same trace as $\Lambda S^{-1}S = \Lambda$. The trace of A equals the trace of Λ = sum of the eigenvalues.

22. $AB - BA = I$ is impossible since the left side has trace = ____. But find an elimination matrix so that $A = E$ and $B = E^T$ give $AB - BA = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ which has trace zero.

23. If $A = S\Lambda S^{-1}$, diagonalize the block matrix $B = \begin{bmatrix} A & 0 \\ 0 & 2A \end{bmatrix}$. Find its eigenvalue and eigenvector (block) matrices.

24. Consider all 4 by 4 matrices A that are diagonalized by the same fixed eigenvector matrix S. Show that the A's form a subspace (cA and $A_1 + A_2$ have this same S). What is this subspace when $S = I$? What is its dimension?

25. Suppose $A^2 = A$. On the left side A multiplies each column of A. Which of our four subspaces contains eigenvectors with $\lambda = 1$? Which subspace contains eigenvectors with $\lambda = 0$? From the dimensions of those subspaces, A has a full set of independent eigenvectors. So a matrix with $A^2 = A$ can be diagonalized.

26. (Recommended) Suppose $Ax = \lambda x$. If $\lambda = 0$ then x is in the nullspace. If $\lambda \neq 0$ then x is in the column space. Those spaces have dimensions $(n - r) + r = n$. So why doesn't every square matrix have n linearly independent eigenvectors?

27. The eigenvalues of A are 1 and 9, and the eigenvalues of B are -1 and 9:

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}.$$

Find a matrix square root of A from $R = S\sqrt{\Lambda}S^{-1}$. Why is there no real matrix square root of B?

28. (Heisenberg's Uncertainty Principle) $AB - BA = I$ can happen for infinite matrices with $A = A^T$ and $B = -B^T$. Then

$$x^T x = x^T ABx - x^T BAx \leq 2\|Ax\|\,\|Bx\|.$$

Explain that last step by using the Schwarz inequality. Then Heisenberg's inequality says that $\|Ax\|/\|x\|$ times $\|Bx\|/\|x\|$ is at least $\frac{1}{2}$. It is impossible to get the position error and momentum error both very small.

29. If A and B have the same λ's with the same independent eigenvectors, their factorizations into ____ are the same. So $A = B$.

30. Suppose the same S diagonalizes both A and B. They have the same eigenvectors in $A = S\Lambda_1 S^{-1}$ and $B = S\Lambda_2 S^{-1}$. Prove that $AB = BA$.

END-OF-PAGE SOLUTION: Problem Set 6.2 (Problems 21–30)

21. trace$(ST) = (aq + bs) + (cr + dt)$ and trace$(TS) = (qa + rc) + (sb + td)$, which are identical.

22. Blank: **0** (while $\text{trace}(I) = n \neq 0$). Take $A = E = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = E^T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$; then $AB - BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

23. $B = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} \Lambda & 0 \\ 0 & 2\Lambda \end{bmatrix} \begin{bmatrix} S^{-1} & 0 \\ 0 & S^{-1} \end{bmatrix}$.

24. $S^{-1}(cA)S = c\Lambda$ (diagonal) and $S^{-1}(A_1 + A_2)S = \Lambda_1 + \Lambda_2$ (diagonal). When $S = I$, the subspace is the space of 4×4 diagonal matrices, which has dimension 4.

25. $A(Ax) = Ax \implies$ the column space $C(A)$ (dimension r) contains eigenvectors with $\lambda = 1$. The nullspace $N(A)$ (dimension $n - r$) contains eigenvectors with $\lambda = 0$. Since $C(A) \cap N(A) = \{0\}$ (if $v = Ax$ and $Av = 0$, then $v = A^2x = Av = 0$), the $r + (n - r) = n$ basis vectors are independent.

26. The nullspace and column space can overlap ($N(A) \cap C(A) \neq \{0\}$), and the column space $C(A)$ might not have a basis of eigenvectors of A (e.g., when A has repeated eigenvalues with GM $<$ AM).

27. $R = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. For B, $\det B = -9 < 0$, whereas any real matrix R satisfies $\det(R^2) = (\det R)^2 \geq 0$.

28. Since $A = A^T$ and $B = -B^T$, $x^TABx = (Ax)^T(Bx) \leq \|Ax\| \, \|Bx\|$ and $-x^TBAx = (Bx)^T(Ax) \leq \|Ax\| \, \|Bx\|$ by the Cauchy-Schwarz inequality. Adding gives $x^Tx \leq 2\|Ax\| \, \|Bx\|$.

29. Blank: $S\Lambda S^{-1}$.

30. $AB = S\Lambda_1 S^{-1} S\Lambda_2 S^{-1} = S\Lambda_1\Lambda_2 S^{-1} = S\Lambda_2\Lambda_1 S^{-1} = BA$ because diagonal matrices commute ($\Lambda_1\Lambda_2 = \Lambda_2\Lambda_1$).

31. (a) If $A = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ then the determinant of $A - \lambda I$ is $(\lambda - a)(\lambda - d)$. Check the "Cayley-Hamilton Theorem" that $(A - aI)(A - dI) =$ zero matrix.

(b) Test the Cayley-Hamilton Theorem on Fibonacci's $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. The theorem predicts that $A^2 - A - I = 0$, since the polynomial $\det(A - \lambda I)$ is $\lambda^2 - \lambda - 1$.

32. Substitute $A = S\Lambda S^{-1}$ into the product $(A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_n I)$ and explain why this produces the zero matrix.

33. Find the eigenvalues and eigenvectors and the kth power of A. For this "adjacency matrix" the i, j entry of A^k counts the k-step paths from i to j.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

34. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $AB = BA$, show that $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is also a diagonal matrix. B has the same eigen____ as A but different eigen____. These diagonal matrices B form a two-dimensional subspace of matrix space. $AB - BA = 0$ gives four equations for the unknowns a, b, c, d—find the rank of the 4 by 4 matrix.

35. The powers A^k approach zero if all $|\lambda_i| < 1$ and they blow up if any $|\lambda_i| > 1$. Peter Lax gives these striking examples in his book *Linear Algebra*:

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} \qquad C = \begin{bmatrix} 5 & 7 \\ -3 & -4 \end{bmatrix} \qquad D = \begin{bmatrix} 5 & 6.9 \\ -3 & -4 \end{bmatrix}.$$

Find the eigenvalues $\lambda = e^{i\theta}$ of B and C to show $B^4 = I$ and $C^3 = -I$.

36. *(Challenge)* The nth power of rotation through θ is rotation through $n\theta$:

$$A^n = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^n = \begin{bmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{bmatrix}.$$

Prove that neat formula by diagonalizing $A = S\Lambda S^{-1}$. The eigenvectors (columns of S) are $(1, i)$ and $(i, 1)$.

37. *(Challenge)* The transpose of $A = S\Lambda S^{-1}$ is $A^T = (S^{-1})^T \Lambda S^T$. The eigenvectors in $A^T y = \lambda y$ are the columns of that matrix $(S^{-1})^T$. They are often called ***left eigenvectors***. How do you multiply matrices to find this formula for A?

Sum of rank-1 matrices $\qquad A = S\Lambda S^{-1} = \lambda_1 x_1 y_1^T + \cdots + \lambda_n x_n y_n^T.$

38. *(Challenge)* The inverse of $A = \text{eye}(n) + \text{ones}(n)$ is $A^{-1} = \text{eye}(n) + C * \text{ones}(n)$. Multiply AA^{-1} to find that number C (depending on n).

END-OF-PAGE SOLUTION: Problem Set 6.2 (Problems 31–38)

31. (a) $(A - aI)(A - dI) = \begin{bmatrix} 0 & b \\ 0 & d - a \end{bmatrix} \begin{bmatrix} a - d & b \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. (b) $A^2 - A - I = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} -$

$\begin{bmatrix}1&1\\1&0\end{bmatrix} - \begin{bmatrix}1&0\\0&1\end{bmatrix} = \begin{bmatrix}0&0\\0&0\end{bmatrix}$.

32. $\prod_{i=1}^{n}(A-\lambda_i I) = S\left[\prod_{i=1}^{n}(\Lambda-\lambda_i I)\right]S^{-1}$. In the product of diagonal matrices $\Lambda-\lambda_i I$, the ith factor has a 0 in position (i,i), so the diagonal product is the zero matrix.

33. $\lambda = 2,-1,0$ with $x_1 = (2,1,1), x_2 = (1,-1,-1), x_3 = (0,1,-1)$. $A^k = \frac{1}{3}\begin{bmatrix}2^{k+1}+(-1)^k & 2^k-(-1)^k & 2^k-(-1)^k\\ 2^k-(-1)^k & 2^{k-1}+2(-1)^k & 2^{k-1}+2(-1)^k\\ 2^k-(-1)^k & 2^{k-1}+2(-1)^k & 2^{k-1}+2(-1)^k\end{bmatrix}$ for $k \geq 1$.

34. $AB = \begin{bmatrix}a&b\\2c&2d\end{bmatrix} = BA = \begin{bmatrix}a&2b\\c&2d\end{bmatrix} \implies b=0, c=0$. Blanks: **vectors**; **values**. The 4 equations are $0=0, -b=0, c=0, 0=0$, so the rank of the 4×4 system is 2.

35. For B: $\lambda^2+1=0 \implies \lambda=\pm i \implies \lambda^4=1 \implies B^4=I$. For C: $\lambda^2-\lambda+1=0 \implies \lambda = e^{\pm i\pi/3} \implies \lambda^3=-1 \implies C^3=-I$.

36. $S = \begin{bmatrix}1&i\\i&1\end{bmatrix}$, $\Lambda = \begin{bmatrix}e^{-i\theta}&0\\0&e^{i\theta}\end{bmatrix}$, $S^{-1} = \frac{1}{2}\begin{bmatrix}1&-i\\-i&1\end{bmatrix}$. Multiplying $S\Lambda^n S^{-1}$ and using $\cos n\theta = \frac{e^{in\theta}+e^{-in\theta}}{2}$, $\sin n\theta = \frac{e^{in\theta}-e^{-in\theta}}{2i}$ yields the rotation matrix for $n\theta$.

37. Multiply the columns $\lambda_i x_i$ of $S\Lambda$ by the rows y_i^T of S^{-1}.

38. Since $\text{ones}(n)^2 = n\ \text{ones}(n)$, $AA^{-1} = I + (1+C+nC)\,\text{ones}(n) = I \implies C = -\dfrac{1}{n+1}$.

CHAPTER 3

Symmetric Matrices (Section 6.4)

Syllabus Equations: $A = A^T$, $\bar{x}^T A x = \lambda \bar{x}^T x$ (real λ), $x_1^T x_2 = 0$ (orthogonal eigenvectors), Spectral Theorem $A = Q\Lambda Q^T = \sum_{i=1}^{n} \lambda_i x_i x_i^T$, $A = LDL^T$ (signs of pivots = signs of eigenvalues), Schur's Theorem $A = QTQ^{-1}$.

3.1 Derivation 1: Real Eigenvalues of a Real Symmetric Matrix

GOAL

Prove that every eigenvalue λ of a real symmetric matrix $A = A^T$ is a real number.

FROM YOUR +2 BASE

Use **T7**: A complex number $\lambda = a + ib$ is real if and only if $\lambda = \bar{\lambda}$. For any complex vector x, $\bar{x}^T x = |x_1|^2 + \cdots + |x_n|^2 > 0$.

Step 1 Start with $Ax = \lambda x$. Take the complex conjugate of every entry. Since A is real ($\bar{A} = A$), we get:

$$A\bar{x} = \bar{\lambda}\bar{x}. \tag{3.1}$$

Step 2 Take the transpose of $A\bar{x} = \bar{\lambda}\bar{x}$ using **T4** and symmetry $A^T = A$:

$$(A\bar{x})^T = \bar{x}^T A^T = \bar{x}^T A = \bar{\lambda}\bar{x}^T. \tag{3.2}$$

Step 3 Multiply $Ax = \lambda x$ on the left by $\bar{x}^T$, and multiply $\bar{x}^T A = \bar{\lambda}\bar{x}^T$ on the right by x:

$$\bar{x}^T A x = \lambda \bar{x}^T x \qquad \text{and} \qquad \bar{x}^T A x = \bar{\lambda}\bar{x}^T x. \tag{3.3}$$

Step 4 The left sides are identical, so subtract the two right sides:

$$(\lambda - \bar{\lambda})\bar{x}^T x = 0. \tag{3.4}$$

Since $x \neq 0$, $\bar{x}^T x = \|x\|^2 > 0$. Therefore $\boxed{\lambda = \bar{\lambda}}$, so λ is real ($b = 0$).

3.2 Derivation 2: Orthogonal Eigenvectors and the Spectral Theorem

GOAL

Prove that eigenvectors of $A = A^T$ for distinct eigenvalues are perpendicular, and derive the Spectral Theorem $A = Q\Lambda Q^T = \sum \lambda_i x_i x_i^T$.

Step 1 Suppose $Ax = \lambda_1 x$, $Ay = \lambda_2 y$, and $\lambda_1 \neq \lambda_2$. Compute $(Ax)^T y$ in two ways using $A^T = A$:

$$\begin{aligned} (Ax)^T y &= (\lambda_1 x)^T y = \lambda_1 x^T y, && \text{use } Ax = \lambda_1 x \\ (Ax)^T y &= x^T A^T y = x^T A y = \lambda_2 x^T y. && \text{use } A^T = A \text{ and } Ay = \lambda_2 y \end{aligned} \tag{3.5}$$

Step 2 Subtract: $(\lambda_1 - \lambda_2) x^T y = 0$. Since $\lambda_1 \neq \lambda_2$, we must have $\boxed{x^T y = 0}$.

Step 3 Normalize the orthogonal eigenvectors to unit length ($\|x_i\| = 1$) and place them in the columns of $S = Q$. Then $Q^T Q = I$, so $Q^{-1} = Q^T$. Substitute $S = Q$ and $S^{-1} = Q^T$ into $A = S\Lambda S^{-1}$:

$$\boxed{A = Q\Lambda Q^{-1} = Q\Lambda Q^T = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} = \lambda_1 x_1 x_1^T + \cdots + \lambda_n x_n x_n^T.} \tag{3.6}$$

3.3 Derivation 3: Matching Signs of Pivots and Eigenvalues

GOAL

Show that the number of positive eigenvalues of a symmetric matrix $A = A^T$ equals the number of positive pivots.

Step 1 Factor A by symmetric elimination without row exchanges into $A = LDL^T$, where L is lower triangular with 1's on the diagonal and $D = \text{diag}(d_1, \ldots, d_n)$ holds the pivots.

Step 2 Move the off-diagonal entries of L continuously to 0 so $L(t) \to I$. For every t, $\det(L(t)DL(t)^T) = \det(D) = d_1 \cdots d_n \neq 0$.

Step 3 Because the matrix is never singular during this deformation, no real eigenvalue $\lambda_i(t)$ can cross zero. When $L = I$, the matrix is D, whose eigenvalues are the pivots $d_1, \ldots, d_n$. Therefore the signs of λ_i match the signs of d_i.

3.4 Section 6.4 In-Text Examples and Worked Examples

BOOK EXAMPLE 6.4.1 2 by 2 Spectral Factorization

Find the λ's and orthonormal x's when $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and write $Q^{-1}AQ = \Lambda$.

BOOK EXAMPLE 6.4.2 General 2 by 2 Symmetric Eigenvectors

Verify that $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ has orthogonal eigenvectors $x_1 = \begin{bmatrix} b \\ \lambda_1 - a \end{bmatrix}$ and $x_2 = \begin{bmatrix} \lambda_2 - c \\ b \end{bmatrix}$.

BOOK EXAMPLE 6.4.3 Conjugate Pair of Complex Eigenvalues

Verify $Ax = \lambda x$ and $A\bar{x} = \bar{\lambda}\bar{x}$ for the rotation matrix $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

BOOK EXAMPLE 6.4.4 Signs of Pivots versus Eigenvalues

Compare the signs of the pivots and eigenvalues for the symmetric matrix $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ and the nonsymmetric matrix $B = \begin{bmatrix} 1 & 6 \\ -1 & -4 \end{bmatrix}$.

BOOK EXAMPLE 6.4 A Worked Example 6.4 A: Reflection Across the θ-Line

What matrix A has eigenvalues $\lambda = 1, -1$ and eigenvectors $x_1 = (\cos\theta, \sin\theta)$ and $x_2 = (-\sin\theta, \cos\theta)$? Which of these properties can be predicted in advance: $A = A^T$, $A^2 = I$, $\det A = -1$, $+$ and $-$ pivot, $A^{-1} = A$?

BOOK EXAMPLE 6.4 B Worked Example 6.4 B: Discrete Sine and Cosine Transforms

Find the eigenvalues of A_3 and B_4, and check the orthogonality of their eigenvectors:

$$A_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad B_4 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

END-OF-PAGE SOLUTION: Section 6.4 Examples (1–4, 6.4 A, 6.4 B)

Ex. 6.4.1: $\lambda_1 = 0, \lambda_2 = 5$. Orthogonal eigenvectors are $(2, -1)$ and $(1, 2)$ with length $\sqrt{5}$. Divide by $\sqrt{5}$: $Q = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$ and $Q^T AQ = \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix} = \Lambda$.

Ex. 6.4.2: $x_1^T x_2 = b(\lambda_2 - c) + (\lambda_1 - a)b = b(\lambda_1 + \lambda_2 - a - c) = 0$ because $\lambda_1 + \lambda_2 = \text{trace}(A) = a + c$.

Ex. 6.4.3: $\lambda_1 = \cos\theta + i\sin\theta$ has $x_1 = (1, -i)$, and $\lambda_2 = \bar{\lambda}_1 = \cos\theta - i\sin\theta$ has $x_2 = \bar{x}_1 = (1, i)$. Both have $|\lambda| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$.

Ex. 6.4.4: For $A = A^T$: pivots are 1 and $1 - 9 = -8$ (one $+$, one $-$); eigenvalues are 4 and

-2 (one $+$, one $-$). For $B \neq B^T$: pivots are 1 and $-4+6=2$ (both $+$); eigenvalues are -1 and -2 (both $-$).

Ex. 6.4 A: All five properties hold! Using **T9**: $A = Q\Lambda Q^T = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \begin{bmatrix} c^2-s^2 & 2cs \\ 2cs & s^2-c^2 \end{bmatrix} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$.

Ex. 6.4 B: A_3 has $\lambda = 2-\sqrt{2}, 2, 2+\sqrt{2}$ and orthogonal eigenvector matrix $S = \begin{bmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{bmatrix}$. B_4 has $\lambda = 0, 2-\sqrt{2}, 2, 2+\sqrt{2}$ and orthogonal eigenvector matrix $C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \sqrt{2}-1 & -1 & 1-\sqrt{2} \\ 1 & 1-\sqrt{2} & -1 & \sqrt{2}-1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$.

3.5 Problem Set 6.4 (Verbatim Questions and Solutions)

1. Write A as $M+N$, symmetric matrix plus skew-symmetric matrix:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 3 & 0 \\ 8 & 6 & 5 \end{bmatrix} = M + N \qquad (M^T = M, N^T = -N).$$

For any square matrix, $M = \frac{A+A^T}{2}$ and $N =$ ____ add up to A.

2. If C is symmetric prove that A^TCA is also symmetric. (Transpose it.) When A is 6 by 3, what are the shapes of C and A^TCA?

3. Find the eigenvalues and the unit eigenvectors of $A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$.

4. Find an orthogonal matrix Q that diagonalizes $A = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$. What is Λ?

5. Find an orthogonal matrix Q that diagonalizes this symmetric matrix:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}.$$

6. Find *all* orthogonal matrices that diagonalize $A = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$.

7. (a) Find a symmetric matrix $\begin{bmatrix} 1 & b \\ b & 1 \end{bmatrix}$ that has a negative eigenvalue.
(b) How do you know it must have a negative pivot?
(c) How do you know it can't have two negative eigenvalues?

8. If $A^3 = 0$ then the eigenvalues of A must be ____. Give an example that has $A \neq 0$. But if A is symmetric, diagonalize it to prove that A must be zero.

9. If $\lambda = a + ib$ is an eigenvalue of a real matrix A, then its conjugate $\bar{\lambda} = a - ib$ is also an eigenvalue. (If $Ax = \lambda x$ then also $A\bar{x} = \bar{\lambda}\bar{x}$.) Prove that every real 3 by 3 matrix has at least one real eigenvalue.

10. Here is a quick "proof" that the eigenvalues of all real matrices are real:

False proof $\quad Ax = \lambda x \quad$ gives $\quad x^TAx = \lambda x^Tx \quad$ so $\quad \lambda = \dfrac{x^TAx}{x^Tx} \quad$ is real.

Find the flaw in this reasoning—a hidden assumption that is not justified. You could test those steps on the 90° rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ with $\lambda = i$ and $x = (i, 1)$.

END-OF-PAGE SOLUTION: Problem Set 6.4 (Problems 1–10)

1. $M = \begin{bmatrix} 1 & 3 & 6 \\ 3 & 3 & 3 \\ 6 & 3 & 5 \end{bmatrix}$, $N = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -3 \\ 2 & 3 & 0 \end{bmatrix}$. Blank: $\dfrac{A - A^T}{2}$.

2. $(A^TCA)^T = A^TC^T(A^T)^T = A^TCA$. C is 6×6 and A^TCA is 3×3.
3. $\lambda = 4, -2, 0$. Unit eigenvectors: $u_1 = \frac{1}{\sqrt{6}}(2,1,1)$, $u_2 = \frac{1}{\sqrt{3}}(1,-1,-1)$, $u_3 = \frac{1}{\sqrt{2}}(0,1,-1)$.
4. $\det(A - \lambda I) = \lambda^2 - 5\lambda - 50 = (\lambda - 10)(\lambda + 5) = 0 \implies \Lambda = \begin{bmatrix} 10 & 0 \\ 0 & -5 \end{bmatrix}$ and $Q = \frac{1}{\sqrt{5}}\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$.
5. $\det(A - \lambda I) = -\lambda^3 + 9\lambda = 0 \implies \lambda = 3, -3, 0$. $Q = \begin{bmatrix} 2/3 & 1/3 & -2/3 \\ -1/3 & 2/3 & -2/3 \\ 2/3 & -2/3 & -1/3 \end{bmatrix}$.
6. $\lambda = 25, 0$ with unit eigenvectors $\pm(3/5, 4/5)$ and $\pm(-4/5, 3/5)$. All 8 orthogonal matrices are formed by choosing signs and column order in $Q = \frac{1}{5}\begin{bmatrix} \pm 3 & \mp 4 \\ \pm 4 & \pm 3 \end{bmatrix}$ or $\frac{1}{5}\begin{bmatrix} \mp 4 & \pm 3 \\ \pm 3 & \pm 4 \end{bmatrix}$.
7. (a) Choose any $|b| > 1$, e.g., $b = 2$ gives $\lambda = 3, -1$. (b) The signs of the pivots match the signs of the eigenvalues for a symmetric matrix (and the first pivot is $1 > 0$). (c) The first pivot is $1 > 0$ (or $\text{trace}(A) = 2 > 0$).
8. Blank: **0**. Example: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. If $A = A^T$, then $A = Q\Lambda Q^T$. Since $\Lambda = 0$, $A = Q0Q^T = 0$.
9. Non-real eigenvalues come in conjugate pairs $\lambda, \bar{\lambda}$. A 3×3 matrix has 3 eigenvalues (roots of a cubic polynomial), so it can have at most one complex conjugate pair, leaving at least one real eigenvalue.
10. The eigenvector x can be complex! When $x = (i, 1)$, $x^Tx = i^2 + 1^2 = 0$, so dividing by x^Tx divides by zero (and x^TAx need not be real). We must use $\bar{x}^Tx$ instead of x^Tx.

11. Write A and B in the form $\lambda_1 x_1 x_1^T + \lambda_2 x_2 x_2^T$ of the spectral theorem $Q\Lambda Q^T$:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} \quad \text{(keep } \|x_1\| = \|x_2\| = 1\text{)}.$$

12. Every 2 by 2 symmetric matrix is $\lambda_1 x_1 x_1^T + \lambda_2 x_2 x_2^T = \lambda_1 P_1 + \lambda_2 P_2$. Explain $P_1 + P_2 = x_1 x_1^T + x_2 x_2^T = I$ from columns times rows of Q. Why is $P_1 P_2 = 0$?

13. What are the eigenvalues of $A = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix}$? Create a 4 by 4 skew-symmetric matrix ($A^T = -A$) and verify that all its eigenvalues are imaginary.

14. (Recommended) This matrix M is skew-symmetric and also ____. Then all its eigenvalues are pure imaginary and they also have $|\lambda| = 1$. ($\|Mx\| = \|x\|$ for every x so $\|\lambda x\| = \|x\|$ for eigenvectors.) Find all four eigenvalues from the trace of M:

$$M = \frac{1}{\sqrt{3}} \begin{bmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 1 \\ -1 & 1 & 0 & -1 \\ -1 & -1 & 1 & 0 \end{bmatrix} \quad \text{can only have eigenvalues } i \text{ or } -i.$$

15. Show that A (**symmetric but complex**) has only one line of eigenvectors:

$$A = \begin{bmatrix} i & 1 \\ 1 & -i \end{bmatrix} \quad \text{is not even diagonalizable: eigenvalues } \lambda = 0, 0.$$

16. Even if A is rectangular, the block matrix $B = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}$ is symmetric:

$$Bx = \lambda x \quad \text{is} \quad \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \lambda \begin{bmatrix} y \\ z \end{bmatrix} \quad \text{which is} \quad \begin{matrix} Az = \lambda y \\ A^T y = \lambda z. \end{matrix}$$

(a) Show that $-\lambda$ is also an eigenvalue, with the eigenvector $(y, -z)$.
(b) Show that $A^T A z = \lambda^2 z$, so that λ^2 is an eigenvalue of $A^T A$.
(c) If $A = I$ (2 by 2) find all four eigenvalues and eigenvectors of B.

17. If $A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ in Problem 16, find all three eigenvalues and eigenvectors of B.

18. ***Another proof that eigenvectors are perpendicular when*** $A = A^T$**.** Two steps:
1. Suppose $Ax = \lambda x$ and $Ay = 0y$ and $\lambda \neq 0$. Then y is in the nullspace and x is in the column space. They are perpendicular because ____. Go carefully—why are these subspaces orthogonal?
2. If $Ay = \beta y$, apply this argument to $A - \beta I$. The eigenvalue of $A - \beta I$ moves to zero and the eigenvectors stay the same—so they are perpendicular.

19. Find the eigenvector matrix S for A and for B. Show that S doesn't collapse at $d = 1$, even though $\lambda = 1$ is repeated. Are the eigenvectors perpendicular?

$$A = \begin{bmatrix} 0 & d & 0 \\ d & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} -d & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & d \end{bmatrix} \quad \text{have} \quad \lambda = 1, d, -d.$$

20. Write a 2 by 2 *complex* matrix with $\bar{A}^T = A$ (a "Hermitian matrix"). Find λ_1 and λ_2 for your complex matrix. Adjust equations (1) and (2) to show that *the eigenvalues of a Hermitian matrix are real*.

END-OF-PAGE SOLUTION: Problem Set 6.4 (Problems 11–20)

11. $A = 4\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} + 2\begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$. $B = 25\begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix} + 0\begin{bmatrix} 16/25 & -12/25 \\ -12/25 & 9/25 \end{bmatrix}$.

12. $QQ^T = \begin{bmatrix} x_1 & x_2 \end{bmatrix}\begin{bmatrix} x_1^T \\ x_2^T \end{bmatrix} = x_1x_1^T + x_2x_2^T = I$. Also $P_1P_2 = x_1(x_1^Tx_2)x_2^T = x_1(0)x_2^T = 0$.

13. $\lambda = \pm ib$. A 4×4 block-diagonal matrix $\text{diag}(A, A)$ has eigenvalues $ib, -ib, ib, -ib$.

14. Blank: **orthogonal** ($M^TM = I$). Since $\text{trace}(M) = 0$, the four eigenvalues are $i, i, -i, -i$.

15. $\det(A - \lambda I) = (\lambda - i)(\lambda + i) - 1 = \lambda^2 = 0 \implies \lambda = 0, 0$. Then $Ax = 0 \implies ix_1 + x_2 = 0 \implies x = c(1, -i)$ (only one line).

16. (a) $B\begin{bmatrix} y \\ -z \end{bmatrix} = \begin{bmatrix} -Az \\ A^Ty \end{bmatrix} = \begin{bmatrix} -\lambda y \\ \lambda z \end{bmatrix} = -\lambda\begin{bmatrix} y \\ -z \end{bmatrix}$. (b) $A^T(Az) = A^T(\lambda y) = \lambda(A^Ty) = \lambda^2 z$. (c) $\lambda = 1, 1, -1, -1$ with eigenvectors $(1, 0, 1, 0), (0, 1, 0, 1), (1, 0, -1, 0), (0, 1, 0, -1)$.

17. $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ has $\lambda = \sqrt{2}, -\sqrt{2}, 0$ with eigenvectors $(1, 1, \sqrt{2}), (1, 1, -\sqrt{2}), (1, -1, 0)$.

18. Blank: **the nullspace is perpendicular to the row space** $C(A^T)$**, and since** $A = A^T$**, the row space equals the column space** $C(A)$.

19. For A: $S = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (perpendicular, independent for all d). For B: $S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 2d \end{bmatrix}$ (independent for $d \neq 0$, not perpendicular).

20. Example: $A = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}$ has $\lambda = 3, 1$. In (1), $\bar{A}\bar{x} = \bar{\lambda}\bar{x}$ transposes to $\bar{x}^T\bar{A}^T = \bar{x}^TA = \bar{\lambda}\bar{x}^T$, so $\bar{x}^TAx = \lambda\bar{x}^Tx = \bar{\lambda}\bar{x}^Tx \implies \lambda = \bar{\lambda}$.

21. *True* (with reason) or *false* (with example). "Orthonormal" is not assumed.
(a) A matrix with real eigenvalues and eigenvectors is symmetric.
(b) A matrix with real eigenvalues and orthogonal eigenvectors is symmetric.
(c) The inverse of a symmetric matrix is symmetric.
(d) The eigenvector matrix S of a symmetric matrix is symmetric.

22. (A paradox for instructors) If $AA^T = A^TA$ then A and A^T share the same eigenvectors (true). A and A^T always share the same eigenvalues. Find the flaw in this conclusion: They must have the same S and Λ. Therefore A equals A^T.

23. (Recommended) Which of these classes of matrices do A and B belong to: Invertible, orthogonal, projection, permutation, diagonalizable, Markov?

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \qquad B = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Which of these factorizations are possible for A and B: LU, QR, $S\Lambda S^{-1}$, $Q\Lambda Q^T$?

24. What number b in $\begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$ makes $A = Q\Lambda Q^T$ possible? What number makes $A = S\Lambda S^{-1}$ impossible? What number makes A^{-1} impossible?

25. Find all 2 by 2 matrices that are orthogonal and also symmetric. Which two numbers can be eigenvalues?

26. This A is nearly symmetric. But its eigenvectors are far from orthogonal:

$$A = \begin{bmatrix} 1 & 10^{-15} \\ 0 & 1+10^{-15} \end{bmatrix} \quad \text{has eigenvectors} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad [?].$$

What is the angle between the eigenvectors?

27. (MATLAB) Take two symmetric matrices with different eigenvectors, say $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 1 \\ 1 & 0 \end{bmatrix}$. Graph the eigenvalues $\lambda_1(A+tB)$ and $\lambda_2(A+tB)$ for $-8 < t < 8$. How close do they come?

28. *(Challenge)* From $\det(A - \lambda I) = 0$, find the eigenvalues of the 2 by 2 "Hermitian" matrix $A = \begin{bmatrix} 4 & 2+i \\ 2-i & 0 \end{bmatrix} = \bar{A}^T$.

29. *(Challenge)* **Normal matrices** have $\bar{A}^TA = A\bar{A}^T$. Show that A has n **orthonormal eigenvectors** ($A = Q\Lambda\bar{Q}^T$) if and only if A is **normal**.

30. *(Challenge)* If $\lambda_{\max}$ is the largest eigenvalue of a symmetric matrix A, no diagonal entry can be larger than $\lambda_{\max}$. What is the first entry a_{11} of $A = Q\Lambda Q^T$? Show why $a_{11} \le \lambda_{\max}$.

31. *(Challenge)* Suppose $A^T = -A$ (real antisymmetric matrix). Explain these facts about A:
(a) $x^TAx = 0$ for every real vector x. (b) The eigenvalues of A are pure imaginary. (c) The determinant of A is positive or zero (not negative).

32. *(Challenge)* If A is symmetric and all its eigenvalues are $\lambda = 2$, how do you know that A must be $2I$?

END-OF-PAGE SOLUTION: Problem Set 6.4 (Problems 21–32)

21. (a) **False** ($\begin{bmatrix}1 & 1\\ 0 & 2\end{bmatrix}$). (b) **True** (if n orthogonal eigenvectors exist, normalize to Q, then $A = Q\Lambda Q^T$ is symmetric). (c) **True** ($(A^{-1})^T = (A^T)^{-1} = A^{-1}$). (d) **False** ($Q$ is orthogonal, not necessarily symmetric).

22. Complex eigenvalues are conjugates ($\Lambda^H = \bar{\Lambda}$); for example, a rotation matrix Q has $\lambda(Q) = e^{i\theta}$ for x_1 while Q^T has $\lambda(Q^T) = e^{-i\theta}$ for the same x_1 (the eigenvalues are in reversed order!).

23. A: Invertible, orthogonal, permutation, diagonalizable, Markov. Factorizations for A: $QR, S\Lambda S^{-1}, Q\Lambda Q^T$ (LU needs a row exchange, $PA = LU$). B: Projection, diagonalizable, Markov. Factorizations for B: LU, QR (with singular R), $S\Lambda S^{-1}, Q\Lambda Q^T$.

24. $Q\Lambda Q^T$ possible when $b = 1$ (A symmetric). $S\Lambda S^{-1}$ impossible when $\lambda^2 - 2\lambda - b = 0$ has a double root: $b = -1$. A^{-1} impossible when $b = 0$.

25. Eigenvalues must be ± 1. Matrices are I, $-I$, and reflections $\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{bmatrix}$.

26. Second eigenvector is $\begin{bmatrix}1\\ 1\end{bmatrix}$. The angle between $(1,0)$ and $(1,1)$ is 45° ($\pi/4$).

27. $\det(A + tB - \lambda I) = \lambda^2 - (3 + 8t)\lambda + (2 + 16t - t^2) = 0$. Gap is $\sqrt{(3+8t)^2 - 4(2+16t-t^2)} = \sqrt{68t^2 - 16t + 1}$. Minimum at $t = 2/17$ gives minimum gap $\sqrt{1 - 16/17} = 1/\sqrt{17} \approx 0.2425$.

28. $\det(A - \lambda I) = \lambda^2 - 4\lambda - |2+i|^2 = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \implies \lambda = 5, -1$.

29. (a) $\bar{A}^T A = Q\bar{\Lambda}\bar{Q}^T Q\Lambda\bar{Q}^T = Q|\Lambda|^2\bar{Q}^T = A\bar{A}^T$. (b) In Schur's $A = QT\bar{Q}^T$, $\bar{T}^T T = T\bar{T}^T$ forces $|a|^2 + |b|^2 = |a|^2 \implies b = 0$, so $T = \Lambda$ is diagonal.

30. $a_{11} = \lambda_1 q_{11}^2 + \cdots + \lambda_n q_{1n}^2 \le \lambda_{\max}(q_{11}^2 + \cdots + q_{1n}^2) = \lambda_{\max}(1) = \lambda_{\max}$.

31. (a) $(x^T Ax)^T = x^T A^T x = -x^T Ax \implies x^T Ax = 0$. (b) $\bar{z}^T Az = \lambda\|z\|^2$ is pure imaginary, so λ is pure imaginary. (c) Nonzero eigenvalues come in pairs $ib, -ib$ whose product is $(ib)(-ib) = b^2 > 0$ (and if n is odd, $\lambda = 0 \implies \det A = 0$).

32. Since A is symmetric, $A = Q\Lambda Q^T$. With $\Lambda = 2I$, $A = Q(2I)Q^T = 2QQ^T = 2I$.

CHAPTER 4

Similar Matrices and Jordan Form (Section 6.6)

Syllabus Equations: $B = M^{-1}AM$, $\det(B-\lambda I) = \det(A-\lambda I)$, $x_B = M^{-1}x_A$, Jordan Form $M^{-1}AM = J = \operatorname{diag}(J_1,\ldots,J_s)$.

4.1 Derivation 1: Eigenvalues and Eigenvectors of Similar Matrices

GOAL

Prove that for any invertible matrix M, the similar matrix $B = M^{-1}AM$ has the exact same characteristic polynomial and eigenvalues as A, with eigenvectors $M^{-1}x$.

FROM YOUR +2 BASE

Use **T10**: $\det(XYZ) = (\det X)(\det Y)(\det Z)$ and $\det(M^{-1})\det(M) = 1$.

Step 1 (Algebraic Proof) Start with $Ax = \lambda x$ and substitute $A = MBM^{-1}$:

$$MBM^{-1}x = \lambda x \implies \boxed{B(M^{-1}x) = \lambda(M^{-1}x).} \tag{4.1}$$

Thus λ is an eigenvalue of B with eigenvector $x_B = M^{-1}x$.

Step 2 (Characteristic Polynomial Proof) Write $\lambda I = M^{-1}(\lambda I)M$ inside $\det(B-\lambda I)$:

$$\begin{aligned}\det(B-\lambda I) &= \det(M^{-1}AM - M^{-1}\lambda IM) = \det\left(M^{-1}(A-\lambda I)M\right)\\ &= \det(M^{-1})\det(A-\lambda I)\det(M) = \boxed{\det(A-\lambda I).}\end{aligned} \tag{4.2}$$

4.2 Derivation 2: The Jordan Canonical Form $J = M^{-1}AM$

GOAL

Describe the Jordan form J when A has fewer than n linearly independent eigenvectors.

RESULT (Jordan's Theorem)

If an $n \times n$ matrix A has s linearly independent eigenvectors, then A is similar to a block-diagonal Jordan matrix $J = M^{-1}AM$ with s Jordan blocks $J_1,\ldots,J_s$:

$$M^{-1}AM = J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{bmatrix}, \qquad J_i = \begin{bmatrix} \lambda_i & 1 & \\ & \lambda_i & \ddots \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}. \tag{4.3}$$

Each block J_i has one eigenvalue λ_i on its diagonal, 1's just above the diagonal, and one eigenvector. Two matrices A and B are similar if and only if they share the same Jordan

form J.

4.3 Section 6.6 In-Text Examples and Worked Examples

BOOK EXAMPLE 6.6.1 Similar Matrices with $\lambda = 1$ and 0

Compute $M^{-1}AM$ for the projection matrix $A = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ using $M_1 = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

BOOK EXAMPLE 6.6.2 Family of Similar Matrices with $\lambda = 0, 0$

Find all matrices $B = M^{-1}AM$ similar to the Jordan block $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ using $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc = 1$.

BOOK EXAMPLE 6.6.3 3 by 3 Jordan Block and Transpose Similarity

For $J = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$, find the rank of $J - 5I$ and find M such that $J^T = M^{-1}JM$.

BOOK EXAMPLE 6.6.4 Differential Equation $du/dt = Ju$

Solve $\dfrac{du}{dt} = Ju = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ by back substitution.

BOOK EXAMPLE 6.6 A Worked Example 6.6 A: Triangular Pascal Matrix

Check that the 4×4 Pascal matrix A and A^{-1} have the same eigenvalues, and find a diagonal matrix D with alternating signs that gives $A^{-1} = D^{-1}AD$:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}.$$

BOOK EXAMPLE 6.6 B Worked Example 6.6 B: The QR Method for Eigenvalues

Factor $A = \begin{bmatrix} c & s \\ s & 0 \end{bmatrix} = QR$ (where $c^2 + s^2 = 1$) and reverse the order to find the similar matrix $A_1 = RQ$.

END-OF-PAGE SOLUTION: Section 6.6 Examples (1–4, 6.6 A, 6.6 B)

Ex. 6.6.1: With $M_1 = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, $M_1^{-1}AM_1 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. With $M_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $M_2^{-1}AM_2 = \begin{bmatrix} .5 & -.5 \\ -.5 & .5 \end{bmatrix}$. All have $\lambda = 1, 0$.

Ex. 6.6.2: $B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} cd & d^2 \\ -c^2 & -cd \end{bmatrix}$ (any rank-1 matrix with zero trace and zero determinant, excluding $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$).

Ex. 6.6.3: $J - 5I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ has rank 2 (GM $= 1$, eigenvector $x = (1,0,0)$). Using the reverse identity $M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = M^{-1}$, we get $M^{-1}JM = J^T$.

Ex. 6.6.4: $dz/dt = 5z \implies z(t) = z(0)e^{5t}$. Next, $dy/dt = 5y + z(0)e^{5t} \implies y(t) = (y(0) + tz(0))e^{5t}$. Finally, $dx/dt = 5x + y \implies x(t) = \left(x(0) + ty(0) + \frac{1}{2}t^2 z(0)\right)e^{5t}$.

Ex. 6.6 A: Both A and A^{-1} have $\lambda = 1,1,1,1$. Choose $D = \operatorname{diag}(-1,1,-1,1) = D^{-1}$. Multiplying $D^{-1}AD$ negates (i,j) entries where $i+j$ is odd, producing A^{-1}.

Ex. 6.6 B: $A = \begin{bmatrix} c & s \\ s & -c \end{bmatrix}\begin{bmatrix} 1 & cs \\ 0 & s^2 \end{bmatrix} = QR$. Reversing gives $A_1 = RQ = Q^{-1}AQ = \begin{bmatrix} c+cs^2 & s^3 \\ s^3 & -cs^2 \end{bmatrix}$, which is similar to A and cubically reduces the off-diagonal entry from s to s^3.

4.4 Problem Set 6.6 (Verbatim Questions and Solutions)

1. If $C = F^{-1}AF$ and also $C = G^{-1}BG$, what matrix M gives $B = M^{-1}AM$? Conclusion: If C is similar to A and also to B then ____.

2. If $A = \text{diag}(1,3)$ and $B = \text{diag}(3,1)$ show that A and B are similar (find an M).

3. Show that A and B are similar by finding M so that $B = M^{-1}AM$:

(a) $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ (c) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$.

4. If a 2 by 2 matrix A has eigenvalues 0 and 1, why is it similar to $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$? Deduce from Problem 1 that all 2 by 2 matrices with those eigenvalues are similar.

5. Which of these six matrices are similar? Check their eigenvalues.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

6. There are sixteen 2 by 2 matrices whose entries are 0's and 1's. Similar matrices go into the same family. How many families? How many matrices (total 16) in each family?

7. (a) If x is in the nullspace of A show that $M^{-1}x$ is in the nullspace of $M^{-1}AM$.
(b) The nullspaces of A and $M^{-1}AM$ have the same (vectors)(basis)(dimension).

8. Suppose $Ax = \lambda x$ and $Bx = \lambda x$ with the same λ's and x's. With n independent eigenvectors we have $A = B$: Why? Find $A \neq B$ when both have eigenvalues $0, 0$ but only one line of eigenvectors $(x_1, 0)$.

9. By direct multiplication find A^2 and A^3 and A^5 when $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Guess the form of A^k. Set $k = 0$ to find A^0 and $k = -1$ to find A^{-1}.

10. By direct multiplication, find J^2 and J^3 when $J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$. Guess the form of J^k. Set $k = 0$ to find J^0. Set $k = -1$ to find J^{-1}.

11. Solve $du/dt = Ju$ for J in Problem 10, starting from $u(0) = (5,2)$. Remember $te^{\lambda t}$.

12. These Jordan matrices have eigenvalues $0, 0, 0, 0$. They have two eigenvectors (one from each block). But the block sizes don't match and they are *not* similar:

$$J = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad K = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right].$$

For any matrix M, compare JM with MK. If they are equal show that M is not invertible.

END-OF-PAGE SOLUTION: Problem Set 6.6 (Problems 1–12)

1. $B = GCG^{-1} = GF^{-1}AFG^{-1} = (FG^{-1})^{-1}A(FG^{-1})$, so $M = FG^{-1}$. Blank: A **is similar to** B.

2. $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = M^{-1}$ exchanges the rows and columns, giving $M^{-1}AM = B$.

3. (a) $M = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. (b) $M = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. (c) $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

4. Distinct eigenvalues $0 \neq 1$ guarantee 2 independent eigenvectors, so $S^{-1}AS = \Lambda$. By Problem 1, any two matrices similar to Λ are similar to each other.

5. Matrix 1 has $\lambda = 1, 1$. Matrix 2 has $\lambda = 1, -1$. Matrices 3, 4, 5, and 6 all have $\lambda = 1, 0$ and are similar to each other!

6. 7 families: (1) $\lambda = 0, 0$ zero matrix [1 matrix]; (2) $\lambda = 0, 0$ rank 1 [2 matrices]; (3) $\lambda = 1, 0$ [6 matrices]; (4) $\lambda = 1, 1$ identity [1 matrix]; (5) $\lambda = 1, 1$ not diagonalizable [2 matrices]; (6) $\lambda = 1, -1$ [3 matrices]; (7) $\lambda = 2, 0$ [1 matrix]. Total $= 1 + 2 + 6 + 1 + 2 + 3 + 1 = 16$.

7. (a) $(M^{-1}AM)(M^{-1}x) = M^{-1}Ax = M^{-1}0 = 0$. (b) **dimension**.

8. $A = S\Lambda S^{-1} = B$. When GM $= 1$: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$.

9. $A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, $A^5 = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$, $A^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$. $A^0 = I$, $A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$.

10. $J^2 = \begin{bmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{bmatrix}$, $J^3 = \begin{bmatrix} \lambda^3 & 3\lambda^2 \\ 0 & \lambda^3 \end{bmatrix}$, $J^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}$. $J^0 = I$, $J^{-1} = \begin{bmatrix} \lambda^{-1} & -\lambda^{-2} \\ 0 & \lambda^{-1} \end{bmatrix}$.

11. $u_2(t) = 2e^{\lambda t}$ and $u_1(t) = (5 + 2t)e^{\lambda t}$.

12. Notice $J^2 = 0$ whereas $K^2 \neq 0$. If $M^{-1}JM = K$, then $K^2 = M^{-1}J^2M = 0$, a contradiction (specifically, $JM = MK$ forces the entire third row of M to be zero, so $\det M = 0$).

13. Based on Problem 12, what are the five Jordan forms when $\lambda = 0, 0, 0, 0$?

14. Prove that A^T *is always similar to* A (we know the λ's are the same):
1. For one Jordan block J_i: Find M_i so that $M_i^{-1} J_i M_i = J_i^T$ (see Example 3).
2. For any J with blocks J_i: Build M_0 from blocks so that $M_0^{-1} J M_0 = J^T$.
3. For any $A = MJM^{-1}$: Show that A^T is similar to J^T and so to J and to A.

15. Prove that $\det(A - \lambda I) = \det(M^{-1}AM - \lambda I)$.

16. Which pairs are similar? Choose a, b, c, d to prove that the other pairs aren't:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \begin{bmatrix} b & a \\ d & c \end{bmatrix} \qquad \begin{bmatrix} c & d \\ a & b \end{bmatrix} \qquad \begin{bmatrix} d & c \\ b & a \end{bmatrix}.$$

17. True or false, with a good reason:
(a) A symmetric matrix can't be similar to a nonsymmetric matrix.
(b) An invertible matrix can't be similar to a singular matrix.
(c) A can't be similar to $-A$ unless $A = 0$.
(d) A can't be similar to $A + I$.

18. If B is invertible, prove that AB is similar to BA. *They have the same eigenvalues.*

19. If A is 6 by 4 and B is 4 by 6, AB and BA have different sizes. But with blocks

$$M^{-1}FM = \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} = G.$$

(a) What sizes are the four blocks (the same four sizes in each matrix)?
(b) This equation is $M^{-1}FM = G$, so F and G have the same 10 eigenvalues. F has the 6 eigenvalues of AB plus 4 zeros; G has the 4 eigenvalues of BA plus 6 zeros. *AB has the same eigenvalues as BA plus ____ zeros.*

20. Why are these statements all true?
(a) If A is similar to B then A^2 is similar to B^2.
(b) A^2 and B^2 can be similar when A and B are not similar (try $\lambda = 0, 0$).
(c) $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ is similar to $\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$. (d) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ is not similar to $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.
(e) If we exchange rows 1 and 2 of A, and then exchange columns 1 and 2, the **eigenvalues stay the same**. In this case $M =$ ____.

21. If J is the 5 by 5 Jordan block with $\lambda = 0$, find J^2 and count its eigenvectors and find its Jordan form (there will be two blocks).

22. *(Challenge)* If an n by n matrix A has all eigenvalues $\lambda = 0$, prove that $A^n =$ zero matrix.

23. *(Challenge)* For the shifted QR method in the Worked Example 6.6 B, show that A_2 is similar to A_1.

24. *(Challenge)* If A is similar to A^{-1}, must all the eigenvalues equal 1 or -1?

END-OF-PAGE SOLUTION: Problem Set 6.6 (Problems 13–24)

13. Block sizes of J: (1) one 4×4 block; (2) 3×3 and 1×1; (3) 2×2 and 2×2 (J in Prob. 12); (4) $2 \times 2, 1 \times 1, 1 \times 1$; (5) four 1×1 blocks (zero matrix).
14. 1. $M_i =$ reverse identity of size k_i. 2. $M_0 = \text{diag}(M_1, \ldots, M_s)$. 3. $A^T = (M^{-1})^T J^T M^T = (M^{-1})^T M_0^{-1} J M_0 M^T$, so A^T is similar to J and thus to A.

15. $\det(M^{-1}AM - \lambda I) = \det(M^{-1}(A - \lambda I)M) = \det(M^{-1})\det(A - \lambda I)\det(M) = \det(A - \lambda I)$.

16. The 1st and 4th are similar via $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; the 2nd and 3rd are similar via the same M. The two pairs have different traces ($a + d$ vs. $b + c$) and opposite determinants.

17. (a) **False** (Example 1). (b) **True** (det is preserved). (c) **False** ($\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has $\lambda = \pm i$). (d) **True** ($\text{trace}(A + I) = \text{trace}(A) + n \neq \text{trace}(A)$).

18. $B(AB)B^{-1} = BA$, so AB is similar to BA with $M = B^{-1}$.

19. (a) Upper-left is 6×6, upper-right is 6×4, lower-left is 4×6, lower-right is 4×4. (b) Blank: **2**.

20. (a) $B^2 = (M^{-1}AM)(M^{-1}AM) = M^{-1}A^2M$. (b) $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ have $A^2 = B^2 = 0$. (c) Distinct eigenvalues $3, 4 \implies$ both diagonalize to $\text{diag}(3, 4)$. (d) $3I$ is similar only to $3I$. (e) $M = P_{12}$ (permutation matrix exchanging rows/columns 1 and 2, since $P_{12}^{-1} = P_{12}$).

21. J^2 has 1's on the second superdiagonal (three 1's, rank 3, nullity 2). It has 2 independent eigenvectors, and its Jordan form has one 3×3 block and one 2×2 block.

22. Each Jordan block J_i of size $k_i \leq n$ with $\lambda = 0$ satisfies $J_i^n = 0$, so $A^n = MJ^nM^{-1} = 0$.

23. $Q_1^{-1}A_1Q_1 = Q_1^{-1}(Q_1R_1 - cs^2I)Q_1 = R_1Q_1 - cs^2I = A_2$.

24. **No!** The eigenvalues only need to come in reciprocal pairs λ and $1/\lambda$ (for example, $A = \text{diag}(2, 1/2)$ is similar to $A^{-1} = \text{diag}(1/2, 2)$).

CHAPTER 5

Singular Value Decomposition (Section 6.7)

Syllabus Equations: $Av_i = \sigma_i u_i$, $AV = U\Sigma$, $A = U\Sigma V^T = \sum_{i=1}^{r} \sigma_i u_i v_i^T$, $A^TA = V\Sigma^T\Sigma V^T$, $AA^T = U\Sigma\Sigma^T U^T$, $\|Ax\| \leq \sigma_1\|x\|$, HITS equations $x^1 = A^Ty^0$ and $y^1 = Ax^0$.

5.1 Derivation 1: The Singular Value Decomposition $A = U\Sigma V^T$

GOAL

For any $m \times n$ matrix A of rank r, construct orthonormal bases $V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ of $\mathbb{R}^n$ and $U = \begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix}$ of $\mathbb{R}^m$ such that $AV = U\Sigma$ and $A = U\Sigma V^T$.

FROM YOUR +2 BASE

Use Chapter 3: The $n \times n$ matrix A^TA is symmetric ($(A^TA)^T = A^TA$), so it has real eigenvalues and an orthonormal set of eigenvectors $v_1, \ldots, v_n$. Also $\|Av_i\|^2 = (Av_i)^T(Av_i) = v_i^TA^TAv_i$.

Step 1 Let $v_1, \ldots, v_r$ be the unit eigenvectors of A^TA with positive eigenvalues $\sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_r^2 > 0$, and let $v_{r+1}, \ldots, v_n$ be an orthonormal basis for the nullspace $N(A)$ ($\lambda = 0$):

$$A^TAv_i = \sigma_i^2 v_i \qquad (i = 1, \ldots, r). \tag{5.1}$$

Step 2 Multiply on the left by v_i^T to find the length of Av_i:

$$v_i^TA^TAv_i = \sigma_i^2 v_i^Tv_i \implies \|Av_i\|^2 = \sigma_i^2 \implies \|Av_i\| = \sigma_i > 0. \tag{5.2}$$

Step 3 Define $u_i = \dfrac{Av_i}{\sigma_i}$ for $i = 1, \ldots, r$. Then $\|u_i\| = 1$ and $Av_i = \sigma_i u_i$. Check that the u_i's are orthogonal eigenvectors of AA^T:

$$u_i^Tu_j = \frac{(Av_i)^T(Av_j)}{\sigma_i\sigma_j} = \frac{v_i^T(A^TAv_j)}{\sigma_i\sigma_j} = \frac{\sigma_j^2}{\sigma_i\sigma_j}v_i^Tv_j = 0 \qquad (i \neq j), \tag{5.3}$$

$$(AA^T)u_i = AA^T\left(\frac{Av_i}{\sigma_i}\right) = A\left(\frac{A^TAv_i}{\sigma_i}\right) = A(\sigma_i v_i) = \sigma_i^2 u_i. \tag{5.4}$$

Step 4 Complete $u_1, \ldots, u_r$ with an orthonormal basis $u_{r+1}, \ldots, u_m$ of $N(A^T)$. Put v_i into V and u_i into U. Then $Av_i = \sigma_i u_i$ ($i \leq r$) and $Av_i = 0$ ($i > r$) becomes:

$$\boxed{AV = U\Sigma \implies A = U\Sigma V^{-1} = U\Sigma V^T = \sigma_1 u_1 v_1^T + \cdots + \sigma_r u_r v_r^T.} \tag{5.5}$$

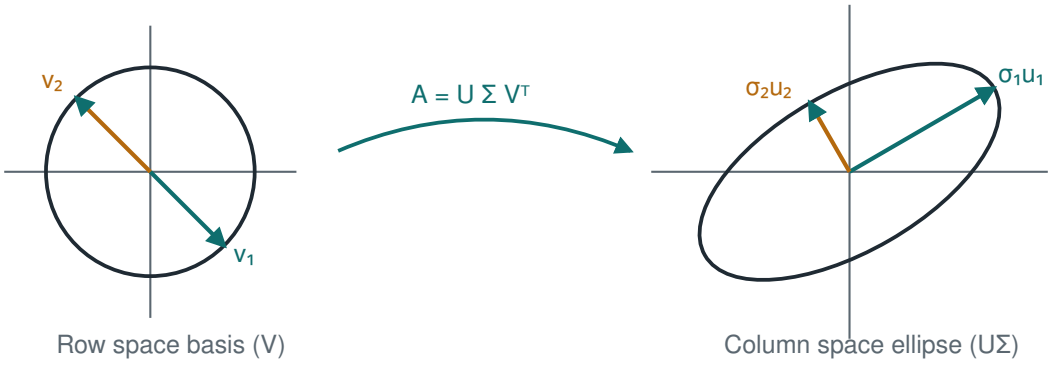


Fig. 5.1 Geometric meaning of the SVD: V^T rotates the unit circle, Σ stretches it by σ_1, σ_2 into an ellipse, and U rotates the ellipse.

RESULT (Orthonormal Bases for the Four Fundamental Subspaces)

- First r columns of V ($v_1, \ldots, v_r$): orthonormal basis for the **row space** $C(A^T)$.
- Last $n - r$ columns of V ($v_{r+1}, \ldots, v_n$): orthonormal basis for the **nullspace** $N(A)$.
- First r columns of U ($u_1, \ldots, u_r$): orthonormal basis for the **column space** $C(A)$.
- Last $m - r$ columns of U ($u_{r+1}, \ldots, u_m$): orthonormal basis for the **left nullspace** $N(A^T)$.

5.2 Section 6.7 In-Text Examples and Worked Examples

BOOK EXAMPLE 6.7.1 When SVD Equals $S\Lambda S^{-1}$

When is $U\Sigma V^T$ (singular values) the same as $S\Lambda S^{-1}$ (eigenvalues)?

BOOK EXAMPLE 6.7.2 SVD of a Rank-One Matrix

If $A = xy^T$ with unit vectors x and y, what is the SVD of A?

BOOK EXAMPLE 6.7.3 SVD of an Invertible 2 by 2 Matrix

Find the singular value decomposition of $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

BOOK EXAMPLE 6.7.4 SVD of a Singular 2 by 2 Matrix

Find the SVD of the singular matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$. The rank is $r = 1$.

BOOK EXAMPLE 6.7.5 Ill-Conditioned Triangular Matrix

Estimate the singular values σ_1 and σ_2 of $A = \begin{bmatrix} 1 & 0 \\ C & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & 0 \\ -C & 1 \end{bmatrix}$.

BOOK EXAMPLE 6.7 A Worked Example 6.7 A: Three Fundamental Matrix Splittings

Identify by name these decompositions $A = c_1b_1 + \cdots + c_rb_r$ of an m by n matrix of rank r:
1. *Orthogonal* columns $c_1, \ldots, c_r$ and *orthogonal* rows $b_1, \ldots, b_r$.
2. *Orthogonal* columns $c_1, \ldots, c_r$ and *triangular* rows $b_1, \ldots, b_r$.
3. *Triangular* columns $c_1, \ldots, c_r$ and *triangular* rows $b_1, \ldots, b_r$.

BOOK EXAMPLE 6.7 C Worked Example 6.7 C: Largest Singular Value Bound

Show that $\sigma_1 \geq |\lambda|_{\max}$ and $\sigma_1 \geq |a_{ij}|_{\max}$.

END-OF-PAGE SOLUTION: Section 6.7 Examples (1–5, 6.7 A, 6.7 C)

Ex. 6.7.1: A must be a symmetric positive semidefinite (or positive definite) matrix $A = Q\Lambda Q^T$ with all $\lambda_i \geq 0$.

Ex. 6.7.2: Reduced SVD is xy^T with $u_1 = x$, $v_1 = y$, and $\sigma_1 = 1$. Complete u_1 to U and v_1 to V.

Ex. 6.7.3: $A^TA = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$ has $\lambda_1 = 8 = \sigma_1^2, \lambda_2 = 2 = \sigma_2^2$, and $v_1 = \frac{1}{\sqrt{2}}(1,1), v_2 = \frac{1}{\sqrt{2}}(-1,1)$. Then $Av_1 = (2\sqrt{2}, 0) = 2\sqrt{2}u_1 \implies u_1 = (1,0), \sigma_1 = 2\sqrt{2}$; $Av_2 = (0, \sqrt{2}) = \sqrt{2}u_2 \implies u_2 = (0,1), \sigma_2 = \sqrt{2}$. Thus $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$.

Ex. 6.7.4: Row space unit vector $v_1 = \frac{1}{\sqrt{2}}(1,1)$; nullspace $v_2 = \frac{1}{\sqrt{2}}(1,-1)$. Column space unit vector $u_1 = \frac{1}{\sqrt{5}}(2,1)$; left nullspace $u_2 = \frac{1}{\sqrt{5}}(1,-2)$. Since $Av_1 = \frac{1}{\sqrt{2}}(4,2) = \sqrt{10}u_1$, $\sigma_1 = \sqrt{10}$ and $\sigma_2 = 0$.

Ex. 6.7.5: First column length gives $\sigma_1 \geq \sqrt{1+C^2} \geq \max(1, |C|)$. Since $\sigma_1\sigma_2 = |\det A| = 1$, $\sigma_2 = 1/\sigma_1 \leq 1/|C|$.
Ex. 6.7 A: 1. Singular Value Decomposition $A = U\Sigma V^T$. 2. Gram-Schmidt QR $A = QR = QHR$. 3. Gaussian Elimination $A = LU = LDU$. Moreover, $|\det A| = |\prod \sigma_i| = |\prod d_i| = |\prod h_i|$.
Ex. 6.7 C: Since orthogonal matrices preserve length, $\|Ax\| = \|U\Sigma V^T x\| = \|\Sigma V^T x\| \leq \sigma_1 \|V^T x\| = \sigma_1 \|x\|$. For an eigenvector $Ax = \lambda x$, $|\lambda| \, \|x\| \leq \sigma_1 \|x\| \implies |\lambda| \leq \sigma_1$. For standard basis vector e_j, $\|Ae_j\| \leq \sigma_1 \implies |a_{ij}| \leq \sigma_1$.

5.3 Problem Set 6.7 (Verbatim Questions and Solutions)

1. Find the eigenvalues and unit eigenvectors v_1, v_2 of A^TA. Then find $u_1 = Av_1/\sigma_1$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad A^TA = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \quad \text{and} \quad AA^T = \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix}.$$

Verify that u_1 is a unit eigenvector of AA^T. Complete the matrices U, Σ, V.

2. Write down orthonormal bases for the four fundamental subspaces of this A.

3. (a) Why is the trace of A^TA equal to the sum of all a_{ij}^2?
(b) For every rank-one matrix, why is $\sigma_1^2 =$ sum of all a_{ij}^2?

4. Find the eigenvalues and unit eigenvectors of A^TA and AA^T. Keep each $Av = \sigma u$:

$$\textbf{Fibonacci matrix} \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Construct the singular value decomposition and verify that A equals $U\Sigma V^T$.

5. Use the **svd** part of the MATLAB demo **eigshow** to find those v's graphically.

6. Compute A^TA and AA^T and their eigenvalues and unit eigenvectors for V and U:

$$\textbf{Rectangular matrix} \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check $AV = U\Sigma$ (this will decide $\pm$ signs in U). Σ has the same shape as A.

7. What is the closest rank-one approximation to that 2 by 3 matrix?

8. A square invertible matrix has $A^{-1} = V\Sigma^{-1}U^T$. This says that the *singular values of* A^{-1} *are* $1/\sigma(A)$. Show that $\sigma_{\max}(A^{-1})\,\sigma_{\max}(A) \geq 1$.

9. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are orthonormal bases for $\mathbb{R}^n$. Construct the matrix A that transforms each v_j into u_j to give $Av_1 = u_1, \ldots, Av_n = u_n$.

10. Construct the matrix with rank one that has $Av = 12u$ for $v = \frac{1}{2}(1,1,1,1)$ and $u = \frac{1}{3}(2,2,1)$. Its only singular value is $\sigma_1 =$ ____.

11. Suppose A has *orthogonal* columns $w_1, w_2, \ldots, w_n$ of lengths $\sigma_1, \sigma_2, \ldots, \sigma_n$. What are U, Σ, and V in the SVD?

12. Suppose A is a 2 by 2 symmetric matrix with unit eigenvectors u_1 and u_2. If its eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = -2$, what are the matrices U, Σ, V^T in its SVD?

13. If $A = QR$ with an orthogonal matrix Q, the SVD of A is almost the same as the SVD of R. Which of the three matrices U, Σ, V is changed because of Q?

14. Suppose A is invertible (with $\sigma_1 > \sigma_2 > 0$). Change A *by as small a matrix as possible* to produce a singular matrix A_0. Hint: U and V do not change.

15. Why doesn't the SVD for $A + I$ just use $\Sigma + I$?

16. *(Challenge)* Find the steady state eigenvector of the Markov web matrix $A = \begin{bmatrix} 0 & .05 & .15 & .1 \\ .1 & 0 & .25 & .3 \\ .2 & .15 & 0 & .6 \\ .7 & .8 & .6 & 0 \end{bmatrix}$.

17. *(Challenge)* For the 4×3 first difference matrix A, verify that A^TA has eigenvalues $\lambda = 2-\sqrt{2}, 2, 2+\sqrt{2}$ with sine eigenvectors V, $AV = U\Sigma$ has orthogonal cosine columns, and find u_4 in $N(A^T)$.

END-OF-PAGE SOLUTION: Problem Set 6.7 (Problems 1–17)

1. $\lambda(A^TA) = 50, 0 \implies \sigma_1 = \sqrt{50} = 5\sqrt{2}$. $v_1 = \frac{1}{\sqrt{5}}(1,2), v_2 = \frac{1}{\sqrt{5}}(2,-1)$; $u_1 = \frac{Av_1}{5\sqrt{2}} = \frac{1}{\sqrt{10}}(1,3), u_2 = \frac{1}{\sqrt{10}}(3,-1)$. $U = \frac{1}{\sqrt{10}}\begin{bmatrix}1 & 3\\ 3 & -1\end{bmatrix}, \Sigma = \begin{bmatrix}\sqrt{50} & 0\\ 0 & 0\end{bmatrix}, V = \frac{1}{\sqrt{5}}\begin{bmatrix}1 & 2\\ 2 & -1\end{bmatrix}$.

2. Row space: $\{v_1\}$. Nullspace: $\{v_2\}$. Column space: $\{u_1\}$. Left nullspace: $\{u_2\}$.

3. (a) The (j,j) diagonal entry of A^TA is the squared length of column j: $\sum_i a_{ij}^2$. Adding down the diagonal gives $\sum_{i,j} a_{ij}^2$. (b) For rank 1, $\sigma_2 = \cdots = \sigma_n = 0$, so $\text{trace}(A^TA) = \sigma_1^2 = \sum a_{ij}^2$.

4. $A^TA = \begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$ has $\sigma_1^2 = \frac{3+\sqrt{5}}{2} = \lambda_1^2$ ($\sigma_1 = \frac{1+\sqrt{5}}{2}$) and $\sigma_2^2 = \frac{3-\sqrt{5}}{2} = \lambda_2^2$ ($\sigma_2 = \frac{\sqrt{5}-1}{2}$).
$V = \frac{1}{\sqrt{\lambda_1^2+1}}\begin{bmatrix}\lambda_1 & -1\\ 1 & \lambda_1\end{bmatrix}, U = \frac{1}{\sqrt{\lambda_1^2+1}}\begin{bmatrix}\lambda_1 & 1\\ 1 & -\lambda_1\end{bmatrix}, \Sigma = \begin{bmatrix}\lambda_1 & 0\\ 0 & -\lambda_2\end{bmatrix}$.

5. v_1 and v_2 are the perpendicular unit vectors where $Av_1 \perp Av_2$.

6. $AA^T = \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ has $\sigma_1 = \sqrt{3}, \sigma_2 = 1$. $U = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $\Sigma = \begin{bmatrix}\sqrt{3} & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$, $V = \begin{bmatrix}1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3}\\ 2/\sqrt{6} & 0 & -1/\sqrt{3}\\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3}\end{bmatrix}$.

7. $\sigma_1 u_1 v_1^T = \sqrt{3}\left(\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ 1\end{bmatrix}\right)\left(\frac{1}{\sqrt{6}}\begin{bmatrix}1 & 2 & 1\end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix}1 & 2 & 1\\ 1 & 2 & 1\end{bmatrix}$.

8. $\sigma_{\max}(A^{-1}) = \dfrac{1}{\sigma_{\min}(A)}$, so $\sigma_{\max}(A^{-1})\sigma_{\max}(A) = \dfrac{\sigma_{\max}(A)}{\sigma_{\min}(A)} \geq 1$.

9. $A = UV^T = u_1v_1^T + \cdots + u_nv_n^T$.

10. $A = 12uv^T = 2\begin{bmatrix}2\\ 2\\ 1\end{bmatrix}\begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}$. Blank: $\sigma_1 = \mathbf{12}$.

11. $U = \begin{bmatrix}w_1/\sigma_1 & \cdots & w_n/\sigma_n\end{bmatrix}$, $\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_n)$, and $V = I$.

12. Since $\sigma_i > 0$, flip the sign of the second vector in V: $U = \begin{bmatrix}u_1 & u_2\end{bmatrix}$, $\Sigma = \begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}$, $V^T = \begin{bmatrix}u_1^T\\ -u_2^T\end{bmatrix}$.

13. If $R = U_R\Sigma V^T$, then $A = (QU_R)\Sigma V^T$, so only U changes (to QU_R).

14. Set the smallest singular value σ_2 to 0: $A_0 = \sigma_1 u_1 v_1^T = A - \sigma_2 u_2 v_2^T$.

15. Because U and V^T are different ($V^T \neq U^{-1}$), so $U(\Sigma + I)V^T = A + UV^T \neq A + I$.

16. Solving $(A - I)p = 0$ with $\sum p_i = 1$ gives $p = \frac{1}{362}(33, 62, 111, 156) \approx (.091, .171, .307, .431)$.

17. (a)-(b) Direct multiplication of A^TAV and AV confirms $\sigma_1 = \sqrt{2-\sqrt{2}}, \sigma_2 = \sqrt{2}, \sigma_3 = \sqrt{2+\sqrt{2}}$. (c) Since the columns of A add to zero, $A^T(1,1,1,1) = 0 \implies u_4 = \frac{1}{2}(1,1,1,1)$.

Module 1: Unique Solved Problems & Master Table

Master Table of Eigenvalues and Eigenvectors [Book p. 374]

Matrix Class	**Eigenvalues** λ_i	**Eigenvectors** x_i
Symmetric: $A^T = A$	Real λ's	Orthogonal $x_i^T x_j = 0$
Orthogonal: $Q^T = Q^{-1}$	All $\lvert\lambda\rvert = 1$	Orthogonal $\bar{x}_i^T x_j = 0$
Skew-symmetric: $A^T = -A$	Pure imaginary λ's	Orthogonal $\bar{x}_i^T x_j = 0$
Complex Hermitian: $\bar{A}^T = A$	Real λ's	Orthogonal $\bar{x}_i^T x_j = 0$
Positive Definite: $x^T Ax > 0$	All $\lambda > 0$	Orthogonal since $A^T = A$
Markov: $m_{ij} > 0, \sum_{i=1}^n m_{ij} = 1$	$\lambda_{\max} = 1$	Steady state $x > 0$
Similar: $B = M^{-1}AM$	$\lambda(B) = \lambda(A)$	$x(B) = M^{-1}x(A)$
Projection: $P = P^2 = P^T$	$\lambda = 1;\ 0$	Column space; nullspace
Plane Rotation	$e^{i\theta}$ and $e^{-i\theta}$	$x = (1, i)$ and $(1, -i)$
Reflection: $I - 2uu^T$	$\lambda = -1;\ 1, \ldots, 1$	u; whole plane $u^\perp$
Rank One: uv^T	$\lambda = v^T u;\ 0, \ldots, 0$	u; whole plane $v^\perp$
Inverse A^{-1} / Shift $A + cI$	$1/\lambda(A)$ / $\lambda(A) + c$	Keep eigenvectors of A
Stable Powers: $A^n \to 0$	All $\lvert\lambda\rvert < 1$	Any eigenvectors
Stable Exponential: $e^{At} \to 0$	All $\mathrm{Re}\,\lambda < 0$	Any eigenvectors
Cyclic Permutation (row 1 last)	$\lambda_k = e^{2\pi ik/n}$	$x_k = (1, \lambda_k, \ldots, \lambda_k^{n-1})$
Tridiagonal: $-1, 2, -1$	$\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$	$x_k = (\sin\frac{k\pi}{n+1}, \sin\frac{2k\pi}{n+1}, \ldots)$
Rectangular SVD: $A = U\Sigma V^T$	$\mathrm{rank}(A) = \mathrm{rank}(\Sigma)$	Eigenvectors of $A^T A, AA^T$ in V, U

BOOK EXAMPLE M1.1 Unique Synthesis Problem: SVD vs. Eigenvalues of $A = xy^T$

For a 2×2 rank-one matrix $A = xy^T$ (with $x, y \in \mathbb{R}^2$ nonzero), compare $A = U\Sigma V^T$ with $A = S\Lambda S^{-1}$ (or Jordan form J when $y^T x = 0$).

END-OF-PAGE SOLUTION: Unique Problem M1.1

1. **SVD ($A = U\Sigma V^T$):** Always exists! Set $u_1 = \frac{x}{\|x\|}$ and $v_1 = \frac{y}{\|y\|}$. Then $\sigma_1 = \|x\|\,\|y\| > 0$ and $\sigma_2 = 0$. Choose $u_2 \perp u_1$ and $v_2 \perp v_1$ to complete the orthogonal matrices U and V.
2. **Eigenvalue Form:** The eigenvalues of $A = xy^T$ are $\lambda_1 = y^T x = \mathrm{trace}(A)$ and $\lambda_2 = 0$.

- **Case 1 ($y^T x \neq 0$):** $\lambda_1 \neq 0$ and $\lambda_2 = 0$ are distinct. A is diagonalizable ($A = S\Lambda S^{-1}$) with $x_1 = x$ (in $C(A)$) and $x_2 = v_2$ (in $N(A)$).
- **Case 2 ($y^T x = 0$):** x is perpendicular to y, so $A^2 = x(y^T x)y^T = 0$. Both eigenvalues are 0 ($\lambda_1 = \lambda_2 = 0$), and $C(A) = N(A)$ is a single line spanned by x. A cannot be diagonalized; its Jordan form is $J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

MODULE 2

Markov Matrices, Population, Economics, and Computer Graphics

CHAPTER 6

Markov Matrices, Population, and Economics (Section 8.3)

Syllabus Equations: Markov conditions $a_{ij} \geq 0$ and $\sum_{i=1}^{n} a_{ij} = 1$, steady state $Au_\infty = u_\infty$ ($\lambda_{\max} = 1$), Perron-Frobenius $Ax = \lambda_{\max} x > 0$, Leslie Matrix, Consumption Matrix $p - Ap = y \implies p = (I - A)^{-1}y = (I + A + A^2 + \dots)y$.

6.1 Derivation 1: Markov Matrix Properties and Steady State

GOAL

Prove that if A has nonnegative entries ($a_{ij} \geq 0$) and every column adds to 1 ($\mathbf{1}^T A = \mathbf{1}^T$), then $\lambda_1 = 1$ is an eigenvalue, no eigenvalue has $|\lambda| > 1$, and $u_k = A^k u_0 \to u_\infty = x_1$ when $A > 0$.

FROM YOUR +2 BASE

Use **T10**: $\det(A - I) = \det(A^T - I)$. If the rows of $A - I$ add to the zero row, $A - I$ is singular.

Step 1 ($\lambda = 1$ is an eigenvalue) Every column of A adds to 1. Therefore every column of $A - I$ adds to $1 - 1 = 0$. Adding all rows of $A - I$ produces the zero row, so the rows are linearly dependent and $\det(A - I) = 0$. Thus $\lambda_1 = 1$ is an eigenvalue.

Step 2 (Probability vectors are preserved) Let $\mathbf{1}^T = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}$. Then $\mathbf{1}^T A = \mathbf{1}^T$. If $u_0 \geq 0$ has components adding to 1 ($\mathbf{1}^T u_0 = 1$), then $u_k = A^k u_0 \geq 0$ and:

$$\mathbf{1}^T A^k u_0 = \mathbf{1}^T u_0 = 1. \tag{6.1}$$

Step 3 ($|\lambda| \leq 1$ and Convergence) Because every entry of A^k remains bounded between 0 and 1, A^k cannot blow up, so $|\lambda_i| \leq 1$ for all i. If $A > 0$ (or $A^m > 0$), then $|\lambda_i| < 1$ for all $i \geq 2$. Expanding $u_0 = x_1 + c_2 x_2 + \cdots + c_n x_n$ (where $\mathbf{1}^T x_1 = 1$ and $\mathbf{1}^T x_i = 0$ for $i \geq 2$) yields:

$$\boxed{u_k = A^k u_0 = x_1 + c_2 \lambda_2^k x_2 + \cdots + c_n \lambda_n^k x_n \xrightarrow{k \to \infty} u_\infty = x_1.} \tag{6.2}$$

6.2 Derivation 2: Perron-Frobenius Theorem for $A > 0$

GOAL

Prove that a strictly positive matrix $A > 0$ has a positive dominant eigenvalue $\lambda_{\max} > 0$ with a strictly positive eigenvector $x > 0$.

Step 1 Consider all numbers $t \geq 0$ such that $Ax \geq tx$ for some nonzero vector $x \geq 0$. Let $t_{\max}$ be the maximum such value.

Step 2 If $Ax \geq t_{\max}x$ were not an equality, multiplying by the strictly positive matrix $A > 0$ would give a strict inequality $A(Ax) > t_{\max}(Ax)$. Setting $y = Ax > 0$ would allow $t_{\max}$ to be increased, contradicting maximality. Therefore $Ax = t_{\max}x$ holds with equality, and $x = Ax/t_{\max} > 0$.

Step 3 For any other eigenvalue $Az = \lambda z$, take absolute values componentwise: $|\lambda|\,|z| = |Az| \leq A|z|$. Thus $|\lambda|$ is a valid candidate t, which proves $\boxed{|\lambda| \leq t_{\max} = \lambda_{\max}}$.

6.3 Derivation 3: Leontief Consumption Matrix and Geometric Series

GOAL

Solve $p - Ap = y$ for nonnegative production $p \geq 0$ given demand $y \geq 0$ and consumption matrix $A \geq 0$.

FROM YOUR +2 BASE

Use **T6**: $1 + x + x^2 + \cdots = (1 - x)^{-1}$ for $|x| < 1$.

Step 1 Write $p - Ap = (I - A)p = y \implies p = (I - A)^{-1}y$.

Step 2 Multiply the matrix series $S = I + A + A^2 + A^3 + \ldots$ by $I - A$:

$$(I - A)(I + A + A^2 + \ldots) = (I + A + A^2 + \ldots) - (A + A^2 + A^3 + \ldots) = I. \tag{6.3}$$

Step 3 If $\lambda_{\max}(A) < 1$, then $A^k \to 0$ and the series converges to a nonnegative matrix:

$$\boxed{(I - A)^{-1} = I + A + A^2 + A^3 + \cdots \geq 0 \iff \lambda_{\max}(A) < 1.} \tag{6.4}$$

6.4 Section 8.3 In-Text Examples

BOOK EXAMPLE 8.3.1 Denver Rental Cars Markov Chain

The fraction of rental cars in Denver starts at $u_0 = (.02, .98)$. Every month, 80% of Denver cars stay in Denver and 5% of outside cars come in, so $A = \begin{bmatrix} .80 & .05 \\ .20 & .95 \end{bmatrix}$. Find u_1, u_2, the eigenvalues and eigenvectors of A, and the long-run steady state u_∞.

BOOK EXAMPLE 8.3.2 Periodic Markov Matrix with No Steady State

Explain why $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has no steady state limit $A^k u_0$.

BOOK EXAMPLE 8.3.3 "Everybody Moves" 3-Group Markov Chain

For $A = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$, compute A^2, find the eigenvalues and steady state x_1, and find u_1, u_2, u_3 starting from $u_0 = (8, 16, 32)$.

BOOK EXAMPLE 8.3.4 Productive Economy ($\lambda_{\max} < 1$)

Show that $A = \begin{bmatrix} .2 & .3 & .4 \\ .4 & .4 & .1 \\ .5 & .1 & .3 \end{bmatrix}$ has $\lambda_{\max} = 0.9 < 1$ and $(I - A)^{-1} \geq 0$.

BOOK EXAMPLE 8.3.5 Unproductive Economy ($\lambda_{\max} > 1$)

Find $\lambda_{\max}$ and $(I - A)^{-1}$ for $A = \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix}$.

END-OF-PAGE SOLUTION: Section 8.3 Examples (1–5)

Ex. 8.3.1: $u_1 = Au_0 = (.065, .935)$; $u_2 = Au_1 = (.09875, .90125)$. Eigenvalues of A are $\lambda_1 = 1$ with $x_1 = (.2, .8)$ and $\lambda_2 = .75$ with $x_2 = (-1, 1)$. Since $u_0 = \begin{bmatrix} .02 \\ .98 \end{bmatrix} = \begin{bmatrix} .2 \\ .8 \end{bmatrix} + (.18)\begin{bmatrix} -1 \\ 1 \end{bmatrix}$, after k steps $u_k = \begin{bmatrix} .2 \\ .8 \end{bmatrix} + (.75)^k(.18)\begin{bmatrix} -1 \\ 1 \end{bmatrix} \to u_\infty = \begin{bmatrix} .2 \\ .8 \end{bmatrix}$.

Ex. 8.3.2: A has $\lambda_1 = 1$ and $\lambda_2 = -1$. Because $|\lambda_2| = 1$, $(-1)^k$ alternates $+1, -1$ and never decays to zero.

Ex. 8.3.3: $A^2 = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} > 0$. Eigenvalues of A are $\lambda_1 = 1, \lambda_2 = \lambda_3 = -1/2$. Steady state is $x_1 = (1/3, 1/3, 1/3)$. Starting from $u_0 = (8, 16, 32)$, $u_1 = (24, 20, 12)$, $u_2 = (16, 18, 22)$, $u_3 = (20, 19, 17) \to (56/3, 56/3, 56/3)$.

Ex. 8.3.4: Every row of A adds to 0.9 $\implies$ $A(1, 1, 1) = 0.9(1, 1, 1)$, so $\lambda_{\max} = 0.9 < 1$. Thus

$(I-A)^{-1} = \frac{1}{93}\begin{bmatrix} 41 & 25 & 27 \\ 33 & 36 & 24 \\ 34 & 23 & 36 \end{bmatrix} > 0.$

Ex. 8.3.5: $\det(A-\lambda I) = \lambda^2 - 4 = 0 \implies \lambda_{\max} = 2 > 1$. $(I-A)^{-1} = \begin{bmatrix} 1 & -4 \\ -1 & 1 \end{bmatrix}^{-1} = -\frac{1}{3}\begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} < 0$ (cannot meet positive demand).

6.5 Problem Set 8.3 (Verbatim Questions and Solutions)

1. Find the eigenvalues of this Markov matrix (their sum is the trace): $A = \begin{bmatrix} .90 & .15 \\ .10 & .85 \end{bmatrix}$. What is the steady state eigenvector for the eigenvalue $\lambda_1 = 1$?

2. Diagonalize the Markov matrix in Problem 1 to $A = S\Lambda S^{-1}$ by finding its other eigenvector. What is the limit of $A^k = S\Lambda^k S^{-1}$ when $\Lambda^k = \begin{bmatrix} 1 & 0 \\ 0 & .75^k \end{bmatrix}$ approaches $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$?

3. What are the eigenvalues and steady state eigenvectors for these Markov matrices?

$$A_1 = \begin{bmatrix} 1 & .2 \\ 0 & .8 \end{bmatrix} \qquad A_2 = \begin{bmatrix} .2 & 1 \\ .8 & 0 \end{bmatrix} \qquad A_3 = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}.$$

4. For every 4 by 4 Markov matrix, what eigenvector of A^T corresponds to the (known) eigenvalue $\lambda = 1$?

5. Every year 2% of young people become old and 3% of old people become dead. (No births.) Find the steady state for

$$\begin{bmatrix} \text{young} \\ \text{old} \\ \text{dead} \end{bmatrix}_{k+1} = \begin{bmatrix} .98 & .00 & 0 \\ .02 & .97 & 0 \\ .00 & .03 & 1 \end{bmatrix} \begin{bmatrix} \text{young} \\ \text{old} \\ \text{dead} \end{bmatrix}_k.$$

6. For a Markov matrix, the sum of the components of x equals the sum of the components of Ax. If $Ax = \lambda x$ with $\lambda \neq 1$, prove that the components of this non-steady eigenvector x add to zero.

7. Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$. Explain why A^k approaches $A^\infty = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}$. Challenge problem: Which Markov matrices produce that steady state $(.6, .4)$?

8. The steady state eigenvector of a permutation matrix is $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$. This is *not* approached when $u_0 = (0, 0, 0, 1)$. What are u_1 and u_2 and u_3 and u_4? What are the four eigenvalues of $P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$, which solve $\lambda^4 = 1$?

9. Prove that the square of a Markov matrix is also a Markov matrix.

10. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a Markov matrix, its eigenvalues are 1 and ____. The steady state eigenvector is $x_1 =$ ____.

END-OF-PAGE SOLUTION: Problem Set 8.3 (Problems 1–10)

1. $\lambda_1 = 1$ and $\lambda_2 = \text{trace}(A) - 1 = 1.75 - 1 = 0.75$. Steady state $x_1 = (0.6, 0.4)$ (or $(3, 2)$).

2. $A = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & .75 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}$. Limit is $\begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}$.

3. A_1: $\lambda = 1, 0.8$, $x_1 = (1,0)$. A_2: $\lambda = 1, -0.8$, $x_1 = (5/9, 4/9)$. A_3: $\lambda = 1, 1/4, 1/4$, $x_1 = (1/3, 1/3, 1/3)$.
4. $x = (1,1,1,1)$.
5. $\lambda = 1, .98, .97$. The steady state for $\lambda = 1$ is $(0,0,1)$ (everyone is eventually dead).
6. $\mathbf{1}^T Ax = \mathbf{1}^T x$ and $\mathbf{1}^T Ax = \lambda \mathbf{1}^T x$. Since $\lambda \neq 1$, $(\lambda - 1)(\mathbf{1}^T x) = 0 \implies \mathbf{1}^T x = x_1 + \cdots + x_n = 0$.
7. $\lambda_1 = 1$ with $x_1 = (.6, .4)$ and $\lambda_2 = 0.5$ with $x_2 = (1, -1)$. Since $0.5^k \to 0$, $A^k \to A^\infty$. Challenge: Any Markov matrix $\begin{bmatrix} 1-2b & 3b \\ 2b & 1-3b \end{bmatrix}$ with $0 < b < 1/2$.
8. $u_1 = (0,0,1,0)$, $u_2 = (0,1,0,0)$, $u_3 = (1,0,0,0)$, $u_4 = (0,0,0,1)$. Eigenvalues are $1, -1, i, -i$.
9. Since $A \geq 0$, $A^2 \geq 0$. Also $\mathbf{1}^T A^2 = (\mathbf{1}^T A)A = \mathbf{1}^T A = \mathbf{1}^T$, so each column of A^2 adds to 1.
10. Blanks: $a + d - 1$ (which equals $a - b = d - c$); $x_1 = \begin{bmatrix} b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 1-a \end{bmatrix}$ (or divided by $b + c$).

11. Complete $A = \begin{bmatrix} .7 & .1 & .2 \\ .1 & .6 & .3 \\ - & - & - \end{bmatrix}$ to a Markov matrix and find the steady state eigenvector. When A is a symmetric Markov matrix, why is $x_1 = (1, \dots, 1)$ its steady state?

12. A Markov differential equation is not $du/dt = Au$ but $du/dt = (A - I)u$. Find the eigenvalues of $B = A - I = \begin{bmatrix} -.2 & .3 \\ .2 & -.3 \end{bmatrix}$. Why does $A - I$ have $\lambda = 0$? When $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ multiply x_1 and x_2, what is the steady state as $t \to \infty$?

13. Each row of the consumption matrix in Example 4 adds to .9. Why does that make $\lambda = .9$ an eigenvalue, and what is the eigenvector?

14. Multiply $I + A + A^2 + A^3 + \dots$ by $I - A$ to show that the series adds to ____. For $A = \begin{bmatrix} 0 & 1/2 \\ 1 & 0 \end{bmatrix}$, find A^2 and A^3 and use the pattern to add up the series.

15. For which of these matrices does $I + A + A^2 + \dots$ yield a nonnegative matrix $(I - A)^{-1}$? Then the economy can meet any demand:

$$A_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 0 & 4 \\ .2 & 0 \end{bmatrix} \qquad A_3 = \begin{bmatrix} .5 & 1 \\ .5 & 0 \end{bmatrix}.$$

If the demands are $y = (2, 6)$, what are the vectors $p = (I - A)^{-1} y$?

16. (Markov again) This matrix has zero determinant. What are its eigenvalues?

$$A = \begin{bmatrix} .4 & .2 & .3 \\ .2 & .4 & .3 \\ .4 & .4 & .4 \end{bmatrix}.$$

Find the limits of $A^k u_0$ starting from $u_0 = (1, 0, 0)$ and then $u_0 = (100, 0, 0)$.

17. If A is a Markov matrix, does $I + A + A^2 + \dots$ add up to $(I - A)^{-1}$?

18. For the Leslie matrix show that $\det(A - \lambda I) = 0$ gives $F_1\lambda^2 + F_2P_1\lambda + F_3P_1P_2 = \lambda^3$.

19. *Sensitivity of eigenvalues:* A matrix change ΔA produces eigenvalue changes $\Delta\Lambda = \operatorname{diag}(S^{-1}\Delta A S)$. Why do the outer terms $S^{-1}A\Delta S$ and $S^{-1}\Delta S\Lambda$ cancel on the diagonal?

20. Suppose $B > A > 0$, meaning that each $b_{ij} > a_{ij} > 0$. How does the Perron-Frobenius discussion show that $\lambda_{\max}(B) > \lambda_{\max}(A)$?

END-OF-PAGE SOLUTION: Problem Set 8.3 (Problems 11–20)

11. Third row is $\begin{bmatrix} .2 & .3 & .5 \end{bmatrix}$. Steady state is $x_1 = (1/3, 1/3, 1/3)$ (or $(1, 1, 1)$). When $A = A^T$, each row is also a column and therefore adds to 1, so $A(1, \dots, 1) = (1, \dots, 1)$.

12. $\lambda_1 = 0, \lambda_2 = -0.5$. $A - I$ is singular because its columns add to zero. As $t \to \infty$, $e^{-0.5t} \to 0$ and $e^{0t} = 1$, so the steady state is $c_1 x_1 = c_1(.6, .4)$.

13. $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \text{row 1 sum} \\ \text{row 2 sum} \\ \text{row 3 sum} \end{bmatrix} = 0.9 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, so $\lambda = 0.9$ with $x = (1, 1, 1)$.

14. Blank: I (so the series is $(I - A)^{-1}$). Here $A^2 = \frac{1}{2}I, A^3 = \frac{1}{2}A$, so $I + A + A^2 + \dots =$

$(1+\frac{1}{2}+\frac{1}{4}+\dots)(I+A) = 2(I+A) = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$.

15. A_1: $\lambda = 0, 0 < 1$ (Yes), $p = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$. A_2: $\lambda = \pm\sqrt{0.8} < 1$ (Yes), $p = 5\begin{bmatrix} 1 & 4 \\ .2 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 130 \\ 32 \end{bmatrix}$. A_3: $\lambda_{\max} = 1$ (No, $I - A_3$ is singular).

16. Markov $\implies \lambda_1 = 1$; $\det A = 0 \implies \lambda_2 = 0$; trace $= 1.2 \implies \lambda_3 = 0.2$. Steady state is $x_1 = (.3, .3, .4)$. From $u_0 = (1, 0, 0)$, limit is $(.3, .3, .4)$; from $u_0 = (100, 0, 0)$, limit is $(30, 30, 40)$.

17. No! A Markov matrix has $\lambda_1 = 1$, so $I - A$ is not invertible and the geometric series diverges.

18. Expanding $\det \begin{bmatrix} F_1 - \lambda & F_2 & F_3 \\ P_1 & -\lambda & 0 \\ 0 & P_2 & -\lambda \end{bmatrix} = 0$ along the first row gives $(F_1 - \lambda)\lambda^2 + F_2P_1\lambda + F_3P_1P_2 = 0$.

19. Since $AS = S\Lambda$, $S^{-1}A = \Lambda S^{-1}$. For any matrix $M = S^{-1}\Delta S$, the diagonal of ΛM equals the diagonal of $M\Lambda$ because $(\Lambda M)_{ii} = \lambda_i m_{ii} = (M\Lambda)_{ii}$. Thus $\text{diag}(\Lambda S^{-1}\Delta S - S^{-1}\Delta S\Lambda) = 0$.

20. Let $Ax = \lambda_{\max}(A)x$ with $x > 0$. Since $B > A$, $Bx > Ax = \lambda_{\max}(A)x$ is a strict inequality, so $t_{\max}(B) = \lambda_{\max}(B) > \lambda_{\max}(A)$.

CHAPTER 7

Homogeneous Coordinates in Computer Graphics (Section 8.7)

Syllabus Equations: Homogeneous point $(x, y, z, 1)$, translation matrix T, scaling matrix S, 3D projection onto a plane $P = I - nn^T$, mirror reflection $M = I - 2nn^T$, affine conjugate $T_- P T_+$.

7.1 Derivation 1: Homogeneous Coordinates for Translation and Scaling

GOAL

Represent both linear transformations $T(v) = Av$ and non-linear shifts (translations) $T(v) = v + v_0$ in $\mathbb{R}^3$ as 4×4 matrix multiplications.

Step 1 Append an extra coordinate 1 to every point $(x, y, z) \in \mathbb{R}^3$ to form the row vector $\begin{bmatrix} x & y & z & 1 \end{bmatrix}$ (in computer graphics, row vectors multiply matrices on the left: $v_{\text{new}} = vM$).

Step 2 To translate by (x_0, y_0, z_0), place $(x_0, y_0, z_0, 1)$ in the bottom row of T:

$$\begin{bmatrix} x & y & z & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ x_0 & y_0 & z_0 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} x + x_0 & y + y_0 & z + z_0 & 1 \end{bmatrix}.} \tag{7.1}$$

Step 3 A difference of two points $(x, y, z, 1) - (0, 0, 0, 1) = (x, y, z, 0)$ is a **vector**, which ends in 0 and is unaffected by translations.

7.2 Derivation 2: 3D Plane Projection $I - nn^T$ and Reflection $I - 2nn^T$

GOAL

Derive the 3×3 projection matrix P and reflection matrix M across a plane $n^T v = 0$ with unit normal vector n ($\|n\| = 1$).

Step 1 The component of v along the unit normal n is $(n^T v)n = (nn^T)v$.

Step 2 Subtract the normal component from v to project onto the plane, and subtract twice the normal component to reflect across the plane:

$$\boxed{Pv = v - (nn^T)v = (I - nn^T)v, \qquad Mv = v - 2(nn^T)v = (I - 2nn^T)v.} \tag{7.2}$$

Step 3 For a plane $n^T v = d$ that does not pass through the origin, translate a point on the plane to the origin by T_-, apply P or M (padded to 4×4 with $0, 0, 0, 1$), and translate back by

$T_+ = T_-^{-1}$:

$$\boxed{P_{\text{affine}} = T_- P T_+, \qquad M_{\text{affine}} = T_- M T_+.} \tag{7.3}$$

7.3 Problem Set 8.7 (Verbatim Questions and Solutions)

1. A typical point in $\mathbb{R}^3$ is $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. The coordinate vectors $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ are $(1,0,0), (0,1,0), (0,0,1)$. The coordinates of the point are (x,y,z). This point in computer graphics is $x\mathbf{i} + y\mathbf{j} + z\mathbf{k} + \mathbf{origin}$. Its homogeneous coordinates are (, , ,). Other coordinates for the same point are (, , ,).

2. A linear transformation T is determined when we know $T(\mathbf{i}), T(\mathbf{j}), T(\mathbf{k})$. For an affine transformation we also need $T(____)$. The input point $(x,y,z,1)$ is transformed to $xT(\mathbf{i}) + yT(\mathbf{j}) + zT(\mathbf{k}) + ____$.

3. Multiply the 4 by 4 matrix T for translation along $(1,4,3)$ and the matrix T_1 for translation along $(0,2,5)$. The product TT_1 is translation along ____.

4. Write down the 4 by 4 matrix S that scales by a constant c. Multiply ST and also TS, where T is translation by $(1,4,3)$. To blow up the picture around the center point $(1,4,3)$, would you use vST or vTS?

5. What scaling matrix S (in homogeneous coordinates, so 3 by 3) would produce a 1 by 1 square page from a standard 8.5 by 11 page?

6. What 4 by 4 matrix would move a corner of a cube to the origin and then multiply all lengths by 2? The corner of the cube is originally at $(1,1,2)$.

7. When the three matrices in equation 1 multiply the unit vector $\mathbf{a}$, show that they give $(\cos\theta)\mathbf{a}$ and $(1-\cos\theta)\mathbf{a}$ and $\mathbf{0}$. Addition gives $\mathbf{a}Q = \mathbf{a}$ and the rotation axis is not moved.

8. If $\mathbf{b}$ is perpendicular to $\mathbf{a}$, multiply by the three matrices in 1 to get $(\cos\theta)\mathbf{b}$ and $\mathbf{0}$ and a vector perpendicular to $\mathbf{b}$. So $Q\mathbf{b}$ makes an angle θ with $\mathbf{b}$. ***This is rotation.***

9. What is the 3 by 3 projection matrix $I - \mathbf{n}\mathbf{n}^T$ onto the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$? In homogeneous coordinates add $0,0,0,1$ as an extra row and column in P.

END-OF-PAGE SOLUTION: Problem Set 8.7 (Problems 1–9)

1. Homogeneous coordinates: $(x,y,z,1)$. Other coordinates: (cx,cy,cz,c) for any $c \neq 0$.

2. Blanks: **origin**; $T(\mathbf{origin})$.

3. $TT_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 6 & 8 & 1 \end{bmatrix}$, which is translation along $(1,6,8)$.

4. $S = \text{diag}(c,c,c,1)$. $ST = \begin{bmatrix} cI_3 & 0 \\ v_0 & 1 \end{bmatrix}$ and $TS = \begin{bmatrix} cI_3 & 0 \\ cv_0 & 1 \end{bmatrix}$. To scale around $(1,4,3)$, first translate $(1,4,3)$ to the origin (T^{-1}), scale by S, and translate back (T): $vT^{-1}ST$. Between vST and vTS, vST scales around the origin and then shifts to $(1,4,3)$.

5. $S = \text{diag}(1/8.5, 1/11, 1)$.

6. $T_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & -2 & -4 & 1 \end{bmatrix}$ (for row vectors vT_S).

7. In Rodrigues' rotation formula $Q = (\cos\theta)I + (1-\cos\theta)\mathbf{a}\mathbf{a}^T + \sin\theta(\mathbf{a}\times)$, $\mathbf{a}^T\mathbf{a} = 1$ and $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, so $\mathbf{a}Q = (\cos\theta)\mathbf{a} + (1-\cos\theta)\mathbf{a} + \mathbf{0} = \mathbf{a}$.

8. Since $\mathbf{a}^T\mathbf{b} = 0$, the middle term $(1-\cos\theta)\mathbf{a}(\mathbf{a}^T\mathbf{b}) = \mathbf{0}$, leaving $(\cos\theta)\mathbf{b} + (\sin\theta)(\mathbf{a} \times \mathbf{b})$.

Because $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{b}\|$ and $(\mathbf{a} \times \mathbf{b}) \perp \mathbf{b}$, the angle with $\mathbf{b}$ is θ.

9. With $\mathbf{n} = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$, $P_{3\times 3} = I - \mathbf{n}\mathbf{n}^T = \frac{1}{9}\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}$.

10. With the same 4 by 4 matrix P, multiply T_-PT_+ to find the projection matrix onto the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$. The translation T_- moves a point on that plane (choose one) to $(0,0,0,1)$. The inverse matrix T_+ moves it back.

11. Project $(3,3,3)$ onto those planes. Use P in Problem 9 and T_-PT_+ in Problem 10.

12. If you project a square onto a plane, what shape do you get?

13. If you project a cube onto a plane, what is the outline of the projection? Make the projection plane perpendicular to a diagonal of the cube.

14. The 3 by 3 mirror matrix that reflects through the plane $\mathbf{n}^T\mathbf{v} = 0$ is $M = I - 2\mathbf{n}\mathbf{n}^T$. Find the reflection of the point $(3,3,3)$ in the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$.

15. Find the reflection of $(3,3,3)$ in the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$. Take three steps T_-MT_+ using 4 by 4 matrices: translate by T_- so the plane goes through the origin, reflect the translated point $(3,3,3,1)T_-$ in that plane, then translate back by T_+.

16. The vector between the origin $(0,0,0,1)$ and the point $(x,y,z,1)$ is the difference $\mathbf{v} =$ ____. In homogeneous coordinates, vectors end in ____. So we add a ____ to a point, not a point to a point.

17. If you multiply only the *last* coordinate of each point to get (x,y,z,c), you rescale the whole space by the number ____. This is because the point (x,y,z,c) is the same as $(\ ,\ ,\ ,1)$.

END-OF-PAGE SOLUTION: Problem Set 8.7 (Problems 10–17)

10. Choose $(0,0,3,1)$ on the plane. Then T_- translates by $(0,0,-3)$ and T_+ translates by $(0,0,3)$: $T_-PT_+ = \begin{bmatrix} 5/9 & -4/9 & -2/9 & 0 \\ -4/9 & 5/9 & -2/9 & 0 \\ -2/9 & -2/9 & 8/9 & 0 \\ 2/3 & 2/3 & 1/3 & 1 \end{bmatrix}$.

11. Onto $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$: $P(3,3,3) = (-1,-1,2)$. Onto $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$: $(-1,-1,2) + \mathbf{n} = (-\frac{1}{3}, -\frac{1}{3}, \frac{7}{3})$.

12. A parallelogram (or a line segment if the square is perpendicular to the plane).

13. A regular hexagon.

14. $M = I - 2\mathbf{n}\mathbf{n}^T = \frac{1}{9}\begin{bmatrix} 1 & -8 & -4 \\ -8 & 1 & -4 \\ -4 & -4 & 7 \end{bmatrix}$. Multiplying by $(3,3,3)$ gives $(-5,-5,1)$.

15. Translate by $(0,0,-3)$: $(3,3,3,1)T_- = (3,3,0,1)$. Reflect: $(3,3,0)M = (-\frac{7}{3}, -\frac{7}{3}, -\frac{8}{3})$. Translate back by $(0,0,3)$: $(-\frac{7}{3}, -\frac{7}{3}, \frac{1}{3}, 1)$ (equivalently $(3,3,3) - 2(\mathbf{n}^T(3,3,3) - 1)\mathbf{n} = (3,3,3) - 8(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) = (-\frac{7}{3}, -\frac{7}{3}, \frac{1}{3})$).

16. Blanks: $(x,y,z,0)$; 0; **vector**.

17. Blanks: $1/c$; $(x/c, y/c, z/c, 1)$.

Module 2: Unique Solved Problems & Formula Synthesis

Module 2 Key Formulas (Markov, Perron-Frobenius, Leontief, Homogeneous Graphics)

- **Markov Steady State:** $a_{ij} \geq 0$, $\sum_{i=1}^{n} a_{ij} = 1 \implies \lambda_1 = 1$, $u_k = A^k u_0 \to u_\infty = x_1$ (normalized so $\sum(x_1)_i = 1$).
- **Eigenvalue Sensitivity:** $\Delta\Lambda = \text{diag}(S^{-1}\Delta A S)$, or $\Delta\lambda_i = y_i^T(\Delta A)x_i$ where $y_i^T x_i = 1$.
- **Leontief Production:** $p - Ap = y \implies p = (I-A)^{-1}y = (I + A + A^2 + \dots)y \geq 0 \iff \lambda_{\max}(A) < 1$.
- **3D Plane Projection & Mirror Reflection ($\|\mathbf{n}\| = 1$):** $P = I - \mathbf{n}\mathbf{n}^T$, $M = I - 2\mathbf{n}\mathbf{n}^T = 2P - I$.

BOOK EXAMPLE M2.1 Unique Synthesis Problem: Leslie Population Sensitivity Analysis

For a 2-age-group Leslie matrix $A = \begin{bmatrix} 0.4 & 1.2 \\ 0.5 & 0 \end{bmatrix}$, find the asymptotic growth rate $\lambda_{\max}$, the right and left eigenvectors x_1 and y_1^T, and compute the first-order change $\Delta\lambda_{\max}$ if the survival rate $P_1 = 0.5$ drops by 10% to 0.45.

END-OF-PAGE SOLUTION: Unique Problem M2.1

1. $\det(A - \lambda I) = \lambda^2 - 0.4\lambda - 0.6 = (\lambda - 1)(\lambda + 0.6) = 0 \implies \lambda_{\max} = 1.0$ (constant total population).

2. Right eigenvector for $\lambda_1 = 1$: $(A - I)x_1 = 0 \implies x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Left eigenvector: $y_1^T(A - I) = 0 \implies y_1^T = \frac{1}{3.2}\begin{bmatrix} 1 & 1.2 \end{bmatrix}$ (normalized so $y_1^T x_1 = \frac{2+1.2}{3.2} = 1$).

3. Perturbation $\Delta A = \begin{bmatrix} 0 & 0 \\ -0.05 & 0 \end{bmatrix}$ gives $\Delta\lambda_{\max} \approx y_1^T(\Delta A)x_1 = \frac{1}{3.2}\begin{bmatrix} 1 & 1.2 \end{bmatrix}\begin{bmatrix} 0 \\ -0.10 \end{bmatrix} = -\frac{0.12}{3.2} = -0.0375$. Thus $\lambda_{\max}^{\text{new}} \approx 0.9625 < 1$, which leads to population extinction!

MODULE 3

Complex Vectors, Hermitian Matrices, and the Fast Fourier Transform

CHAPTER 8

Complex Numbers and Polar Form (Section 10.1)

Syllabus Equations: $i^2 = -1$, $z = a + ib = r(\cos\theta + i\sin\theta) = re^{i\theta}$, $\bar{z} = a - ib = re^{-i\theta}$, $|z|^2 = z\bar{z} = a^2 + b^2 = r^2$, De Moivre's Theorem $z^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta}$, nth roots of unity $w = e^{2\pi i/n}$ $(1 + w + \cdots + w^{n-1} = 0)$.

8.1 Derivation 1: Conjugate, Modulus, Division, and Polar Powers

GOAL

Derive complex division $1/z$, Euler's polar form $z = re^{i\theta}$, and the nth roots of 1 from **T7** and **T8**.

Step 1 (Division) Multiply numerator and denominator of $\dfrac{1}{a+ib}$ by the conjugate $\bar{z} = a - ib$:

$$\boxed{\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{a-ib}{a^2+b^2} = \frac{1}{r}e^{-i\theta}.} \tag{8.1}$$

Step 2 (Products and Powers) Substitute $x = i\theta$ into the Taylor series $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ using $i^2 = -1, i^3 = -i, i^4 = 1$:

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \dots\right) = \cos\theta + i\sin\theta. \tag{8.2}$$

Multiplying $z = re^{i\theta}$ by itself n times multiplies the modulus r^n and adds the angle $n\theta$:

$$\boxed{z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta).} \tag{8.3}$$

Step 3 (nth Roots of 1) Solve $z^n = 1 = e^{2\pi ik}$ for $k = 0, 1, \dots, n-1$. Set $w = e^{2\pi i/n}$:

$$\boxed{z_k = w^k = e^{2\pi ik/n} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \qquad k = 0, 1, \dots, n-1.} \tag{8.4}$$

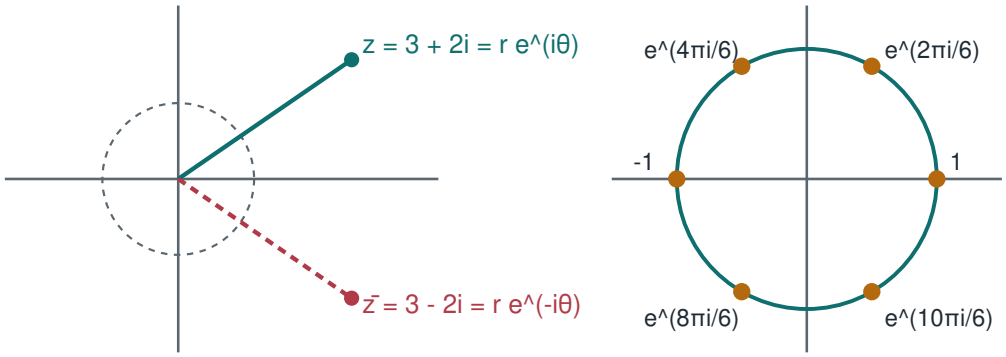


Fig. 8.1 Left: $z = a + ib$ and its conjugate $\bar{z} = a - ib$. Right: The 6th roots of unity equally spaced on the unit circle.

BOOK EXAMPLE 10.1.1 Modulus and Angle of $1+i$ and $1-i$

Find r and θ for $z = 1 + i$ and also for the conjugate $\bar{z} = 1 - i$, and compute $(1+i)^8$.

END-OF-PAGE SOLUTION: Example 10.1.1

$|z| = |\bar{z}| = \sqrt{1^2+1^2} = \sqrt{2}$. For $z = 1+i$, $\theta = \pi/4$ (45°); for $\bar{z} = 1-i$, $\theta = -\pi/4$ (or $7\pi/4$). For $(1+i)^8$: $r^8 = (\sqrt{2})^8 = 2^4 = 16$ and $8\theta = 8(\pi/4) = 2\pi$, so $(1+i)^8 = 16e^{2\pi i} = 16$.

8.2 Problem Set 10.1 (Verbatim Questions and Solutions)

1. Add and multiply each pair of complex numbers:
(a) $2+i, 2-i$ (b) $-1+i, -1+i$ (c) $\cos\theta + i\sin\theta, \cos\theta - i\sin\theta$.

2. Locate these points on the complex plane. Simplify them if necessary:
(a) $2+i$ (b) $(2+i)^2$ (c) $\frac{1}{2+i}$ (d) $|2+i|$.

3. Find the absolute value $r = |z|$ of these four numbers. If θ is the angle for $6-8i$, what are the angles for the other three numbers?
(a) $6-8i$ (b) $(6-8i)^2$ (c) $\frac{1}{6-8i}$ (d) $(6+8i)^2$.

4. If $|z| = 2$ and $|w| = 3$ then $|z \times w| =$ ____ and $|z+w| \leq$ ____ and $|z/w| =$ ____ and $|z-w| \leq$ ____.

5. Find $a+ib$ for the numbers at angles $30°, 60°, 90°, 120°$ on the unit circle. If w is the number at $30°$, check that w^2 is at $60°$. What power of w equals 1?

6. If $z = r\cos\theta + ir\sin\theta$ then $1/z$ has absolute value ____ and angle ____. Its polar form is ____. Multiply $z \times 1/z$ to get 1.

7. The complex multiplication $M = (a+bi)(c+di)$ is a 2 by 2 real multiplication

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \end{bmatrix}.$$

The right side contains the real and imaginary parts of M. Test $M = (1+3i)(1-3i)$.

8. $A = A_1 + iA_2$ is a complex n by n matrix and $b = b_1 + ib_2$ is a complex vector. The solution to $Ax = b$ is $x_1 + ix_2$. Write $Ax = b$ as a real system of size $2n$:

$$\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

9. Write down the complex conjugate of each number by changing i to $-i$:
(a) $2-i$ (b) $(2-i)(1-i)$ (c) $e^{i\pi/2}$ (which is i) (d) $e^{i\pi} = -1$ (e) $\frac{1+i}{1-i}$ (which is also i) (f) $i^{103} =$ ____.

10. The sum $z + \bar{z}$ is always ____. The difference $z - \bar{z}$ is always ____. Assume $z \neq 0$. The product $z \times \bar{z}$ is always ____. The ratio $z/\bar{z}$ always has absolute value ____.

11. Find the eigenvalues $\lambda, \bar{\lambda}$ and eigenvectors $x, \bar{x}$ of $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$.

12. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: (a) If $a = b = d = 1$, the eigenvalues are complex when c is ____. (b) What are the eigenvalues when $ad = bc$? (c) The two eigenvalues (plus sign and minus sign) are not always conjugates of each other. Why not?

END-OF-PAGE SOLUTION: Problem Set 10.1 (Problems 1–12)

1. (a) Sum $= 4$, Product $= 5$. (b) Sum $= -2+2i$, Product $= -2i$. (c) Sum $= 2\cos\theta$, Product $= 1$.
2. (a) $(2,1)$. (b) $3+4i$ at $(3,4)$. (c) $\frac{2-i}{5}$ at $(2/5, -1/5)$. (d) $\sqrt{5}$ at $(\sqrt{5}, 0)$.
3. (a) $r = 10$, angle θ. (b) $r = 100$, angle 2θ. (c) $r = 1/10$, angle $-\theta$. (d) $r = 100$, angle -2θ.
4. Blanks: 6; 5; 2/3; 5.

5. $30°$: $\frac{\sqrt{3}}{2}+\frac{1}{2}i$; $60°$: $\frac{1}{2}+\frac{\sqrt{3}}{2}i$; $90°$: i; $120°$: $-\frac{1}{2}+\frac{\sqrt{3}}{2}i$. $w^{12}=1$.
6. Blanks: $1/r$; $-\theta$; $\frac{1}{r}e^{-i\theta}$.
7. Right side: $\begin{bmatrix} ac-bd \\ bc+ad \end{bmatrix}$. For $(1+3i)(1-3i)$: $\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -3 \end{bmatrix}=\begin{bmatrix} 10 \\ 0 \end{bmatrix}$ $(M=10)$.
8. $\begin{bmatrix} A_1 & -A_2 \\ A_2 & A_1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$.
9. (a) $2+i$. (b) $(2+i)(1+i)=1+3i$. (c) $e^{-i\pi/2}=-i$. (d) -1. (e) $\frac{1-i}{1+i}=-i$. (f) $i^{103}=-i$, so conjugate is $+i$.
10. Blanks: **real** $(2a)$; **pure imaginary** $(2ib)$; **real and positive** $(a^2+b^2>0)$; **1**.
11. $\lambda=a+ib$ with $x=(1,i)$, and $\bar{\lambda}=a-ib$ with $\bar{x}=(1,-i)$.
12. (a) $c<0$. (b) $\lambda_1=a+d$ and $\lambda_2=0$. (c) When $(a+d)^2-4(ad-bc)>0$, both eigenvalues are distinct real numbers, which are not complex conjugates of each other.

13. In Problem 12 the eigenvalues are not real when $(\text{trace})^2 = (a+d)^2$ is smaller than ____. Show that the λ's *are* real when $bc > 0$.

14. Find the eigenvalues and eigenvectors of this permutation matrix:

$$P_4 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \text{has} \quad \det(P_4 - \lambda I) = \text{____}.$$

15. Extend P_4 above to P_6 (five 1's below the diagonal and one in the corner). Find $\det(P_6 - \lambda I)$ and the six eigenvalues in the complex plane.

16. A real skew-symmetric matrix ($A^T = -A$) has pure imaginary eigenvalues. First proof: If $Ax = \lambda x$ then block multiplication gives

$$\begin{bmatrix} 0 & A \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ ix \end{bmatrix} = i\lambda \begin{bmatrix} x \\ ix \end{bmatrix}.$$

This block matrix is symmetric. Its eigenvalues must be ____! So λ is ____.

17. Write these numbers in Euler's form $re^{i\theta}$. Then square each number:
(a) $1 + \sqrt{3}i$ (b) $\cos 2\theta + i \sin 2\theta$ (c) $-7i$ (d) $5 - 5i$.

18. Find the absolute value and the angle for $z = \sin\theta + i\cos\theta$ (careful). Locate this z in the complex plane. Multiply z by $\cos\theta + i\sin\theta$ to get ____.

19. Draw all eight solutions of $z^8 = 1$ in the complex plane. What is the rectangular form $a + ib$ of the root $z = \bar{w} = \exp(-2\pi i/8)$?

20. Locate the cube roots of 1 in the complex plane. Locate the cube roots of -1. Together these are the sixth roots of ____.

21. By comparing $e^{3i\theta} = \cos 3\theta + i \sin 3\theta$ with $(e^{i\theta})^3 = (\cos\theta + i\sin\theta)^3$, find the "triple angle" formulas for $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos\theta$ and $\sin\theta$.

22. Suppose the conjugate $\bar{z}$ is equal to the reciprocal $1/z$. What are all possible z's?

23. (a) Why do e^i and i^e both have absolute value 1? (b) Put stars near e^i and i^e. (c) The number i^e could be $(e^{i\pi/2})^e$ or $(e^{5i\pi/2})^e$. Are those equal?

24. Draw the paths of these numbers from $t = 0$ to $t = 2\pi$ in the complex plane:
(a) e^{it} (b) $e^{(-1+i)t} = e^{-t}e^{it}$ (c) $(-1)^t = e^{t\pi i}$.

END-OF-PAGE SOLUTION: Problem Set 10.1 (Problems 13–24)

13. Blank: $4(ad - bc)$. Discriminant is $(a+d)^2 - 4(ad-bc) = (a-d)^2 + 4bc > 0$ whenever $bc > 0$.
14. $\det(P_4 - \lambda I) = \lambda^4 - 1 = 0 \implies \lambda = 1, -1, i, -i$ with eigenvectors $(1,1,1,1), (1,-1,1,-1), (1,-i,-1,i), (1,i,-1,-i)$.
15. $\det(P_6 - \lambda I) = \lambda^6 - 1 = 0 \implies \lambda_k = e^{2\pi ik/6}$ for $k = 0,1,2,3,4,5$ (which are $\pm 1, \frac{1}{2} \pm i\frac{\sqrt{3}}{2}, -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$).
16. Blanks: **real** (so $i\lambda$ is real); **pure imaginary**.
17. (a) $2e^{i\pi/3}$, square is $4e^{2i\pi/3}$. (b) $e^{2i\theta}$, square is $e^{4i\theta}$. (c) $7e^{-i\pi/2}$ (or $7e^{3i\pi/2}$), square is $49e^{-i\pi} = -49$. (d) $5\sqrt{2}e^{-i\pi/4}$, square is $50e^{-i\pi/2} = -50i$.
18. $r = 1$, angle is $\frac{\pi}{2} - \theta$ ($z = e^{i(\pi/2-\theta)}$). Product: $e^{i(\pi/2-\theta)}e^{i\theta} = e^{i\pi/2} = \mathbf{i}$.

19. $\bar{w} = e^{-i\pi/4} = \cos(-\pi/4) + i\sin(-\pi/4) = \dfrac{\sqrt{2}}{2} - i\dfrac{\sqrt{2}}{2}$.
20. Cube roots of 1: $1, e^{2\pi i/3}, e^{4\pi i/3}$. Cube roots of -1: $-1, e^{\pi i/3}, e^{5\pi i/3}$. Blank: **1**.
21. Expanding $(\cos\theta + i\sin\theta)^3 = (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta)$ gives $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta$.
22. $\bar{z} = 1/z \iff z\bar{z} = |z|^2 = 1 \iff$ all numbers $z = e^{i\theta}$ on the unit circle!
23. (a) $e^i = e^{i(1)}$ has $r = 1, \theta = 1$ rad; $i^e = (e^{i\pi/2})^e = e^{ie\pi/2}$ has $r = 1, \theta = e\pi/2$ rad. (c) **No**, they differ by the factor $e^{2\pi ie} \neq 1$ because e is irrational.
24. (a) Counterclockwise circle of radius 1 (one full turn). (b) Inward spiral starting at $(1, 0)$ and decaying to 0. (c) Counterclockwise circle of radius 1 (piturns, i.e., π full turns).

CHAPTER 9

Hermitian and Unitary Matrices (Section 10.2)

Syllabus Equations: Conjugate transpose $z^H = \bar{z}^T$, length squared $\|z\|^2 = z^H z = \sum |z_i|^2$, complex inner product $u^H v$, adjoint rule $(Au)^H v = u^H (A^H v)$, $(AB)^H = B^H A^H$, Hermitian $A = A^H \implies z^H A z \in \mathbb{R}$, $\lambda \in \mathbb{R}$, $y^H z = 0$, Spectral Theorem $A = U \Lambda U^H$, Unitary $U^H U = I \iff U^{-1} = U^H \implies \|Uz\| = \|z\|$ and $|\lambda| = 1$.

9.1 Derivation 1: Complex Inner Product and Hermitian Matrices

GOAL

Define the complex inner product $u^H v = \bar{u}^T v$ so that $\|z\|^2 = z^H z > 0$, and prove the three fundamental theorems for any Hermitian matrix $A = A^H$: (1) $z^H A z$ is real, (2) every eigenvalue λ is real, and (3) eigenvectors for distinct eigenvalues are orthogonal ($y^H z = 0$).

Step 1 (Length and Inner Product) For $z = (z_1, \ldots, z_n) \in \mathbb{C}^n$, define $z^H = \bar{z}^T = \begin{bmatrix} \bar{z}_1 & \cdots & \bar{z}_n \end{bmatrix}$. Then:

$$\boxed{\|z\|^2 = z^H z = \bar{z}_1 z_1 + \cdots + \bar{z}_n z_n = |z_1|^2 + \cdots + |z_n|^2, \qquad u^H v = \bar{u}_1 v_1 + \cdots + \bar{u}_n v_n.} \tag{9.1}$$

Step 2 ($z^H A z$ is real when $A = A^H$) Take the conjugate transpose of the 1×1 number $z^H A z$:

$$(z^H A z)^H = z^H A^H (z^H)^H = z^H A z. \tag{9.2}$$

Since the scalar $z^H A z$ equals its own complex conjugate, $\boxed{z^H A z \text{ is real}}$.

Step 3 (Real Eigenvalues) If $Az = \lambda z$ ($z \neq 0$), multiply on the left by z^H:

$$z^H A z = \lambda z^H z \implies \boxed{\lambda = \frac{z^H A z}{z^H z} = \frac{\text{real}}{\text{real} > 0} \in \mathbb{R}.} \tag{9.3}$$

Step 4 (Orthogonal Eigenvectors) If $Az = \lambda z$ and $Ay = \beta y$ with $\lambda \neq \beta$, take the conjugate transpose of $Ay = \beta y$ (using $\bar{\beta} = \beta$ and $A^H = A$) to get $y^H A = \beta y^H$. Then:

$$y^H A z = \lambda y^H z \quad \text{and} \quad (y^H A) z = \beta y^H z \implies (\lambda - \beta) y^H z = 0 \implies \boxed{y^H z = 0.} \tag{9.4}$$

9.2 Derivation 2: Unitary Matrices $U^H U = I$

GOAL

Show that a unitary matrix U ($U^H U = I$) preserves lengths and has $|\lambda| = 1$.

Step 1 For any vector $z \in \mathbb{C}^n$, compute the squared length of Uz:

$$\|Uz\|^2 = (Uz)^H(Uz) = z^H U^H U z = z^H I z = z^H z = \|z\|^2 \implies \boxed{\|Uz\| = \|z\|.} \tag{9.5}$$

Step 2 If $Uz = \lambda z$ ($z \neq 0$), then $\|Uz\| = \|\lambda z\| = |\lambda|\,\|z\| = \|z\|$, which forces $\boxed{|\lambda| = 1}$.

9.3 Section 10.2 In-Text Examples

BOOK EXAMPLE 10.2.1 Orthogonal Complex Vectors

Compute the inner product $u^H v$ of $u = \begin{bmatrix} 1 \\ i \end{bmatrix}$ and $v = \begin{bmatrix} i \\ 1 \end{bmatrix}$.

BOOK EXAMPLE 10.2.2 Eigenvalues and Unitary Diagonalization of a Hermitian Matrix

Find the eigenvalues and orthonormal eigenvectors (unitary matrix U) of the Hermitian matrix $A = \begin{bmatrix} 2 & 3-3i \\ 3+3i & 5 \end{bmatrix}$.

BOOK EXAMPLE 10.2.3 The 3 by 3 Fourier Matrix F_3

Show that the 3×3 Fourier matrix $F = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{2\pi i/3} & e^{4\pi i/3} \\ 1 & e^{4\pi i/3} & e^{2\pi i/3} \end{bmatrix}$ is unitary ($F^H F = I$) and find F^{-1}.

END-OF-PAGE SOLUTION: Section 10.2 Examples (1–3)

Ex. 10.2.1: $u^H v = \begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = 1(i) + (-i)(1) = i - i = 0.$

Ex. 10.2.2: $\det(A - \lambda I) = (2-\lambda)(5-\lambda) - |3+3i|^2 = \lambda^2 - 7\lambda + 10 - 18 = (\lambda - 8)(\lambda + 1) = 0 \implies \lambda_1 = 8, \lambda_2 = -1$. For $\lambda_1 = 8$: $(A - 8I)z = 0 \implies z = (1, 1+i)$. For $\lambda_2 = -1$: $(A + I)y = 0 \implies y = (1-i, -1)$. Check orthogonality: $y^H z = \begin{bmatrix} 1+i & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1+i \end{bmatrix} = 0.$ Both have length $\sqrt{3}$, giving $U = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$.

Ex. 10.2.3: Each column has squared length $\frac{1}{3}(1+1+1) = 1$. Inner product of distinct columns is $\frac{1}{3}(1 + e^{2\pi i/3} + e^{4\pi i/3}) = 0$. Therefore F is unitary, and $F^{-1} = F^H = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{-2\pi i/3} & e^{-4\pi i/3} \\ 1 & e^{-4\pi i/3} & e^{-2\pi i/3} \end{bmatrix}$.

9.4 Problem Set 10.2 (Verbatim Questions and Solutions)

1. Find the lengths of $u = (1+i, 1-i, 1+2i)$ and $v = (i, i, i)$. Also find $u^H v$ and $v^H u$.

2. Compute $A^H A$ and AA^H. Those are both ____ matrices: $A = \begin{bmatrix} i & 1 & i \\ 1 & i & i \end{bmatrix}$.

3. Solve $Az = 0$ to find a vector in the nullspace of A in Problem 2. Show that z is orthogonal to the columns of A^H. Show that z is *not* orthogonal to the columns of A^T. ***The good row space is no longer*** $C(A^T)$. ***Now it is*** $C(A^H)$.

4. Problem 3 indicates that the four fundamental subspaces are $C(A)$ and $N(A)$ and ____ and ____. Their dimensions are still r and $n-r$ and r and $m-r$.

5. (a) Prove that $A^H A$ is always a Hermitian matrix.
(b) If $Az = 0$ then $A^H Az = 0$. If $A^H Az = 0$, multiply by z^H to prove that $Az = 0$. The nullspaces of A and $A^H A$ are ____.

6. True or false (give a reason if true or a counterexample if false):
(a) If A is a real matrix then $A + iI$ is invertible.
(b) If A is a Hermitian matrix then $A + iI$ is invertible.
(c) If U is a unitary matrix then $U + iI$ is invertible.

7. When you multiply a Hermitian matrix by a real number c, is cA still Hermitian? Show that iA is skew-Hermitian when A is Hermitian.

8. Which classes of matrices does $P = \begin{bmatrix} 0 & i & 0 \\ 0 & 0 & i \\ i & 0 & 0 \end{bmatrix}$ belong to: invertible, Hermitian, unitary? Compute P^2, P^3, and P^{100}. What are the eigenvalues of P?

9. Find the unit eigenvectors of P in Problem 8, and put them into the columns of a unitary matrix F. What property of P makes these eigenvectors orthogonal?

10. Write down the 3 by 3 circulant matrix $C = 2I + 5P$. It has the same eigenvectors as P in Problem 8. Find its eigenvalues.

11. If U and V are unitary matrices, show that U^{-1} is unitary and also UV is unitary.

12. How do you know that the determinant of every Hermitian matrix is real?

13. The matrix $A^H A$ is not only Hermitian but also positive definite, when the columns of A are independent. Proof: $z^H A^H Az$ is positive if z is nonzero because ____.

14. Diagonalize this Hermitian matrix to reach $A = U\Lambda U^H$: $A = \begin{bmatrix} 0 & 1-i \\ i+1 & 1 \end{bmatrix}$.

15. Diagonalize this skew-Hermitian matrix to reach $K = U\Lambda U^H$. All λ's are ____: $K = \begin{bmatrix} 0 & -1+i \\ 1+i & i \end{bmatrix}$.

END-OF-PAGE SOLUTION: Problem Set 10.2 (Problems 1–15)

1. $\|u\| = \sqrt{2+2+5} = 3$; $\|v\| = \sqrt{1+1+1} = \sqrt{3}$. $u^H v = 2+3i$ and $v^H u = \overline{u^H v} = 2-3i$.

2. Blank: **Hermitian**. $AA^H = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $A^H A = \begin{bmatrix} 2 & 0 & 1+i \\ 0 & 2 & 1+i \\ 1-i & 1-i & 2 \end{bmatrix}$.

3. $z = (1, 1, -1+i)$. Then A^H columns give $(A^H)^H z = Az = 0$, whereas A^T columns give $(A^T)^H z = \bar{A}z = (-2i, -2i)^T \neq 0$.
4. Blanks: $C(A^H)$ **and** $N(A^H)$.
5. (a) $(A^H A)^H = A^H (A^H)^H = A^H A$. (b) $z^H A^H A z = \|Az\|^2 = 0 \implies Az = 0$. Blank: **the same**.
6. (a) **False** ($\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has $\lambda = -i$). (b) **True** (A has only real λ's, so $\lambda \neq -i$). (c) **False** ($U = -iI$ is unitary with $\lambda = -i$).
7. Yes, $(cA)^H = \bar{c}A^H = cA$ when $c \in \mathbb{R}$. Also $(iA)^H = -iA^H = -(iA)$ (skew-Hermitian).
8. P is **invertible** and **unitary** ($P^H P = I$, not Hermitian). $P^2 = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$, $P^3 = -iI$, $P^{100} = (P^3)^{33} P = (-i)^{33} P = -iP$. Eigenvalues satisfy $\lambda^3 = -i \implies \lambda = i, ie^{2\pi i/3}, ie^{4\pi i/3}$.
9. $F = F_3$ (the 3×3 Fourier matrix). P is **unitary** (and therefore normal), which makes its eigenvectors orthogonal.
10. $C = \begin{bmatrix} 2 & 5i & 0 \\ 0 & 2 & 5i \\ 5i & 0 & 2 \end{bmatrix}$ has eigenvalues $2 + 5\lambda(P) = 2 + 5i, 2 + 5ie^{2\pi i/3}, 2 + 5ie^{4\pi i/3}$.
11. $(U^{-1})^H U^{-1} = (U^H)^H U^H = UU^H = I$, and $(UV)^H (UV) = V^H U^H UV = V^H IV = I$.
12. $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$ is the product of real eigenvalues.
13. Blank: $z^H A^H A z = (Az)^H (Az) = \|Az\|^2 > 0$ (since independent columns mean $Az \neq 0$).
14. $\lambda^2 - \lambda - 2 = 0 \implies \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ and $U = \frac{1}{\sqrt{3}} \begin{bmatrix} 1-i & 1 \\ 1 & -1-i \end{bmatrix}$.
15. Blank: **pure imaginary**. $\Lambda = \begin{bmatrix} 2i & 0 \\ 0 & -i \end{bmatrix}$ and $U = \frac{1}{\sqrt{3}} \begin{bmatrix} 1+i & 1 \\ 1 & -1+i \end{bmatrix}$.

16. Diagonalize this orthogonal matrix to reach $Q = U\Lambda U^H$. Now all λ's are ____: $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

17. Diagonalize this unitary matrix V to reach $V = U\Lambda U^H$. Again all λ's are ____: $V = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$.

18. If $v_1, \ldots, v_n$ is an orthonormal basis for $\mathbb{C}^n$, the matrix with those columns is a ____ matrix. Show that any vector z equals $(v_1^H z)v_1 + \cdots + (v_n^H z)v_n$.

19. The functions e^{-ix} and e^{ix} are orthogonal on the interval $0 \le x \le 2\pi$ because their inner product is $\int_0^{2\pi}$ ____ $= 0$.

20. The vectors $v = (1, i, 1), w = (i, 1, 0)$ and $z =$ ____ are an orthogonal basis for ____.

21. If $A = R + iS$ is a Hermitian matrix (R, S real), are its real and imaginary parts symmetric?

22. The (complex) dimension of $\mathbb{C}^n$ is ____. Find a non-real basis for $\mathbb{C}^n$.

23. Describe all 1 by 1 and 2 by 2 Hermitian matrices and unitary matrices.

24. How are the eigenvalues of A^H related to the eigenvalues of the square complex matrix A?

25. If $u^H u = 1$ show that $I - 2uu^H$ is Hermitian and also unitary. The rank-one matrix uu^H is the projection onto what line in $\mathbb{C}^n$?

26. If $A + iB$ is a unitary matrix (A and B are real) show that $Q = \begin{bmatrix} A & -B \\ B & A \end{bmatrix}$ is an orthogonal matrix.

27. If $A + iB$ is Hermitian (A and B are real) show that $\begin{bmatrix} A & -B \\ B & A \end{bmatrix}$ is symmetric.

28. Prove that the inverse of a Hermitian matrix is also Hermitian (transpose $A^{-1}A = I$).

29. Diagonalize this matrix by constructing its eigenvalue matrix Λ and its eigenvector matrix S: $A = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix} = A^H$.

30. A matrix with orthonormal eigenvectors has the form $A = U\Lambda U^{-1} = U\Lambda U^H$. *Prove that* $AA^H = A^H A$. These are exactly the **normal matrices**. Construct a 2 by 2 normal matrix by choosing complex eigenvalues in Λ.

END-OF-PAGE SOLUTION: Problem Set 10.2 (Problems 16–30)

16. Blank: **on the unit circle ($|\lambda| = 1$)**. $\Lambda = \begin{bmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{bmatrix}$ and $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$.

17. Blank: ± 1 **(since V is both unitary and Hermitian)**. $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

18. Blank: **unitary**. Since $UU^H = I$, $z = UU^H z = \sum_{k=1}^n v_k(v_k^H z)$.

19. Blank: $\overline{(e^{-ix})}(e^{ix})\,dx = \int_0^{2\pi} e^{2ix}\,dx = \left[\frac{e^{2ix}}{2i}\right]_0^{2\pi} = 0$.

20. Blanks: $z = (1, -i, -2)$; $\mathbb{C}^3$.

21. $A^H = R^T - iS^T = R + iS \implies R^T = R$ (R is symmetric) and $S^T = -S$ (S is skew-symmetric).

22. Blank: n. Non-real basis: $ie_1, ie_2, \ldots, ie_n$.

23. 1×1 Hermitian: $[a]$ ($a\in\mathbb{R}$); 1×1 Unitary: $[e^{i\theta}]$. 2×2 Hermitian: $\begin{bmatrix} a & b+ic \\ b-ic & d \end{bmatrix}$ ($a,b,c,d\in\mathbb{R}$); 2×2 Unitary: $\begin{bmatrix} a & b \\ -e^{i\phi}\bar{b} & e^{i\phi}\bar{a} \end{bmatrix}$ with $|a|^2+|b|^2=1$.

24. The eigenvalues of A^H are the complex conjugates $\bar{\lambda}_1,\dots,\bar{\lambda}_n$ of the eigenvalues of A.

25. $(I-2uu^H)^H = I-2uu^H$, and $(I-2uu^H)^2 = I-4uu^H+4u(u^Hu)u^H = I$. Projection onto the line spanned by u.

26. $(A-iB^T)(A+iB)=I \implies A^TA+B^TB=I$ and $A^TB-B^TA=0$, which gives $Q^TQ=I$.

27. $A^T=A$ and $B^T=-B$, so $\begin{bmatrix} A & -B \\ B & A \end{bmatrix}^T = \begin{bmatrix} A^T & B^T \\ -B^T & A^T \end{bmatrix} = \begin{bmatrix} A & -B \\ B & A \end{bmatrix}$.

28. $(A^{-1}A)^H = A^H(A^{-1})^H = A(A^{-1})^H = I \implies (A^{-1})^H = A^{-1}$.

29. $\lambda^2-5\lambda+4=0 \implies \Lambda=\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ and $S=U=\frac{1}{\sqrt{3}}\begin{bmatrix} 1-i & -1+i \\ 2 & 1 \end{bmatrix}$.

30. $AA^H = (U\Lambda U^H)(U\bar{\Lambda}U^H) = U\Lambda\bar{\Lambda}U^H = U\bar{\Lambda}\Lambda U^H = A^HA$. Example: $\begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$ has $\lambda=1\pm i$.

CHAPTER 10

The Fast Fourier Transform (Section 10.3)

Syllabus Equations: $n \times n$ Fourier Matrix $(F_n)_{jk} = w^{jk}$ $(w = e^{2\pi i/n})$, $F_n^{-1} = \frac{1}{n}\bar{F}_n$, Radix-2 FFT Factorization $F_{2m} = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_m & 0 \\ 0 & F_m \end{bmatrix} P$, FFT Multiplication Count $\frac{1}{2}n\ell = \frac{1}{2}n\log_2 n$, Circulant Diagonalization $C = FEF^{-1}$.

10.1 Derivation 1: Radix-2 FFT Factorization and $\frac{1}{2}n\log_2 n$ Count

GOAL

Factor the $n \times n$ Fourier matrix F_n (for $n = 2^\ell$) into sparse matrices and prove that the multiplication count drops from n^2 to $\frac{1}{2}n\log_2 n$.

Step 1 (Even-Odd Splitting) Separate the input vector $c = (c_0, c_1, \dots, c_{n-1})$ into its even-indexed components $c' = (c_0, c_2, \dots)$ and odd-indexed components $c'' = (c_1, c_3, \dots)$ using the permutation matrix P.

Step 2 (Half-Size Reduction) Since $w_n^2 = (e^{2\pi i/n})^2 = e^{2\pi i/(n/2)} = w_{n/2}$, the n-point transform splits into two $(n/2)$-point transforms $F_{n/2}c'$ and $F_{n/2}c''$, recombined with the diagonal matrix $D = \mathrm{diag}(1, w_n, w_n^2, \dots, w_n^{n/2-1})$:

$$\boxed{F_n = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_{n/2} & 0 \\ 0 & F_{n/2} \end{bmatrix} \begin{bmatrix} P \end{bmatrix}.} \tag{10.1}$$

Step 3 (Operation Count) For $n = 2^\ell$, there are $\ell = \log_2 n$ recursive levels going from $n = 2^\ell$ down to $n = 1$. At each level, the only complex multiplications come from the diagonal D matrices of total size $n/2$. Across ℓ levels, the total number of multiplications is:

$$\boxed{\text{FFT Multiplications} = \ell \times \frac{n}{2} = \frac{1}{2}n\ell = \frac{1}{2}n\log_2 n.} \tag{10.2}$$

10.2 Problem Set 10.3 (Verbatim Questions and Solutions)

1. Multiply the three matrices in equation (3) for F_4 and compare with F_4. In which six entries do you need to know that $i^2 = -1$?

2. Invert the three factors in equation (3) to find a fast factorization of F^{-1}.

3. F is symmetric. So transpose equation (3) to find a new Fast Fourier Transform!

4. All entries in the factorization of F_6 involve powers of $w_6 =$ sixth root of 1:

$$F_6 = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_3 & \\ & F_3 \end{bmatrix} \begin{bmatrix} P \end{bmatrix}.$$

Write down these matrices with $1, w_6, w_6^2$ in D and $w_3 = w_6^2$ in F_3. Multiply!

5. If $v = (1,0,0,0)$ and $w = (1,1,1,1)$, show that $Fv = w$ and $Fw = 4v$. Therefore $F^{-1}w = v$ and $F^{-1}v =$ ____.

6. What is F^2 and what is F^4 for the 4 by 4 Fourier matrix?

7. Put the vector $c = (1,0,1,0)$ through the three steps of the FFT to find $y = Fc$. Do the same for $c = (0,1,0,1)$.

8. Compute $y = F_8 c$ by the three FFT steps for $c = (1,0,1,0,1,0,1,0)$. Repeat the computation for $c = (0,1,0,1,0,1,0,1)$.

9. If $w = e^{2\pi i/64}$ then w^2 and $\sqrt{w}$ are among the ____ and ____ roots of 1.

10. (a) Draw all the sixth roots of 1 on the unit circle. Prove they add to zero.
(b) What are the three cube roots of 1? Do they also add to zero?

11. The columns of the Fourier matrix F are the *eigenvectors* of the cyclic permutation P. Multiply PF to find the eigenvalues λ_1 to λ_4:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \lambda_3 & \\ & & & \lambda_4 \end{bmatrix}.$$

12. The equation $\det(P - \lambda I) = 0$ is $\lambda^4 = 1$. This shows again that the eigenvalue matrix Λ is ____. Which permutation P has eigenvalues = cube roots of 1?

13. (a) Two eigenvectors of the circulant matrix $C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3$ are $(1,1,1,1)$ and $(1, i, i^2, i^3)$. Find the eigenvalues e_1 and e_2.
(b) $C = F(c_0 I + c_1 \Lambda + c_2 \Lambda^2 + c_3 \Lambda^3) F^{-1} = FEF^{-1}$. That matrix E in parentheses is diagonal. It contains the ____ of C.

14. Find the eigenvalues of the "periodic" $-1, 2, -1$ matrix $C = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$ from $E = 2I - \Lambda - \Lambda^3$.

15. ***Fast convolution.*** To multiply C times a vector x, we can multiply $F(E(F^{-1}x))$ instead. The second way uses only $n \log_2 n + n$ multiplications. How many of those come from E, how many from F, and how many from F^{-1}?

16. Why is row i of $\bar{F}$ the same as row $N - i$ of F (numbered 0 to $N - 1$)?

END-OF-PAGE SOLUTION: Problem Set 10.3 (Problems 1–16)

1. In the lower-right 3×3 block of $F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$, the six entries containing $i^2 = -1, i^3 = -i, i^4 = 1, i^6 = -1, i^9 = i$ use $i^2 = -1$.

2. $F_4^{-1} = P^T \begin{bmatrix} F_2^{-1} & 0 \\ 0 & F_2^{-1} \end{bmatrix} \frac{1}{2} \begin{bmatrix} I & I \\ D^{-1} & -D^{-1} \end{bmatrix}$.

3. $F_n = F_n^T = P^T \begin{bmatrix} F_{n/2} & 0 \\ 0 & F_{n/2} \end{bmatrix} \begin{bmatrix} I & I \\ D & -D \end{bmatrix}$.

4. $D = \mathrm{diag}(1, w_6, w_6^2)$, $F_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & w_6^2 & w_6^4 \\ 1 & w_6^4 & w_6^2 \end{bmatrix}$, and P separates even indices $(0,2,4)$ from odd indices $(1,3,5)$.

5. $F_4(1,0,0,0) =$ column 0 of $F_4 = (1,1,1,1) = w$. $F_4(1,1,1,1) = (4,0,0,0) = 4v$. Blank: $\frac{1}{4}w$.

6. $F_4^2 = 4 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ and $F_4^4 = 16I$.

7. For $c = (1,0,1,0)$: $Pc = (1,1,0,0) \to (2,0,0,0) \to y = (2,0,2,0)$. For $c = (0,1,0,1)$: $Pc = (0,0,1,1) \to (0,0,2,0) \to y = (2,0,-2,0)$.

8. For $c = (1,0,1,0,1,0,1,0)$: $y = (4,0,0,0,4,0,0,0)$. For $c = (0,1,0,1,0,1,0,1)$: $y = (4,0,0,0,-4,0,0,0)$.

9. Blanks: **32nd**; **128th**.

10. (a) $(1-w_6)(1+w_6+\cdots+w_6^5) = 1 - w_6^6 = 0$ and $w_6 \neq 1$, so the sum is 0. (b) $1, e^{2\pi i/3}, e^{4\pi i/3}$; yes, they add to 0.

11. $\lambda_1 = 1, \lambda_2 = i, \lambda_3 = i^2 = -1, \lambda_4 = i^3 = -i$.

12. Blank: $\mathrm{diag}(1, i, -1, -i)$. The 3×3 cyclic permutation $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

13. (a) $e_1 = c_0 + c_1 + c_2 + c_3$ and $e_2 = c_0 + c_1 i + c_2 i^2 + c_3 i^3$. (b) Blank: **eigenvalues** ($E = Fc$).

14. $e_1 = 2 - 1 - 1 = 0$; $e_2 = 2 - i - i^3 = 2$; $e_3 = 2 - (-1) - (-1) = 4$; $e_4 = 2 - (-i) - (-i)^3 = 2$. Eigenvalues are $0, 2, 4, 2$.

15. E uses n multiplications; F uses $\frac{1}{2} n \log_2 n$; F^{-1} uses $\frac{1}{2} n \log_2 n$. Total $= n \log_2 n + n$.

16. Because $\bar{w}^{ik} = w^{-ik} = w^{(N-i)k}$ (since $w^{Nk} = (w^N)^k = 1^k = 1$).

Module 3: Unique Solved Problems & Formula Synthesis

Module 3 Key Formulas (Complex Numbers, Hermitian/Unitary, Circulants, and FFT)

- **Complex Inner Product & Adjoint:** $u^H v = \bar{u}^T v$, $\|z\|^2 = z^H z$, $(Au)^H v = u^H (A^H v)$, $(AB)^H = B^H A^H$.
- **Hermitian ($A^H = A$):** $z^H A z \in \mathbb{R}$, all $\lambda_i \in \mathbb{R}$, $A = U \Lambda U^H$.
- **Unitary ($U^H U = I$):** $U^{-1} = U^H$, $\|Uz\| = \|z\|$, all $|\lambda_i| = 1$.
- **Fourier & Circulant Matrices:** $(F_n)_{jk} = e^{2\pi i jk/n}$, $F_n^{-1} = \frac{1}{n} F_n^H$. Any circulant $C = \sum_{k=0}^{n-1} c_k P^k$ is diagonalized by F_n: $C = F_n E F_n^{-1}$ with eigenvalues $E = \text{diag}(F_n c)$.

BOOK EXAMPLE M3.1 Unique Synthesis Problem: Solving a Circulant System via 4-Point FFT

Solve the 4×4 periodic system $Cx = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 0 \\ -4 \\ 0 \end{bmatrix}$ using the Fourier matrix F_4.

END-OF-PAGE SOLUTION: Unique Problem M3.1

1. From Problem 10.3.14, $C = F_4 E F_4^{-1}$ where $E = \text{diag}(0, 2, 4, 2)$ holds the eigenvalues of C corresponding to the columns f_0, f_1, f_2, f_3 of F_4.

2. Transform the right-hand side $b = (4, 0, -4, 0)^T$: notice that $b = 2 \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix} = 2f_1 + 2f_3$.

3. Since $Cf_1 = 2f_1$ ($\lambda_2 = 2$) and $Cf_3 = 2f_3$ ($\lambda_4 = 2$), divide each Fourier component by its eigenvalue $\lambda = 2$, and add any multiple of the nullspace eigenvector $f_0 = (1, 1, 1, 1)^T$ ($\lambda_1 = 0$):

$$x = \frac{2}{2} f_1 + \frac{2}{2} f_3 + c\, f_0 = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$